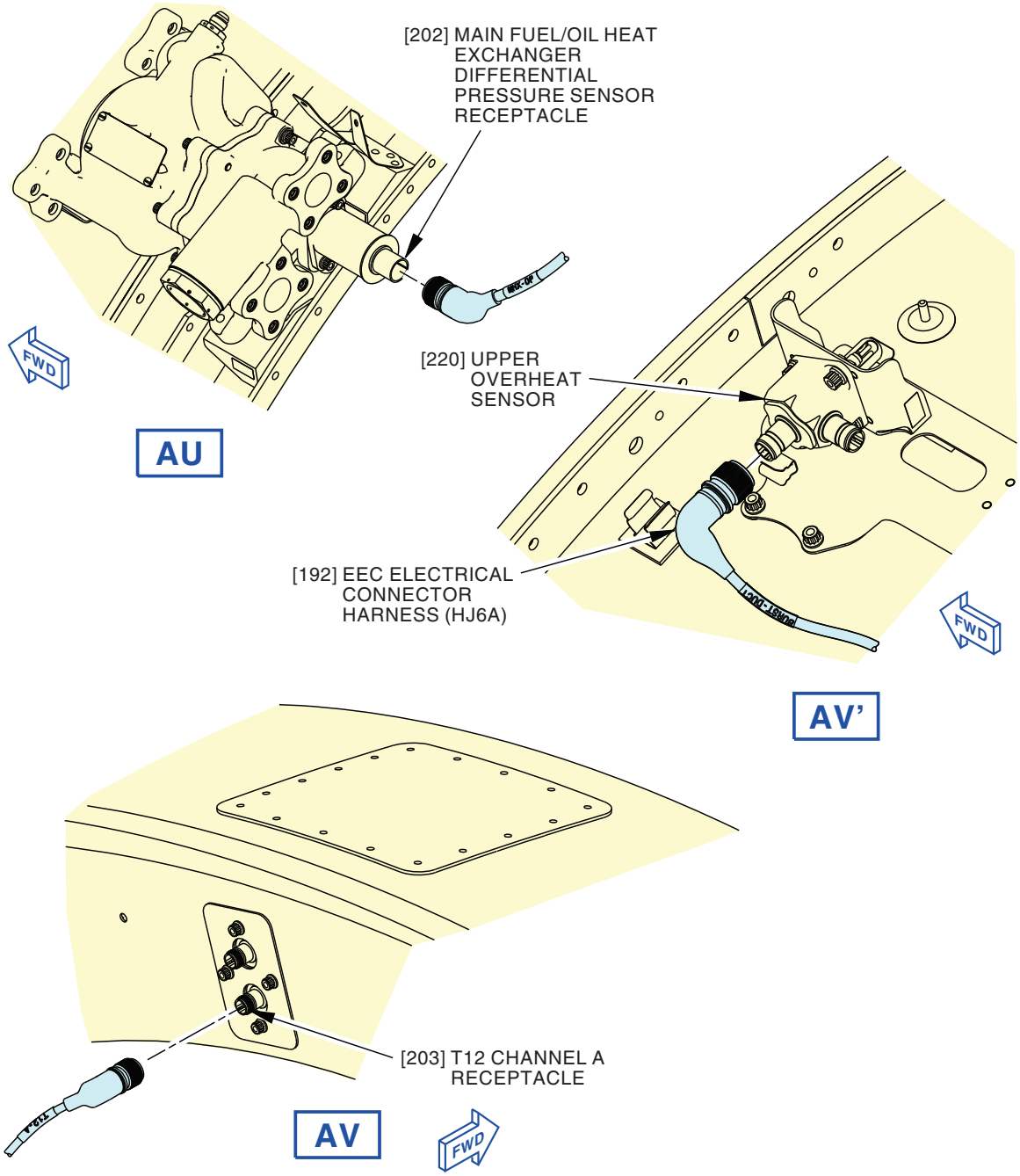
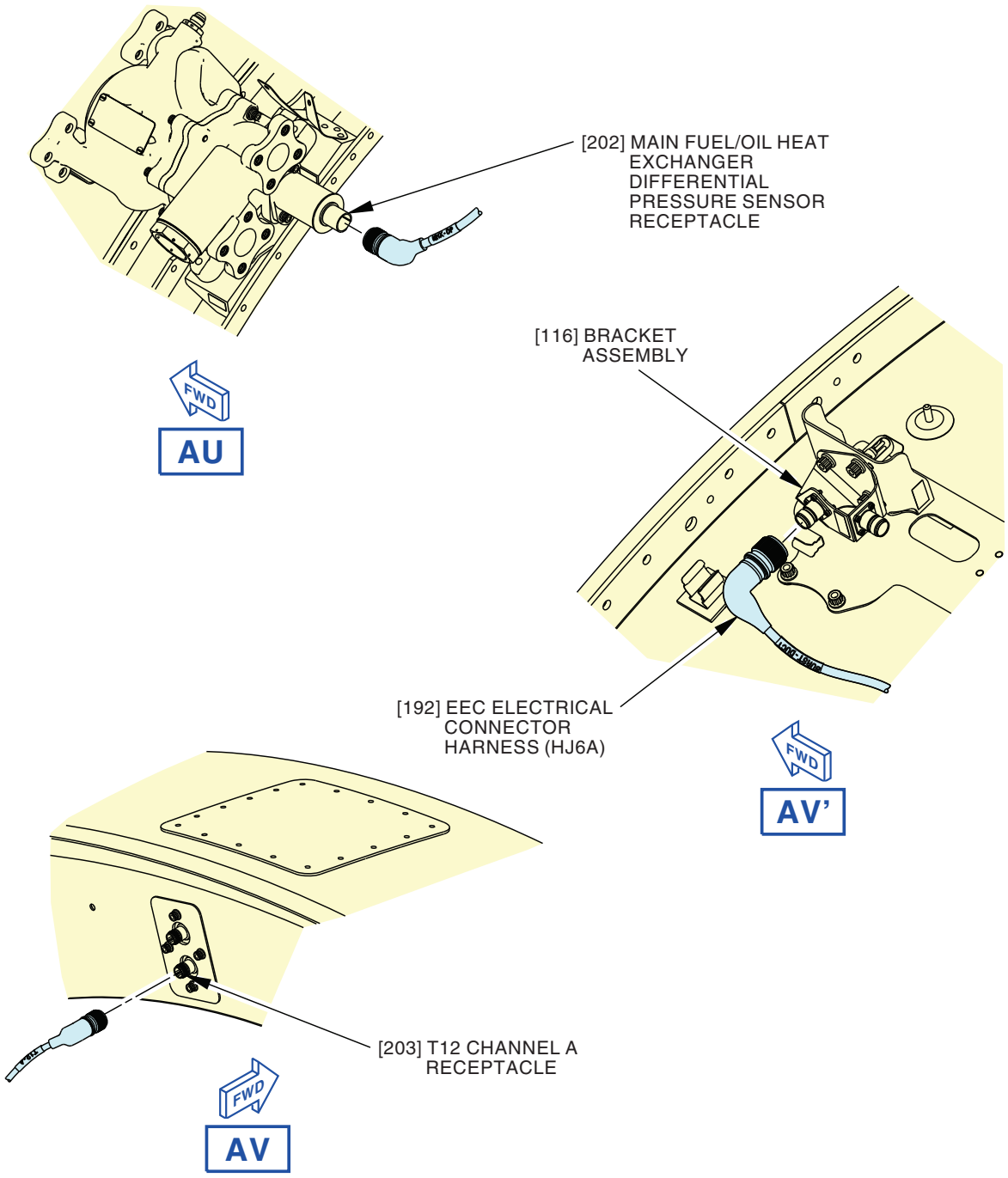


ICN-LEAP-1B-732201-B-F0301-A0037-A-001-01
2354875 S00061516091_V2

EEC Electrical Harness (HJ6A) Installation
Figure 409/73-22-01-990-855-G00 (Sheet 4 of 9)

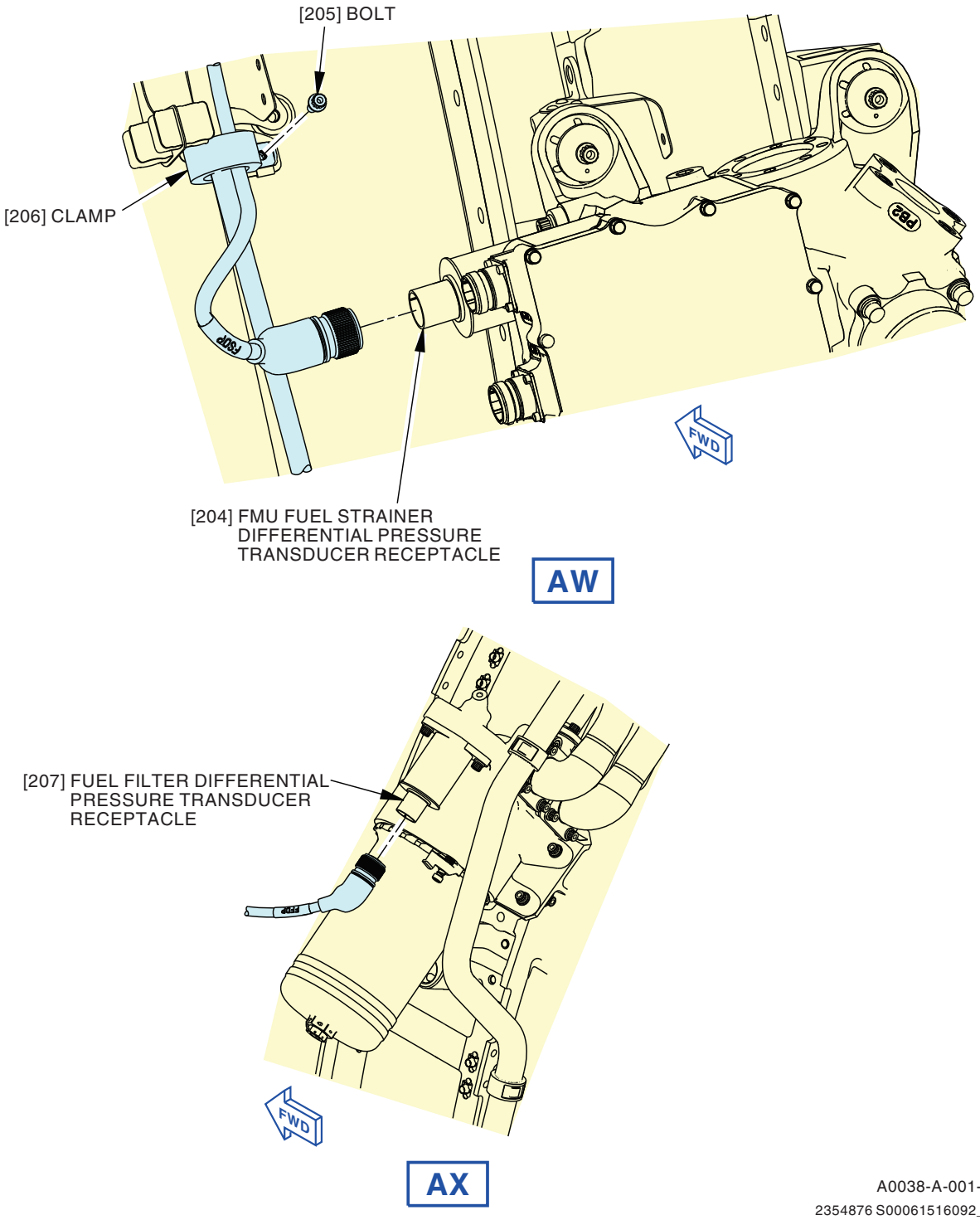
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2833208 S0000658227_V1

EEC Electrical Harness (HJ6A) Installation
Figure 409/73-22-01-990-855-G00 (Sheet 5 of 9)

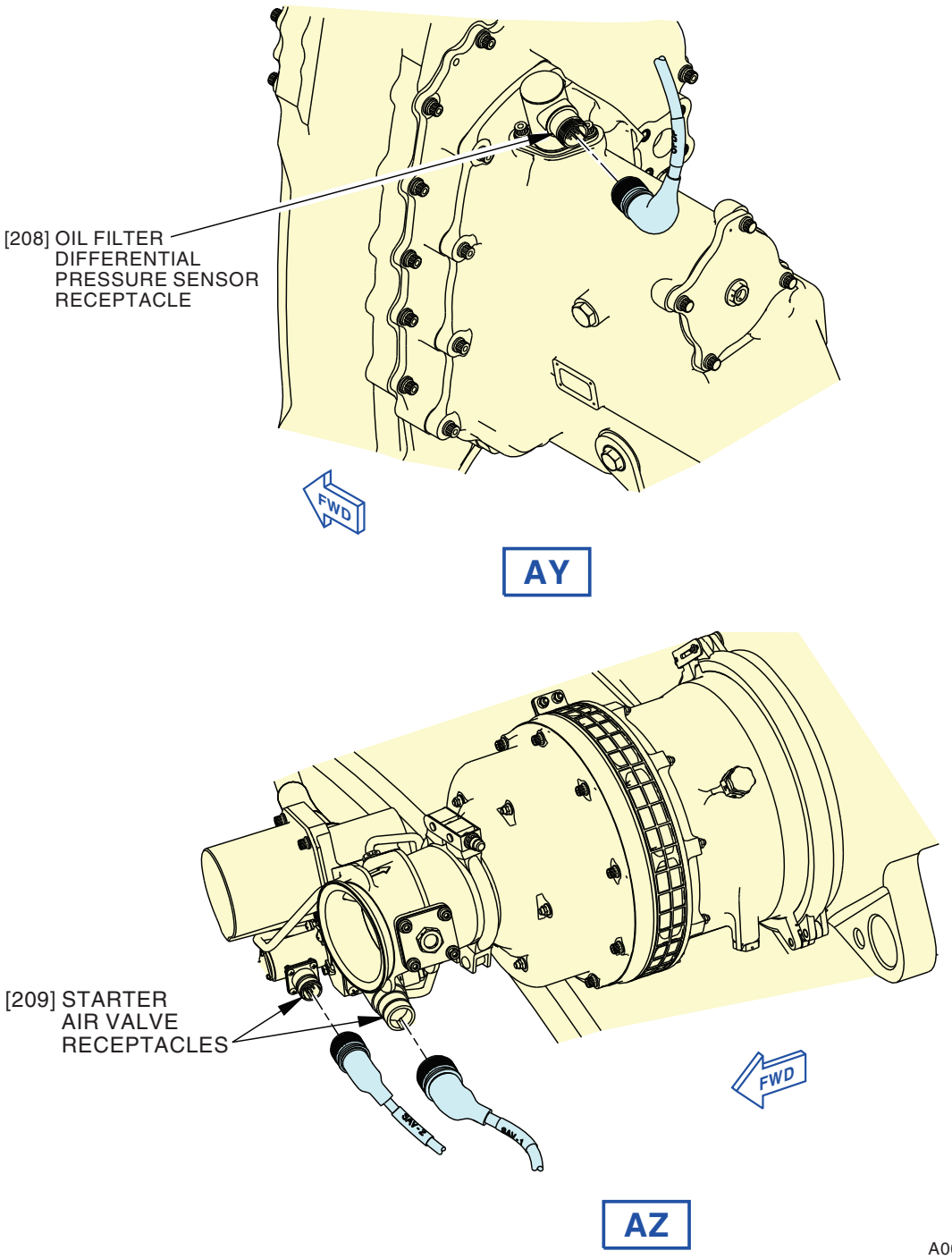
ECCN 9E991 BOEING PROPRIETARY - See title page for details



A0038-A-001-01
2354876 S00061516092_V1

EEC Electrical Harness (HJ6A) Installation
Figure 409/73-22-01-990-855-G00 (Sheet 6 of 9)

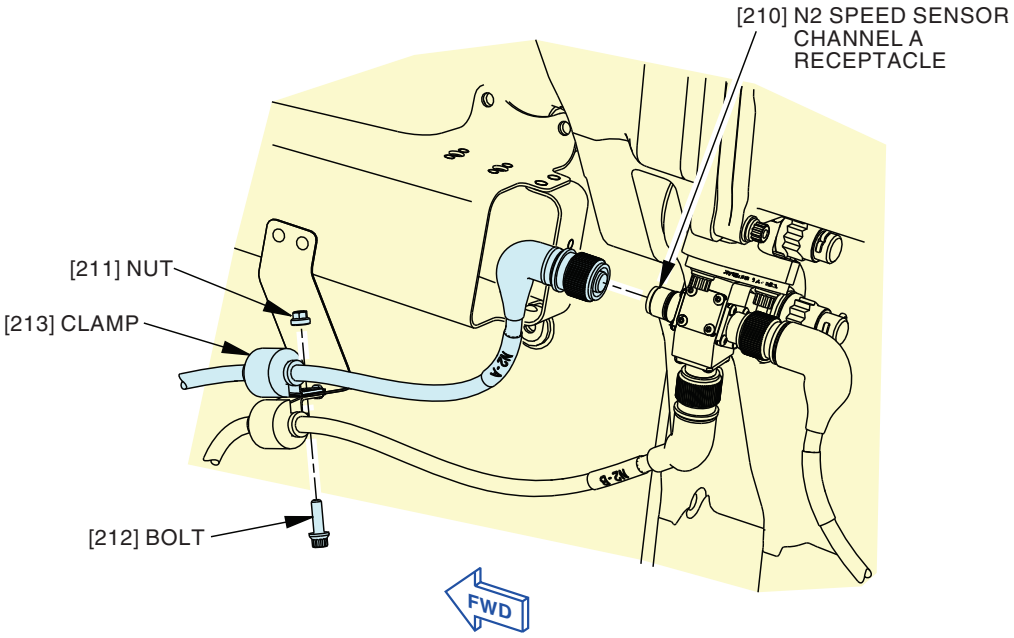
ECCN 9E991 BOEING PROPRIETARY - See title page for details

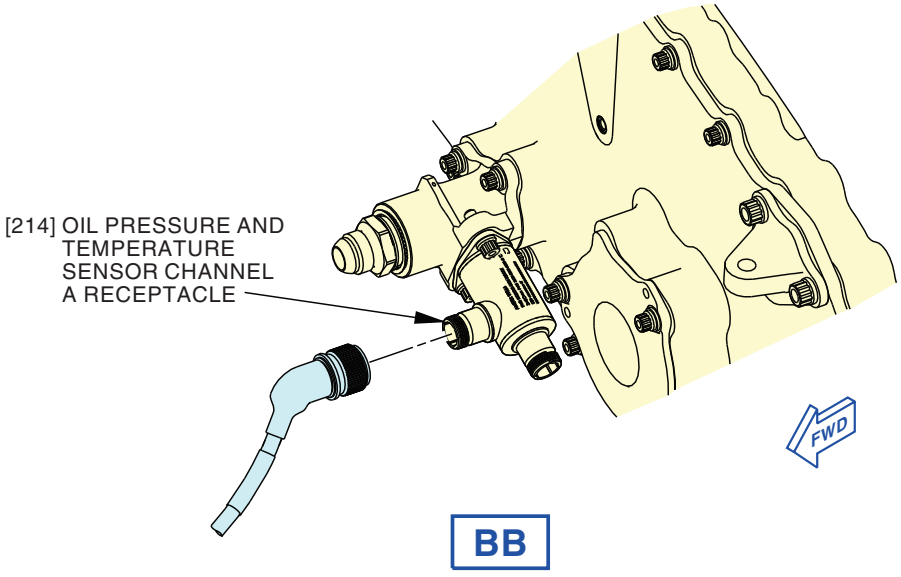


A0039-A-001-01
2354877 S00061516093_V1

EEC Electrical Harness (HJ6A) Installation
Figure 409/73-22-01-990-855-G00 (Sheet 7 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

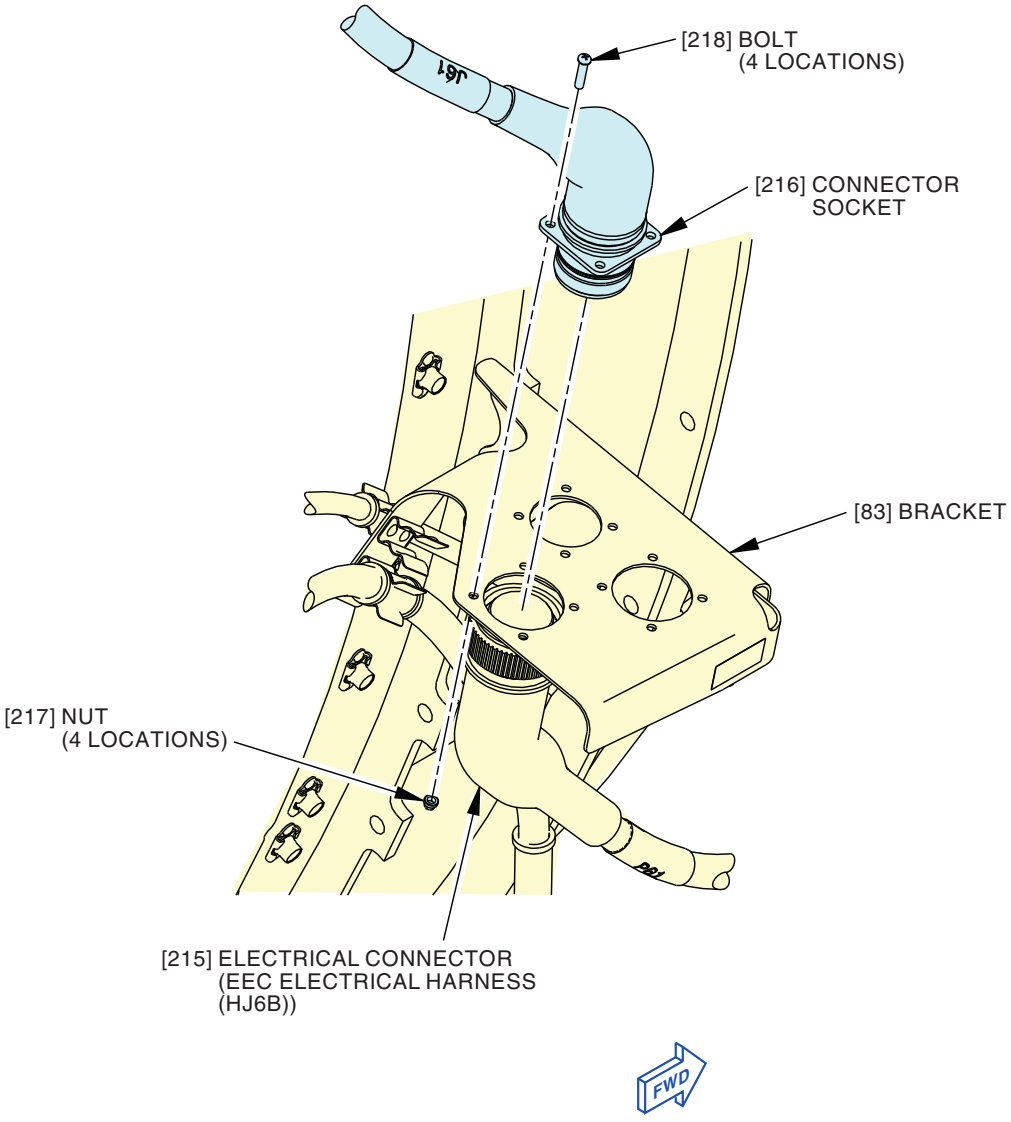




A0040-A-001-01
2354878 S00061516094_V1

EEC Electrical Harness (HJ6A) Installation
Figure 409/73-22-01-990-855-G00 (Sheet 8 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



BC

A0041-A-001-01
2354879 S00061516095_V1

EEC Electrical Harness (HJ6A) Installation
Figure 409/73-22-01-990-855-G00 (Sheet 9 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 73-22-01-400-805-G00

11. Electronic Engine Control (EEC) Electrical Harness (HJ6A) Installation

(Figure 409)

A. General

(1) This task gives the instructions to install the electronic engine control (EEC) electrical harness (HJ6A).

(2) The electronic engine control (EEC) electrical harness is referred to as the EEC electrical harness.

(3) You must do the tests that are given in the power plant test reference table after you install the applicable engine control electrical harnesses.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 2
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Electronic Engine Control (EEC) Electrical Harness (HJ6A) Installation

(Figure 409)

SUBTASK 73-22-01-420-010-G00

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION

USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

(1) Install the EEC electrical harness (HJ6A) [192] as follows:

(a) Make sure that the mating faces are clean and free of unwanted materials.

(b) Remove the protective covers on the connectors and receptacles.

(c) Install the EEC electrical harness (HJ6A) [192] in the correct position with the 44 clips [194] along its routing.

(d) Install the bolt [212] and clamp [213] that attach the EEC electrical harness (HJ6A) [192] to the engine.

(e) Tighten the nut [211] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

(f) Install the bolt [205] and clamp [206] that attach the EEC electrical harness (HJ6A) [192] to the engine.

(g) Tighten the bolt [205] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

(h) Install the bolt [200] and clamp [201] that attach the EEC electrical harness (HJ6A) [192] to the engine.

(i) Tighten the bolt [200] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

(j) Install the bolt [197] and clamp [198] that attach the EEC electrical harness (HJ6A) [192] to the engine.

(k) Tighten the bolt [197] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

(l) Install the bolt [195] and clamp [196] that attach the EEC electrical harness (HJ6A) [192] to the engine.

(m) Tighten the bolt [195] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

(n) Connect the connector socket [216] of the EEC electrical harness (HJ6A) [192] as follows:

1) Make sure that the mating faces are clean and free of unwanted materials.

2) Install the four nuts [217] and four bolts [218] to attach the connector socket [216] to the bracket [83].

3) Tighten the four nuts [217] to 10.2 in-lb (1.15 N·m) - 11.1 in-lb (1.25 N·m).

4) Connect the HJ6B engine wiring harness electrical connector [215] to the connector socket [216].

NOTE: If the HJ6B engine wiring harness has been removed, do not do this step.

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION

USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

(o) Use the teflon-jawed pliers, STD-664 to connect the EEC electrical harness (HJ6A) [192] as follows (TASK 70-00-01-910-803-G00):

1) Connect the EEC electrical harness (HJ6A) [192] connector to the oil P/T sensor channel A receptacle [214].
2) Connect the EEC electrical harness (HJ6A) [192] connector to the N2 speed sensor channel A receptacle [210].
3) Connect the EEC electrical harness (HJ6A) [192] connector to the SAV receptacle [209].
4) Connect the EEC electrical harness (HJ6A) [192] connector to the oil filter DP sensor receptacle [208].
5) Connect the EEC electrical harness (HJ6A) [192] connector to the fuel filter DP transducer receptacle [207].
6) Connect the EEC electrical harness (HJ6A) [192] connector to the FMU fuel strainer DP transducer receptacle [204].
7) Connect the EEC electrical harness (HJ6A) [192] connector to the T12 Channel A receptacle [203].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0178

8) Connect the EEC electrical harness (HJ6A) [192] to the upper overheat sensor [220].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0178 AND PRE LEAP-1B SERVICE BULLETIN 73-0019

9) Connect the EEC electrical harness (HJ6A) [192] to the bracket assembly [116].

PTF ALL

10) Connect the EEC electrical harness (HJ6A) [192] connector to the main f/o heat exchanger DP sensor receptacle [202].
11) Connect the EEC electrical harness (HJ6A) [192] connector to the PSS channel A receptacle [199].
12) Connect the EEC electrical harness (HJ6A) [192] connector to the EEC channel A J6 receptacle [219].

(p) Make sure that the work area is clean of all tools and other items.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 73-22-01-410-026-G00

WARNING

OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 73-22-01-860-037-G00

(2) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	4	C04008	ENGINE 1 ALTN PWR CHAN B
A	5	C04007	ENGINE 1 ALTN PWR CHAN A
B	3	C01312	ENGINE 1 RUN/PWR

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	5	C01313	ENGINE 2 RUN/PWR
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	3	C04018	ENG 1 STBY ALTN PWR RELAY CHAN A
E	4	C04019	ENG 1 STBY ALTN PWR CHAN A

SUBTASK 73-22-01-860-038-G00

(3) For the right engine, remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 73-22-01-860-039-G00

(4) For the left engine, remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 73-22-01-860-040-G00

(5) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

H. Electronic Engine Control Electrical (EEC) Electrical Harness (HJ6A) Installation Test

SUBTASK 73-22-01-730-005-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

TASK 73-22-01-000-806-G00

12. Electronic Engine Control (EEC) Electrical Harness (HJ6B) Removal

(Figure 410)

A. General

(1) This task gives the instructions on how to remove the electronic engine control (EEC) electrical harness (HJ6B).

(2) The electronic engine control (EEC) electrical harness is referred to as the EEC electrical harness.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
73-21-09-000-801-G00	Data Entry Plug Removal (P/B 401)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Removal

SUBTASK 73-22-01-860-041-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-22-01-860-042-G00

(2) Open these applicable circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	4	C04008	ENGINE 1 ALTN PWR CHAN B
A	5	C04007	ENGINE 1 ALTN PWR CHAN A
B	3	C01312	ENGINE 1 RUN/PWR

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	5	C01313	ENGINE 2 RUN/PWR
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	3	C04018	ENG 1 STBY ALTN PWR RELAY CHAN A
E	4	C04019	ENG 1 STBY ALTN PWR CHAN A

SUBTASK 73-22-01-860-043-G00

(3) For the right engine, open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 73-22-01-860-044-G00

(4) For the left engine, open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

SUBTASK 73-22-01-010-028-G00

(5) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Electronic Engine Control (EEC) Electrical Harness (HJ6B) Removal

(Figure 410)

SUBTASK 73-22-01-020-012-G00

(1) Remove the data entry plug [193] (TASK 73-21-09-000-801-G00).

SUBTASK 73-22-01-020-013-G00

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION

USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(2) Use the teflon-jawed pliers, STD-664 to disconnect the EEC electrical harness (HJ6B) [240] as follows (TASK 70-00-01-910-803-G00):

(a) Disconnect the EEC electrical harness (HJ6B) [240] connector from the EEC channel B J6 receptacle [241].

(b) Disconnect the EEC electrical harness (HJ6B) [240] connector from the PSS channel B receptacle [244].

(c) Disconnect the EEC electrical harness (HJ6B) [240] connector from the ODMS unit conditioner receptacle [247].

(d) Disconnect the EEC electrical harness (HJ6B) [240] connector from the EEC HJ6A harness receptacle [143].

(e) Disconnect the EEC electrical harness (HJ6B) [240] connector from the N2 speed sensor channel B receptacle [250].

(f) Disconnect the EEC electrical harness (HJ6B) [240] connector from the oil P/T sensor channel B receptacle [252].

(g) Disconnect the EEC electrical harness (HJ6B) [240] connector from the T12 channel B receptacle [253].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0178

(h) Disconnect the EEC electrical harness (HJ6B) [240] from the upper overheat sensor [220].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0178 AND PRE LEAP-1B SERVICE BULLETIN 73-0019

(i) Disconnect the EEC electrical harness (HJ6B) [240] from the bracket assembly [116].

PTF ALL

(j) Remove the bolt [242] from the clamp [243] that attaches the EEC electrical harness (HJ6B) [240] to the fan case bracket.

(k) Remove the bolt [245] from the clamp [246] that attaches the EEC electrical harness (HJ6B) [240] to the fan case bracket.

(l) Remove the bolt [248] from the clamp [249] that attaches the EEC electrical harness (HJ6B) [240] to the fan case bracket.

(m) Remove the nut [211] and bolt [212] from the clamp [251] that attaches the EEC electrical harness (HJ6B) [240] to the fan case bracket.

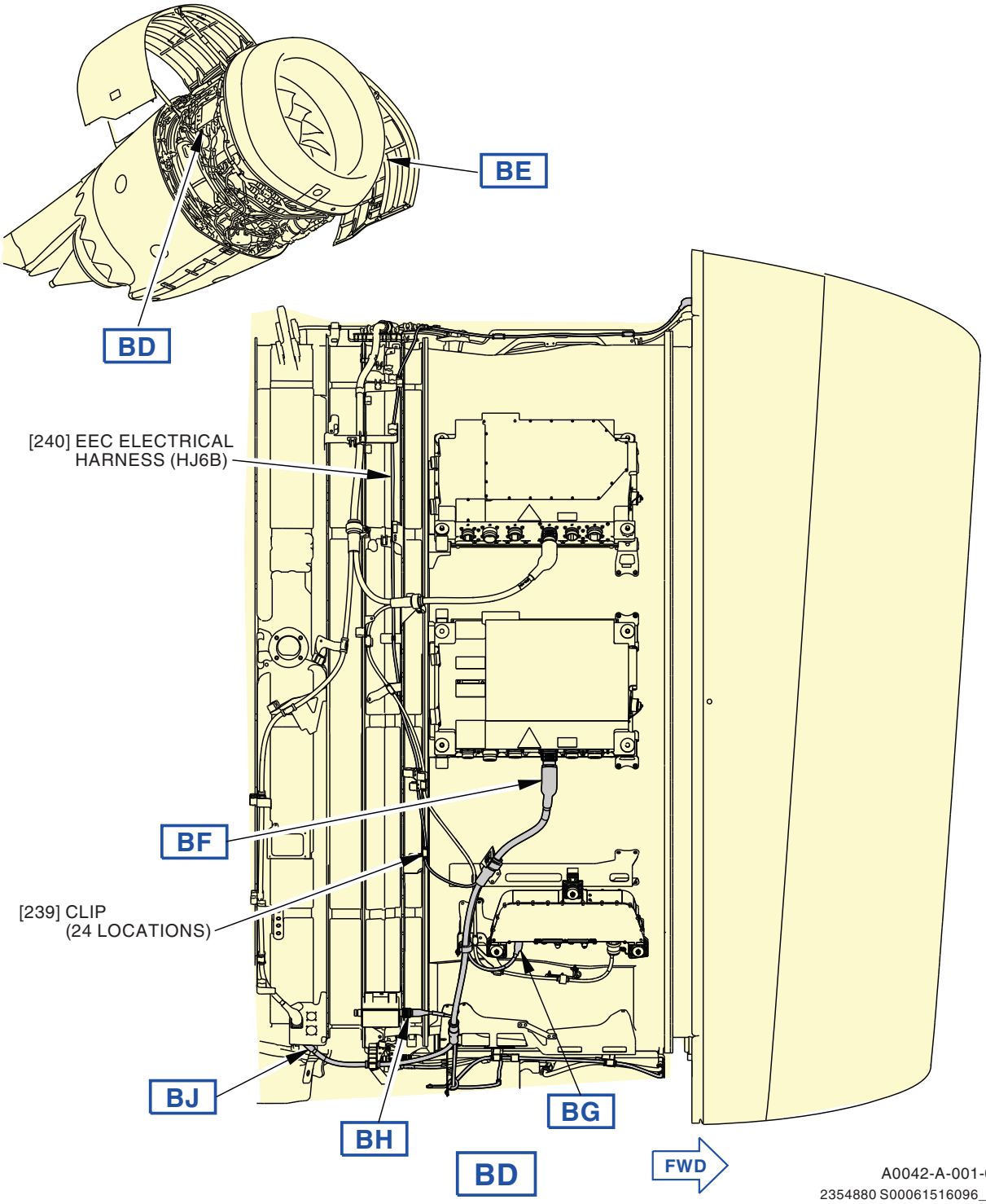
(n) Remove the EEC electrical harness (HJ6B) [240] from the 24 clips [239] along its routing.

(o) Carefully remove the EEC electrical harness (HJ6B) [240] from the engine.

(p) Install the protective covers on all of the harness connectors and the component receptacles.

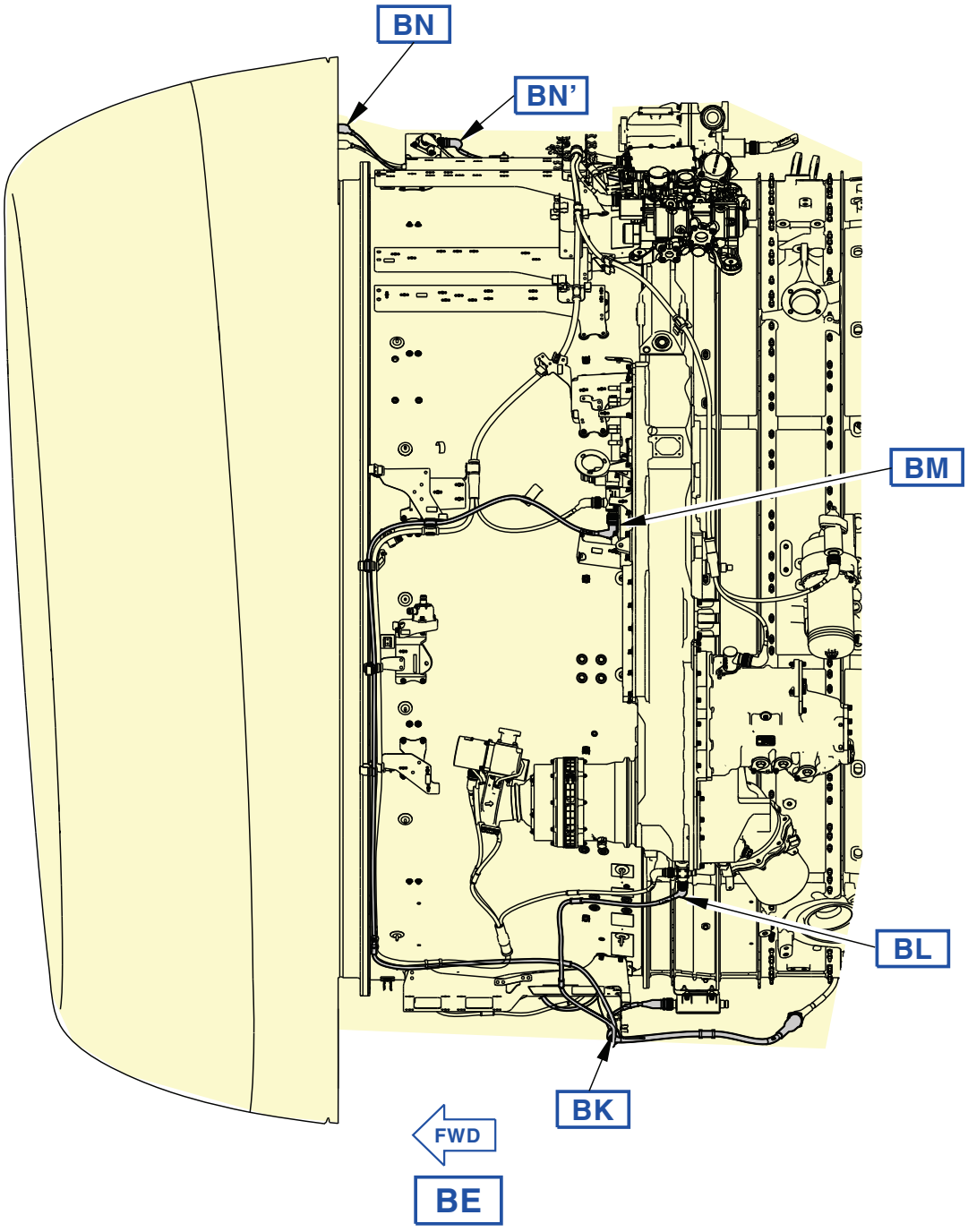
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



EEC Electrical Harness (HJ6B) Installation
Figure 410/73-22-01-990-856-G00 (Sheet 1 of 7)

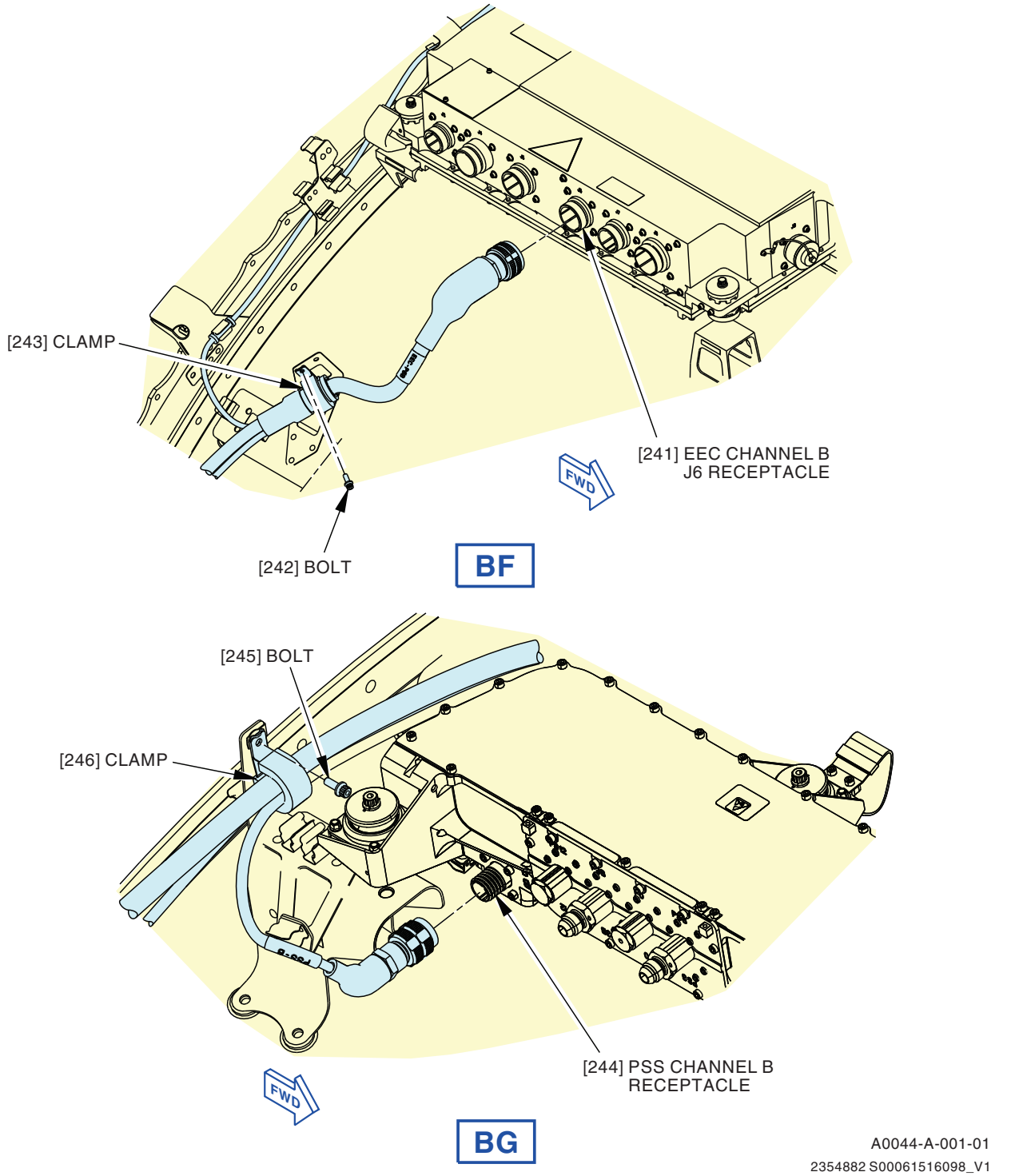
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2354881 S00061516097_V4

EEC Electrical Harness (HJ6B) Installation
Figure 410/73-22-01-990-856-G00 (Sheet 2 of 7)

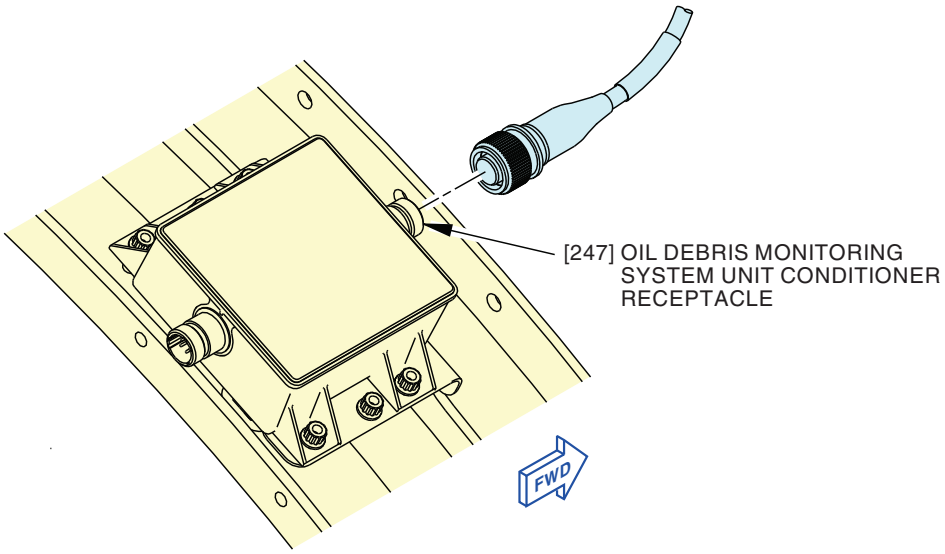
ECCN 9E991 BOEING PROPRIETARY - See title page for details



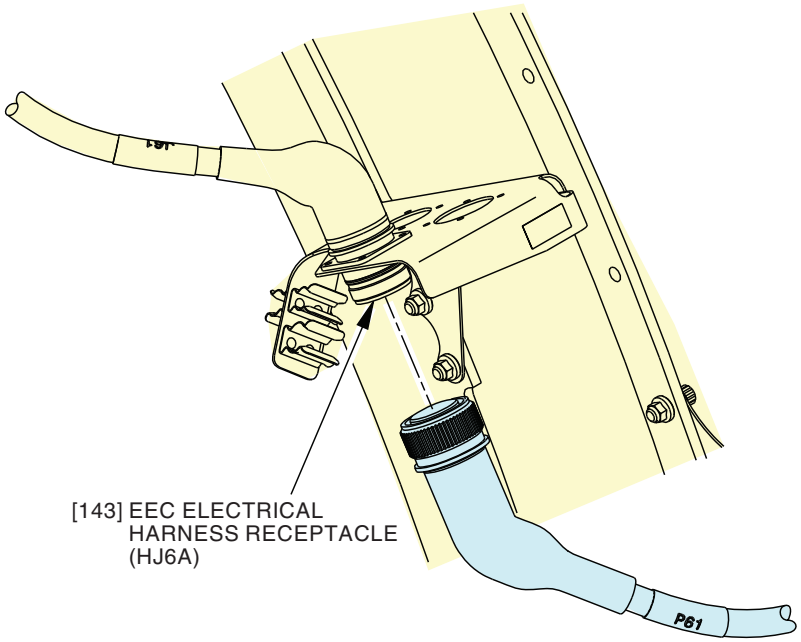
A0044-A-001-01
2354882 S00061516098_V1

EEC Electrical Harness (HJ6B) Installation
Figure 410/73-22-01-990-856-G00 (Sheet 3 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



BH

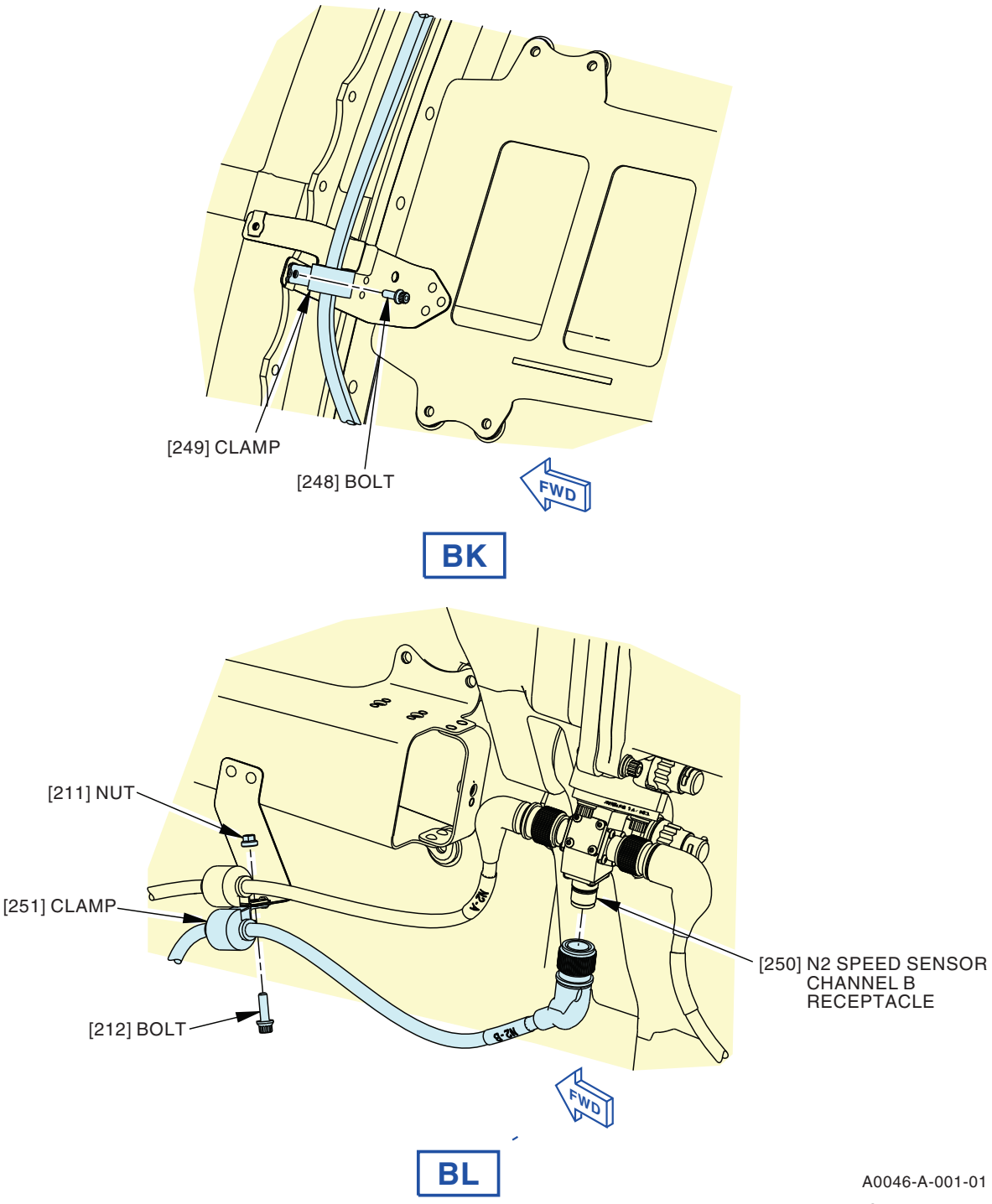


BJ

A0045-A-001-01
2354883 S00061516099_V1

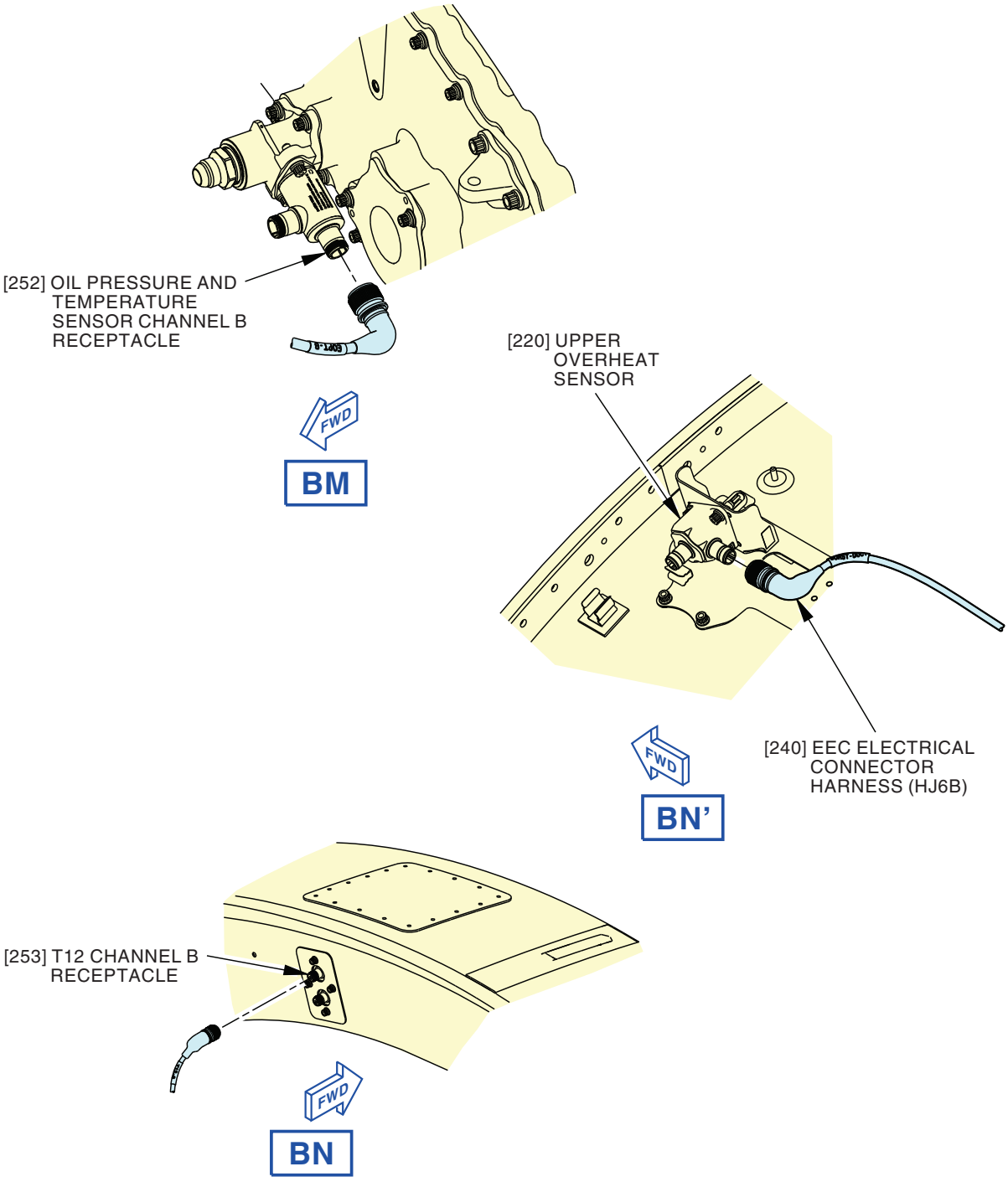
EEC Electrical Harness (HJ6B) Installation
Figure 410/73-22-01-990-856-G00 (Sheet 4 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



EEC Electrical Harness (HJ6B) Installation
Figure 410/73-22-01-990-856-G00 (Sheet 5 of 7)

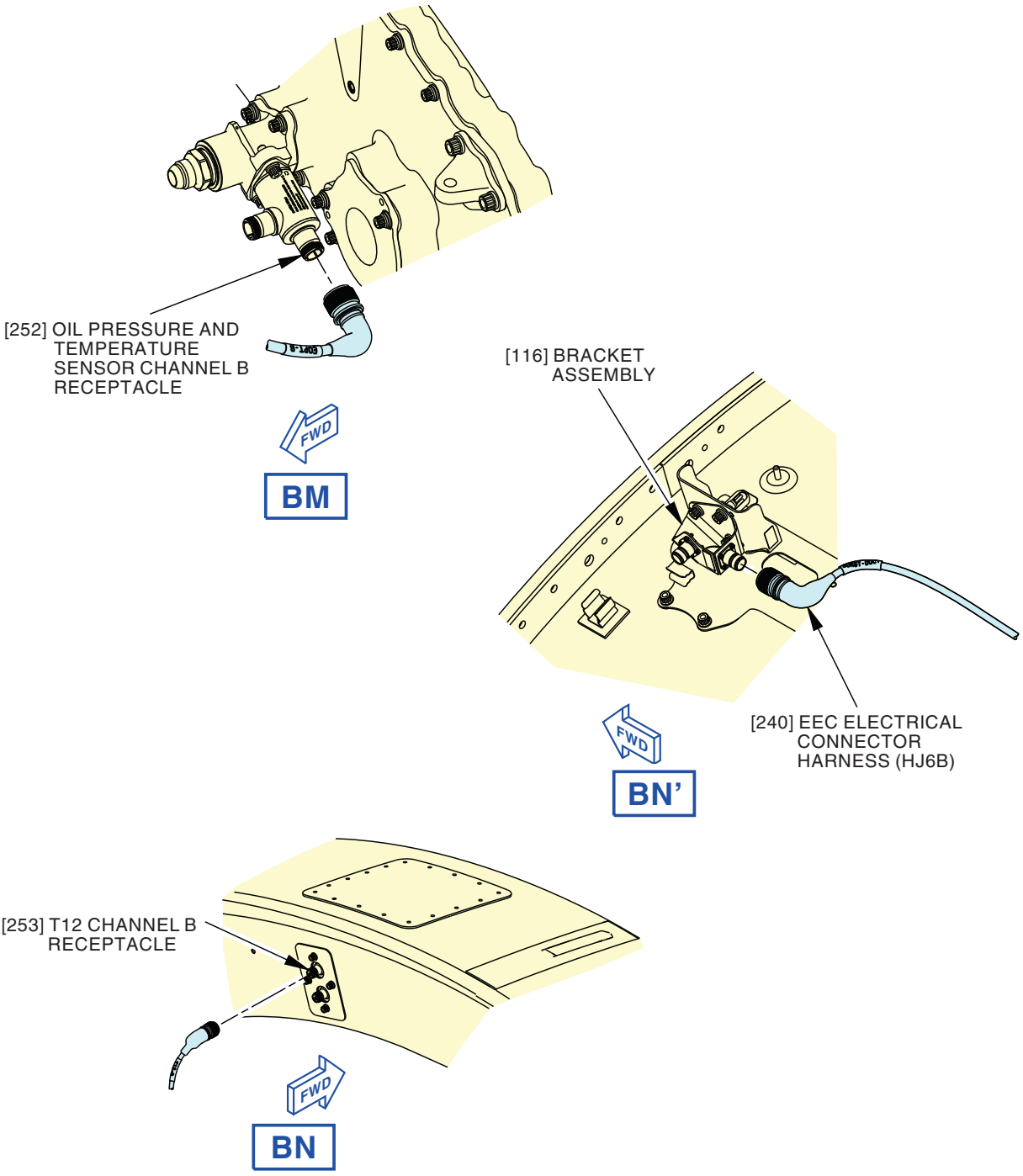
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2354885 S00061516101_V3

EEC Electrical Harness (HJ6B) Installation
Figure 410/73-22-01-990-856-G00 (Sheet 6 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2833353 S0000658229_V1

EEC Electrical Harness (HJ6B) Installation
Figure 410/73-22-01-990-856-G00 (Sheet 7 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 73-22-01-400-806-G00

13. Electronic Engine Control (EEC) Electrical Harness (HJ6B) Installation

(Figure 410)

A. General

(1) This task gives the instructions to install the electronic engine control (EEC) electrical harness (HJ6B).

(2) The electronic engine control (EEC) electrical harness is referred to as the EEC electrical harness.

(3) You must do the tests that are given in the power plant test reference table after you install the applicable engine control electrical harnesses.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
73-21-09-400-801-G00	Data Entry Plug Installation (P/B 401)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Electronic Engine Control Electrical Harness (HJ6B) Installation

(Figure 410)

SUBTASK 73-22-01-420-012-G00

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

(1) Install the EEC electrical harness (HJ6B) [240] as follows (TASK 70-00-01-910-803-G00):

(a) Make sure that the mating faces are clean and free of unwanted materials.

(b) Remove the protective covers on the connectors and receptacles.

(c) Install the EEC electrical harness (HJ6B) [240] in a correct position with the 24 clips [239] along its routing.

(d) Install the bolt [212] and clamp [251] that attach the EEC electrical harness (HJ6B) [240] to the engine.

(e) Tighten the nut [211] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

(f) Install the bolt [248] and clamp [249] that attach the EEC electrical harness (HJ6B) [240] to the engine.

(g) Tighten the bolt [248] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

(h) Install the bolt [245] and clamp [246] that attach the EEC electrical harness (HJ6B) [240] to the engine.

(i) Tighten the bolt [245] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

(j) Install the bolt [242] and clamp [243] that attach the EEC electrical harness (HJ6B) [240] to the engine.

(k) Tighten the bolt [242] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

(l) Use the teflon-jawed pliers, STD-664 to connect the EEC electrical harness (HJ6B) [240] as follows (TASK 70-00-01-910-803-G00):

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0178

1) Connect the EEC electrical harness (HJ6B) [240] to the upper overheat sensor [220].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0178 AND PRE LEAP-1B SERVICE BULLETIN 73-0019

2) Connect the EEC electrical harness (HJ6B) [240] to the bracket assembly [116]

PTF ALL

3) Connect the EEC electrical harness (HJ6B) [240] connector to the T12 channel B receptacle [253].
4) Connect the EEC electrical harness (HJ6B) [240] connector to the oil P/T sensor channel B receptacle [252].
5) Connect the EEC electrical harness (HJ6B) [240] connector to the N2 speed sensor channel B receptacle [250].
6) Connect the EEC electrical harness (HJ6B) [240] connector to the EEC HJ6A harness receptacle [143].
7) Connect the EEC electrical harness (HJ6B) [240] connector to the ODMS unit conditioner receptacle [247].
8) Connect the EEC electrical harness (HJ6B) [240] connector to the PSS channel B receptacle [244].
9) Connect the EEC electrical harness (HJ6B) [240] connector to the EEC channel B J6 receptacle [241].

(m) Make sure that the work area is clean of all tools and other items.

SUBTASK 73-22-01-420-013-G00

(2) Install the data entry plug [193] (TASK 73-21-09-400-801-G00).

G. Put the Airplane Back to Its Usual Condition

SUBTASK 73-22-01-410-027-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 73-22-01-860-045-G00

(2) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	4	C04008	ENGINE 1 ALTN PWR CHAN B
A	5	C04007	ENGINE 1 ALTN PWR CHAN A
B	3	C01312	ENGINE 1 RUN/PWR

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	5	C01313	ENGINE 2 RUN/PWR
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	3	C04018	ENG 1 STBY ALTN PWR RELAY CHAN A
E	4	C04019	ENG 1 STBY ALTN PWR CHAN A

SUBTASK 73-22-01-860-046-G00

(3) For the right engine, remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 73-22-01-860-047-G00

(4) For the left engine, remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

SUBTASK 73-22-01-860-048-G00

(5) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Electronic Engine Control Electrical (EEC) Electrical Harness (HJ6B) Installation Test

SUBTASK 73-22-01-730-006-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

TASK 73-22-01-000-807-G00

14. Electronic Engine Control (EEC) Electrical Harness (HJ7) Removal

(Figure 411)

A. General

(1) This task gives the instructions on how to remove the electronic engine control (EEC) electrical harness (HJ7).

(2) The electronic engine control (EEC) electrical harness is referred to as the EEC electrical harness.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Removal

SUBTASK 73-22-01-860-049-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-22-01-860-050-G00

(2) Open these applicable circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	4	C04008	ENGINE 1 ALTN PWR CHAN B
A	5	C04007	ENGINE 1 ALTN PWR CHAN A
B	3	C01312	ENGINE 1 RUN/PWR

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	5	C01313	ENGINE 2 RUN/PWR
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	3	C04018	ENG 1 STBY ALTN PWR RELAY CHAN A
E	4	C04019	ENG 1 STBY ALTN PWR CHAN A

SUBTASK 73-22-01-860-051-G00

(3) For the right engine, open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 73-22-01-860-052-G00

(4) For the left engine, open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

SUBTASK 73-22-01-010-029-G00

(5) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Electronic Engine Control (EEC) Electrical Harness (HJ7) Removal

(Figure 411)

SUBTASK 73-22-01-020-014-G00

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(1) Use the teflon-jawed pliers, STD-664 to disconnect the EEC electrical harness (HJ7) [285] as follows (TASK 70-00-01-910-803-G00):

(a) Disconnect the EEC electrical harness (HJ7) [285] connector from the EEC channel A J7 receptacle [286].

(b) Disconnect the EEC electrical harness (HJ7) [285] connector from the EEC channel B J7 receptacle [287].

PTF ALL; EEC HARDWARE BLOWER SYSTEM REMOVAL

(c) (e) Remove the bolt bolt [288] from the clamp [289] that attaches the EEC electrical harness (HJ7) [285] to the bracket [293].

(d) Remove the bolt [290] from the clamp [291] that attaches the EEC electrical harness (HJ7) [285] to the bracket [294].

PTF ALL

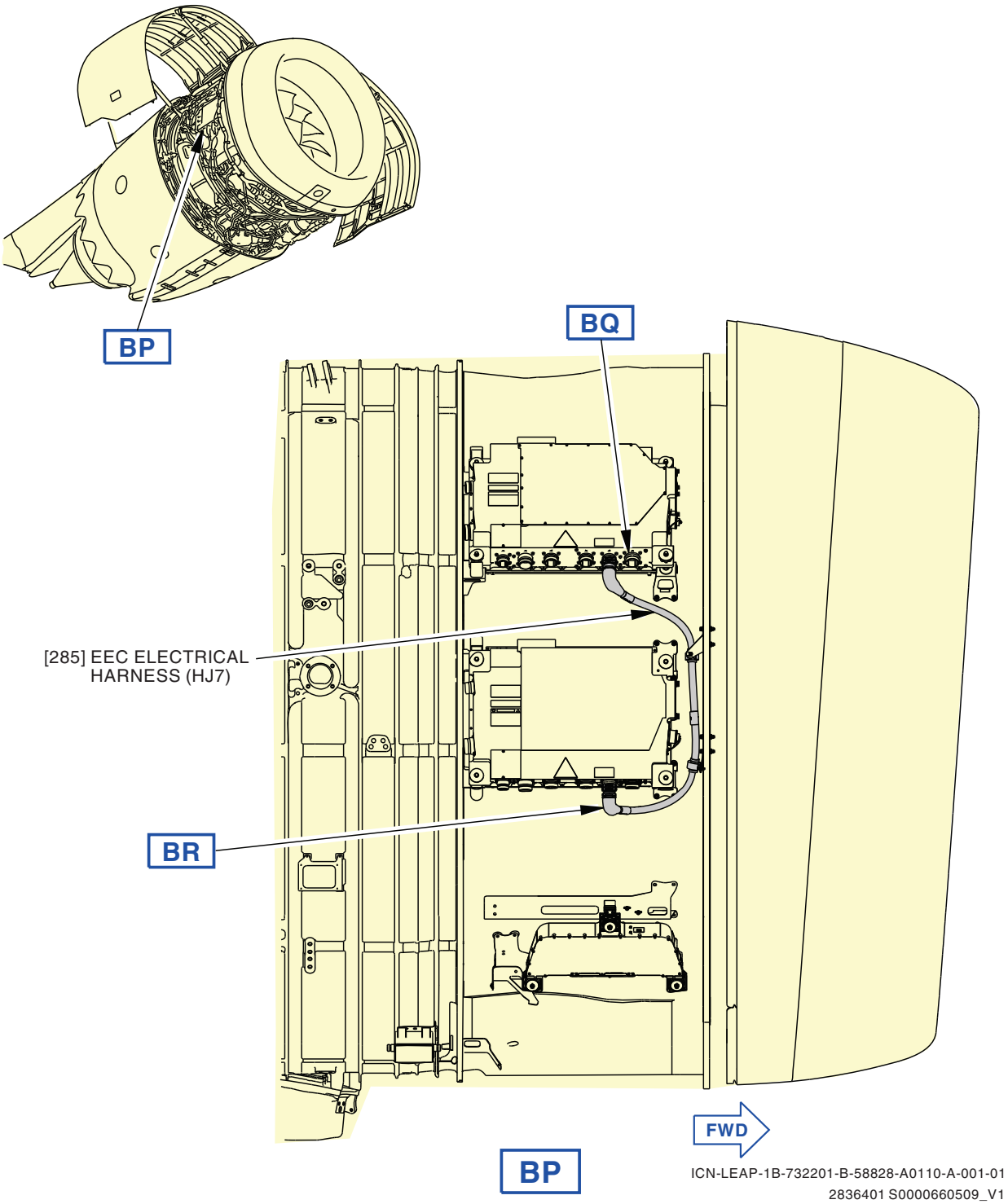
(e) Carefully remove the EEC electrical harness (HJ7) [285] from the engine.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(f) Install the protective covers on all harness connectors and component receptacles.

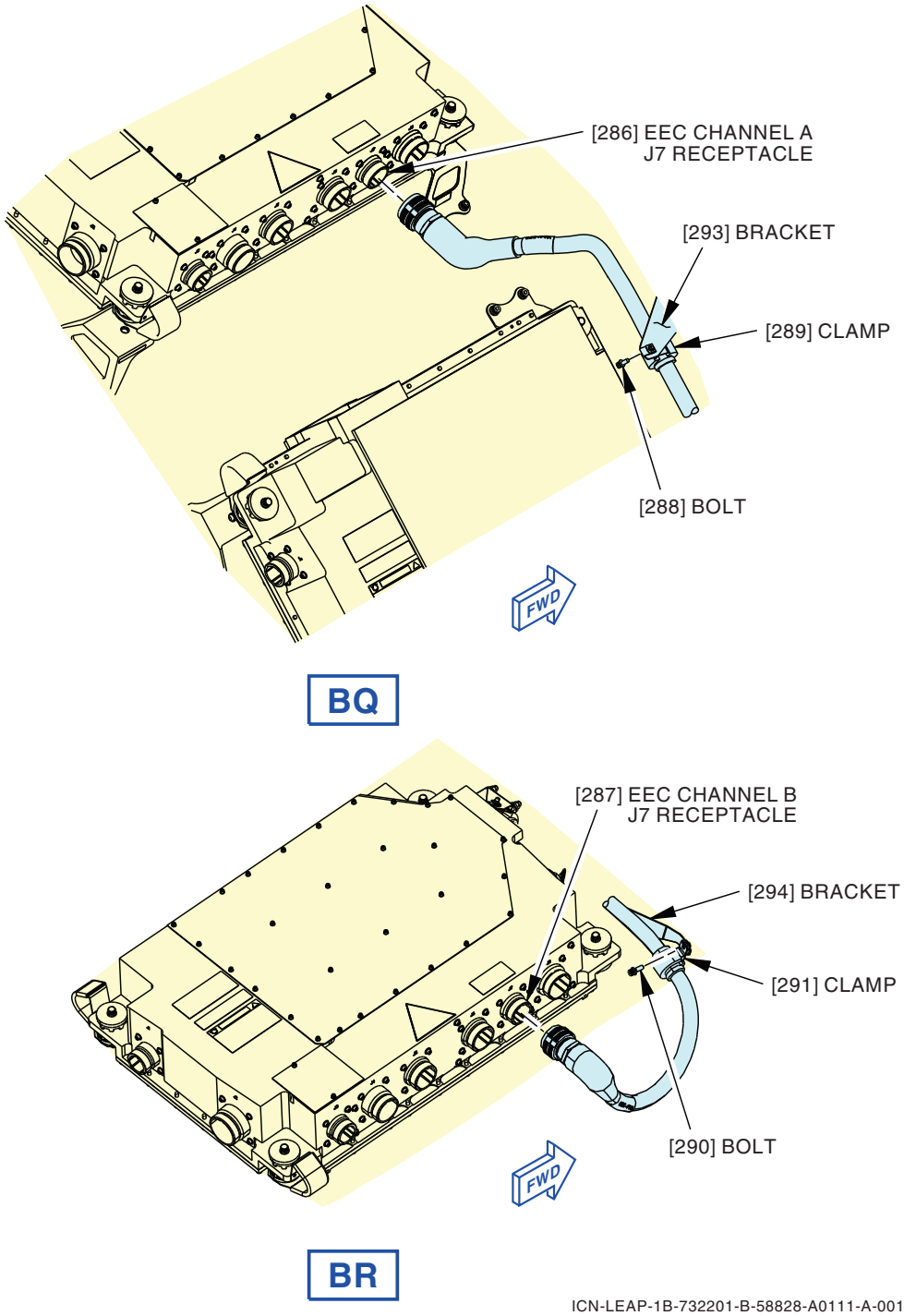
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



EEC Electrical Harness (HJ7) Installation
Figure 411/73-22-01-990-857-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ICN-LEAP-1B-732201-B-58828-A0111-A-001-01
2836411 S0000660610_V1

EEC Electrical Harness (HJ7) Installation
Figure 411/73-22-01-990-857-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 73-22-01-400-807-G00

15. Electronic Engine Control (EEC) Electrical Harness (HJ7) Installation

(Figure 411)

A. General

(1) This task gives the instructions on how to install the electronic engine control (EEC) electrical harness (HJ7).

(2) The electronic engine control (EEC) electrical harness is referred to as the EEC electrical harness.

(3) You must do the tests that are given in the power plant test reference table after you install the applicable engine control electrical harnesses.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Electronic Engine Control (EEC) Electrical Harness (HJ7) Installation

(Figure 411)

SUBTASK 73-22-01-420-014-G00

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION

USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

(1) Install the EEC electrical harness (HJ7) [285] as follows:

(a) Make sure that the mating faces are clean and free of unwanted materials.

(b) Remove the protective covers on the connectors and receptacles.

(c) Install the EEC electrical harness (HJ7) [285] in a correct position along its routing.

PTF ALL; EEC HARDWARE BLOWER SYSTEM REMOVAL

(d) Install the bolt [290] and the clamp [291] that attach the EEC electrical harness (HJ7) [285] to the bracket [294].

PTF ALL

(e) Tighten the bolt [290] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

PTF ALL; EEC HARDWARE BLOWER SYSTEM REMOVAL

(f) Install the bolt [288] and the clamp [289] that attach the EEC electrical harness (HJ7) [285] to the bracket [293].

PTF ALL

(g) Tighten the bolt [288] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

(h) Use the teflon-jawed pliers, STD-664 to connect the EEC electrical harness (HJ7) [285] as follows (TASK 70-00-01-910-803-G00):

1) Connect the EEC electrical harness (HJ7) [285] connector to the EEC channel B J7 receptacle [287].

2) Connect the EEC electrical harness (HJ7) [285] connector to the EEC channel A J7 receptacle [286].

(i) Make sure that the work area is clean of all tools and other items.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 73-22-01-410-028-G00

WARNING

OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 73-22-01-860-053-G00

(2) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	4	C04008	ENGINE 1 ALTN PWR CHAN B
A	5	C04007	ENGINE 1 ALTN PWR CHAN A
B	3	C01312	ENGINE 1 RUN/PWR

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	5	C01313	ENGINE 2 RUN/PWR
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	3	C04018	ENG 1 STBY ALTN PWR RELAY CHAN A
E	4	C04019	ENG 1 STBY ALTN PWR CHAN A

SUBTASK 73-22-01-860-054-G00

(3) For the right engine, remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 73-22-01-860-055-G00

(4) For the left engine, remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 73-22-01-860-056-G00

(5) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

H. Electronic Engine Control Electrical (EEC) Electrical Harness (HJ7) Installation Test

SUBTASK 73-22-01-730-007-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

TASK 73-22-01-000-808-G00

16. Electronic Engine Control (EEC) Electrical Harness (HJ8A) Removal

(Figure 412, Figure 413, and Figure 414)

A. General

(1) This task gives the instructions to remove the electronic engine control (EEC) electrical harness (HJ8A).

(2) The electronic engine control (EEC) electrical harness is referred to as the EEC electrical harness.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
72-23-04-000-801-G00	Fan Frame Module Segment Shrouds Removal (P/B 401)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Removal

SUBTASK 73-22-01-860-057-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-22-01-860-058-G00

(2) Open these applicable circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	4	C04008	ENGINE 1 ALTN PWR CHAN B
A	5	C04007	ENGINE 1 ALTN PWR CHAN A
B	3	C01312	ENGINE 1 RUN/PWR

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	5	C01313	ENGINE 2 RUN/PWR
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	3	C04018	ENG 1 STBY ALTN PWR RELAY CHAN A
E	4	C04019	ENG 1 STBY ALTN PWR CHAN A

SUBTASK 73-22-01-860-059-G00

(3) For the right engine, open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 73-22-01-860-060-G00

(4) For the left engine, open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

SUBTASK 73-22-01-010-030-G00

(5) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Electronic Engine Control (EEC) Electrical Harness (HJ8A) Removal

SUBTASK 73-22-01-020-015-G00

(1) Remove the segment shroud 1 [304] (TASK 72-23-04-000-801-G00).

SUBTASK 73-22-01-020-016-G00

(2) Remove the segment shroud 2 [68] (TASK 72-23-04-000-801-G00).

SUBTASK 73-22-01-020-017-G00

(3) Remove the segment shroud 5 [305] (TASK 72-23-04-000-801-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 73-22-01-020-018-G00

CAUTION	MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION	USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(4) Use the teflon-jawed pliers, STD-664, to disconnect the EEC electrical harness (HJ8A) [303] as follows (TASK 70-00-01-910-803-G00):

(a) Disconnect the EEC electrical harness (HJ8A) [303] connector from the EEC channel A J8 receptacle [306].

(b) Remove the bolt [307] from the clamp [308] that attaches the EEC electrical harness (HJ8A) [303] to the fan case bracket.

(c) Remove the bolt [312] from the clamp [313] that attaches the EEC electrical harness (HJ8A) [303] to the fan case bracket.

(d) Disconnect the connector socket [309] of the EEC electrical harness (HJ8A) [303] as follows:

1) Disconnect the HJ8B engine wiring harness electrical connector [146] (at 5:00 o'clock) from the connector socket [309].

NOTE: If the HJ8B engine wiring harness has been removed, do not do this step.

2) Remove the four screws [311] that attach the connector socket [309] on the bracket [83].

3) Disconnect the connector socket [309] from the bracket [83].

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0410

(e) Disconnect the EEC electrical harness (HJ8A) [303] from the bracket [83] as follows:

1) Disconnect the EEC electrical harness (W2B) [431] from the connector socket of the EEC electrical harness (HJ8A) [303].

NOTE: If the EEC electrical harness (W2B) [431] is removed, this step is not necessary.

2) Remove the four screws [311] that attach the connector socket of the EEC electrical harness (HJ8A) [303] to the bracket [83].

3) Disconnect the connector socket of the EEC electrical harness (HJ8A) [303] from the bracket [83].

PTF ALL

(f) Disconnect the EEC electrical harness (HJ8A) [303] connector from the start bleed/booster anti-ice valve receptacle [314].

(g) Disconnect the EEC electrical harness (HJ8A) [303] connector from the TBV receptacle [315].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0410

(h) Disconnect the EEC electrical harness (HJ8A) [303] connector from the bleed valve [515].

PTF ALL

(i) Disconnect the EEC electrical harness (HJ8A) [303] connector from the HPTACC valve receptacle [328].

(j) Disconnect the EEC electrical harness (HJ8A) [303] connector from the LPTACC valve receptacle [331].

(k) Disconnect the EEC electrical harness (HJ8A) [303] connector from the TCF accelerometer receptacle [334].

(l) Disconnect the EEC electrical harness (HJ8A) [303] connector from the EGT harness 1 (left) receptacle [335].

(m) Disconnect the EEC electrical harness (HJ8A) [303] connector from the T3 sensor channel A receptacle [336].

(n) Remove the bolt [316] that attaches the EEC electrical harness (HJ8A) [303] to the bracket [317].

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0410

(o) Remove the bolt [318] from the clamp [319] that attaches the EEC electrical harness (HJ8A) [303] to the forward stator case bracket.

PTF ALL

(p) Remove the bolt [320] from the clamp [321] that attaches the EEC electrical harness (HJ8A) [303] to the forward stator case bracket.

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0410

(q) Remove the bolt [326] from the clamp [327] that attaches the EEC electrical harness (HJ8A) [303] to the forward stator case bracket.

(r) Remove the bolt [329] from the clamp [330] that attaches the EEC electrical harness (HJ8A) [303] to the forward stator case bracket.

PTF ALL

(s) Remove the bolt [332] from the clamp [333] that attaches the EEC electrical harness (HJ8A) [303] to the forward stator case bracket.

(t) Remove the EEC electrical harness (HJ8A) [303] from the clips [302] along its routing.

(u) Remove blanket [190] from the strut 5 outer fitting [109] (View B, Figure 413).

(v) Remove the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503] from the strut 5 outer fitting [109] as follows:

1) Remove the five bolts [106] that attach the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503] to the strut 5 outer fitting [109].

2) Cut and remove the sealant between the assembly of the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], strut 5 bracket lower block [503], and strut 5 outer fitting [109].

3) Move the assembly of the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503] rearward from the electrical harnesses.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

4) Remove the two nuts [107] and two bolts [108] to disassemble the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503].

5) Cut and remove the sealant on the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], strut 5 bracket lower block [503], and around the harnesses.

6) Remove the harnesses from the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503].

7) Remove any remaining sealant residue.

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0288

(w) Remove the harness support block [504], harness support block [505], and harness support block [506] from the harness strut (6:00 o'clock) [104] as follows:

1) Remove the four bolts [111] and four nuts [110] that attach the harness support block [504], harness support block [505], and harness support block [506] to the harness strut (6:00 o'clock) [104].

2) Cut and remove the sealant between the assembly of the harness support block [504], harness support block [505], harness support block [506], and harness strut (6:00 o'clock) [104].

3) Move the assembly of the harness support block [504], harness support block [505], and harness support block [506] upward and away from the harness strut (6:00 o'clock) [104].

4) Remove the two nuts [112] and two bolts [113] to disassemble the harness support block [504], harness support block [505], and harness support block [506].

5) Cut and remove the sealant on the harness support block [504], harness support block [505], harness support block [506], and around the harnesses.

6) Remove the harnesses from the harness support block [504], harness support block [505], and harness support block [506].

7) Remove any remaining sealant residue.

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0288

(x) Remove the left outer metallic block [507], left inner metallic block [508], right inner metallic block [509], and right outer metallic block [510] from the harness strut (6:00 o'clock) [104] as follows:

1) Remove the four bolts [124] that attach the left outer metallic block [507], left inner metallic block [508], right inner metallic block [509], and right outer metallic block [510] to the harness strut (6:00 o'clock) [104].

2) Cut and remove the sealant between the assembly of the left outer metallic block [507], left inner metallic block [508], right inner metallic block [509], right outer metallic block [510], and harness strut (6:00 o'clock) [104].

3) Move the assembly of the left outer metallic block [507], left inner metallic block [508], right inner metallic block [509], and right outer metallic block [510] upward and away from the harness strut (6:00 o'clock) [104].

4) Remove the four nuts [126] and four bolts [125] to disassemble the left outer metallic block [507], left inner metallic block [508], right inner metallic block [509], and right outer metallic block [510].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0288 (Continued)

5) Cut and remove the sealant on the left outer metallic block [507], left inner metallic block [508], right inner metallic block [509], right outer metallic block [510], and around the harnesses.

6) Remove the harnesses from the left outer metallic block [507], left inner metallic block [508], right inner metallic block [509], and right outer metallic block [510].

7) Remove any remaining sealant residue.

PTF ALL

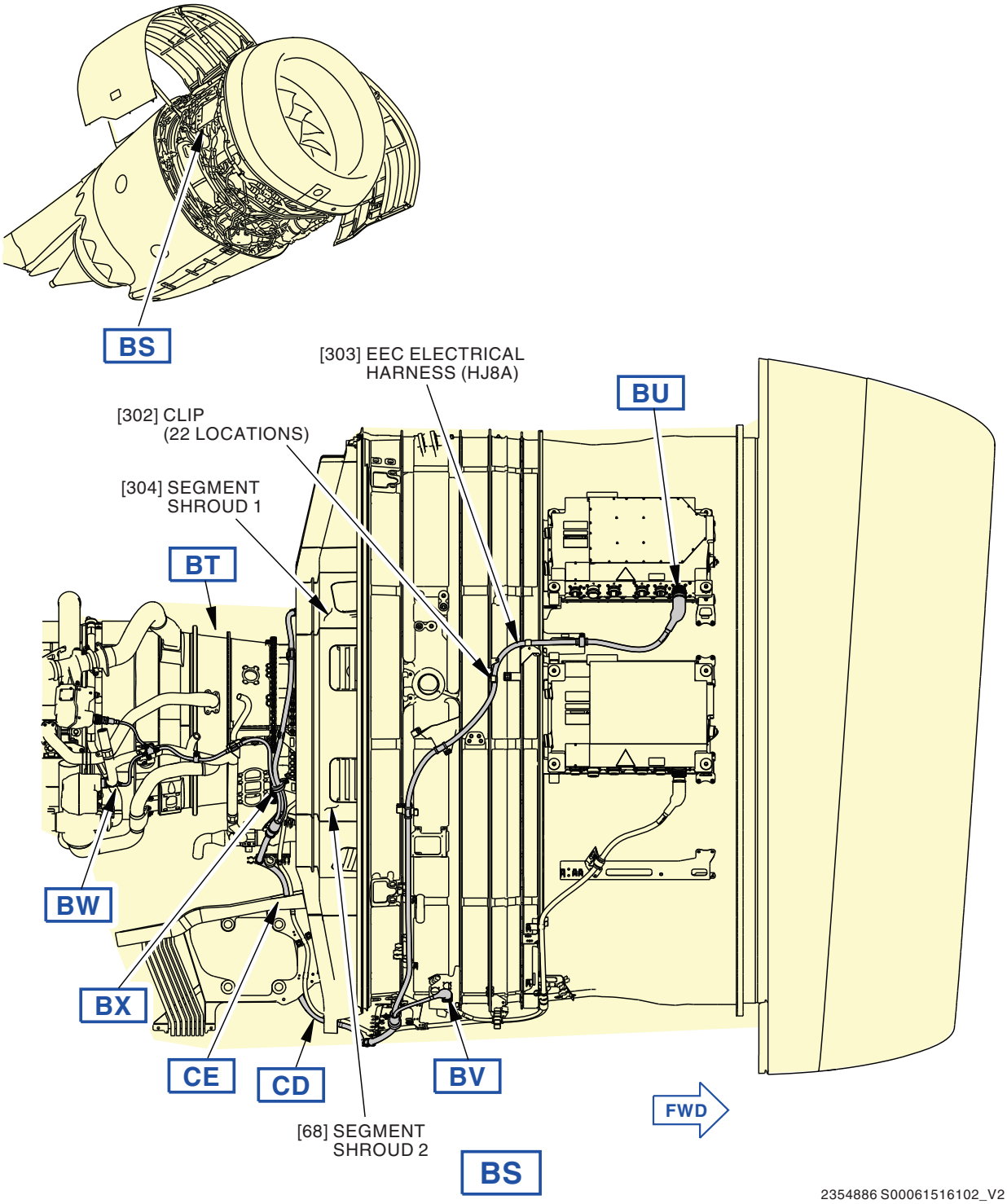
(y) Remove the EEC electrical harness (HJ8A) [303] from the harness strut (6:00 o'clock) [104].

(z) Carefully remove the EEC electrical harness (HJ8A) [303] from the engine.

(aa) Install the protective covers on all of the harness connectors and the component receptacles.

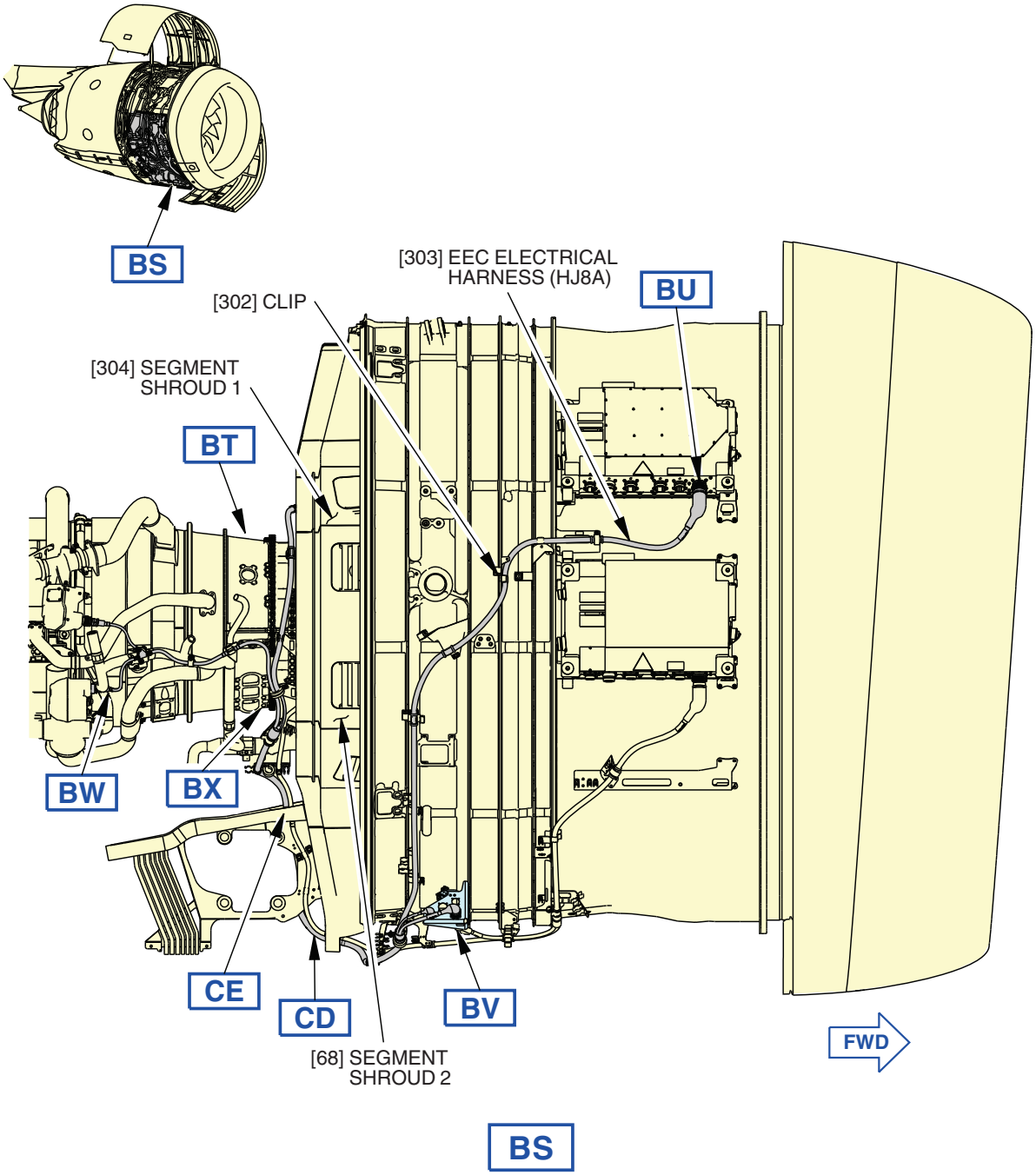
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



EEC Electrical Harness (HJ8A) Installation
Figure 412/73-22-01-990-858-G00 (Sheet 1 of 14)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

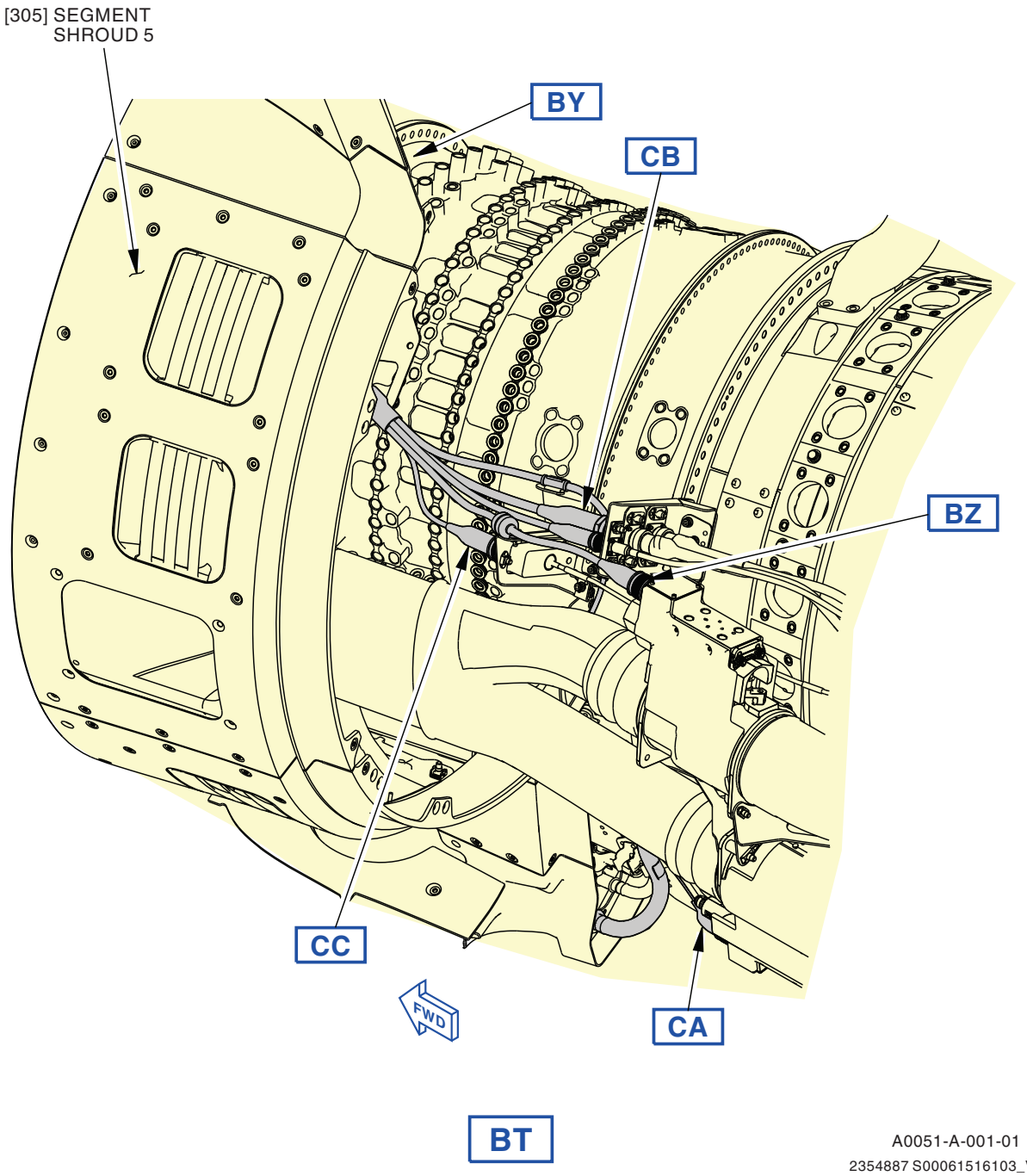
SOME COMPONENTS ARE HIDDEN FOR BETTER CLARITY.

ICNLEAP1B732201D5882800160A00101

3132211 S0000862089_V1

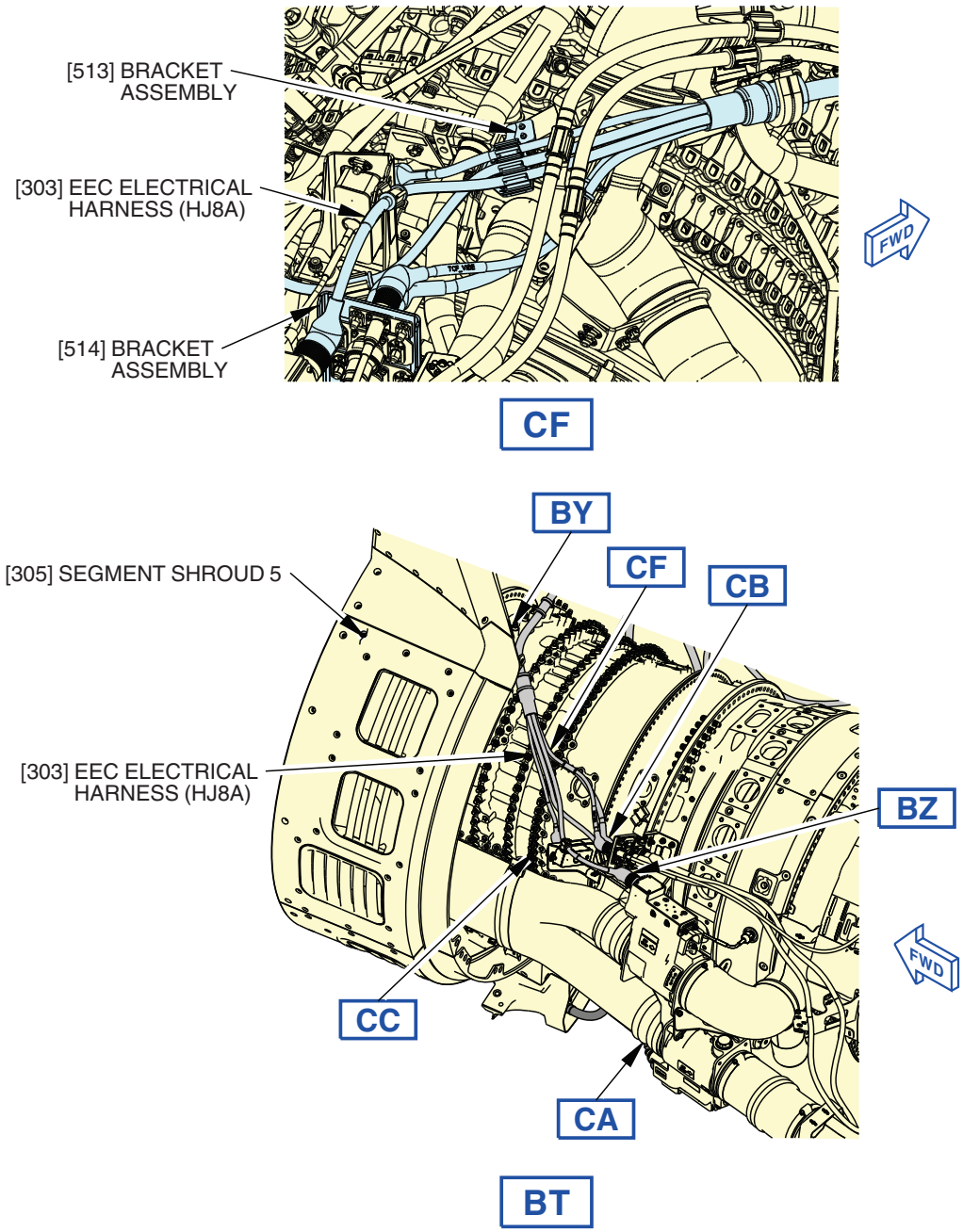
EEC Electrical Harness (HJ8A) Installation
Figure 412/73-22-01-990-858-G00 (Sheet 2 of 14)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



EEC Electrical Harness (HJ8A) Installation
Figure 412/73-22-01-990-858-G00 (Sheet 3 of 14)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

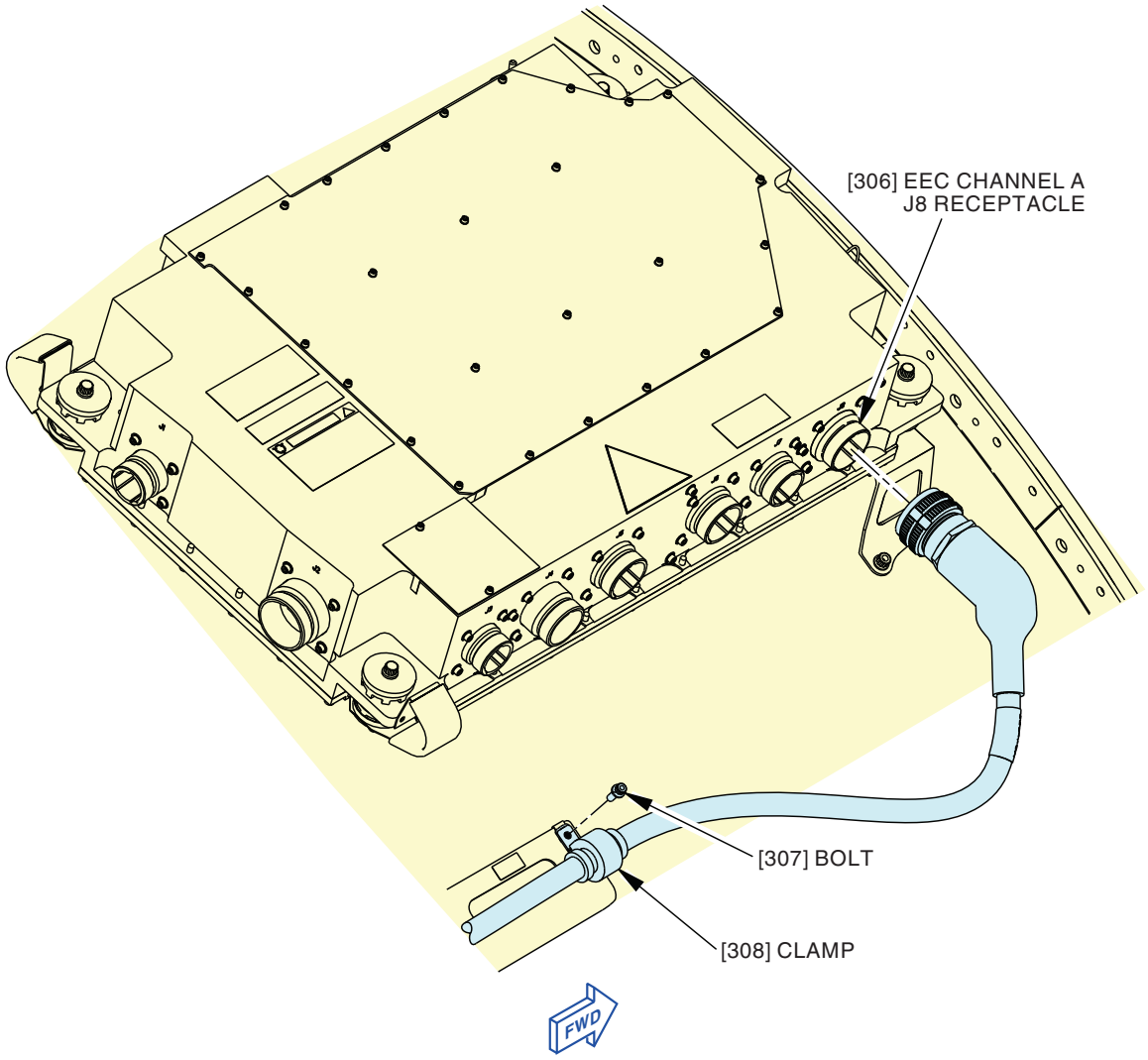
SOME COMPONENTS ARE HIDDEN FOR BETTER CLARITY.

ICNLEAP1B732201D5882800161A00101

3132214 S0000862090_V1

EEC Electrical Harness (HJ8A) Installation
Figure 412/73-22-01-990-858-G00 (Sheet 4 of 14)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

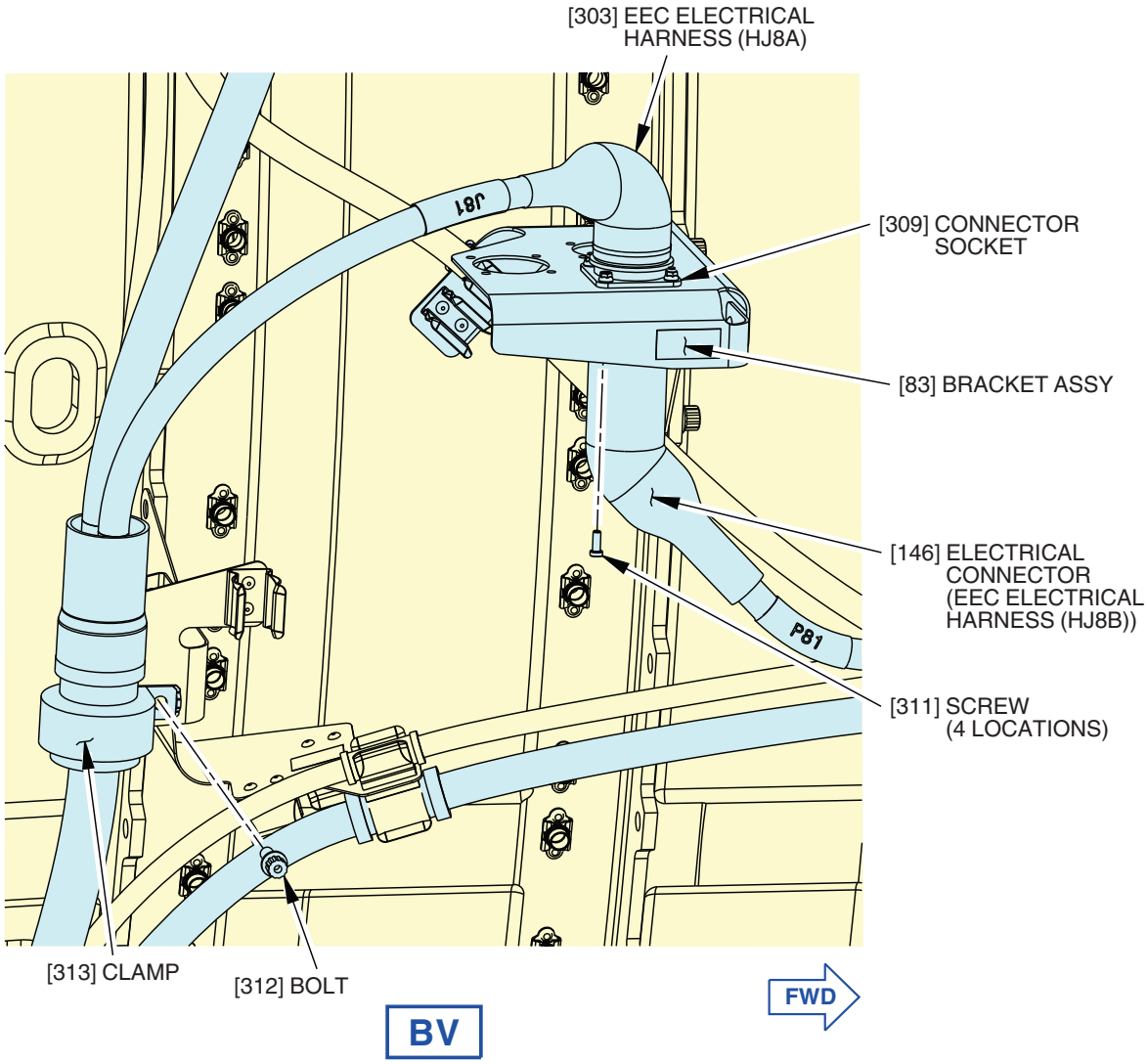


BU

A0052-A-001-01
2354888 S00061516104_V1

EEC Electrical Harness (HJ8A) Installation
Figure 412/73-22-01-990-858-G00 (Sheet 5 of 14)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



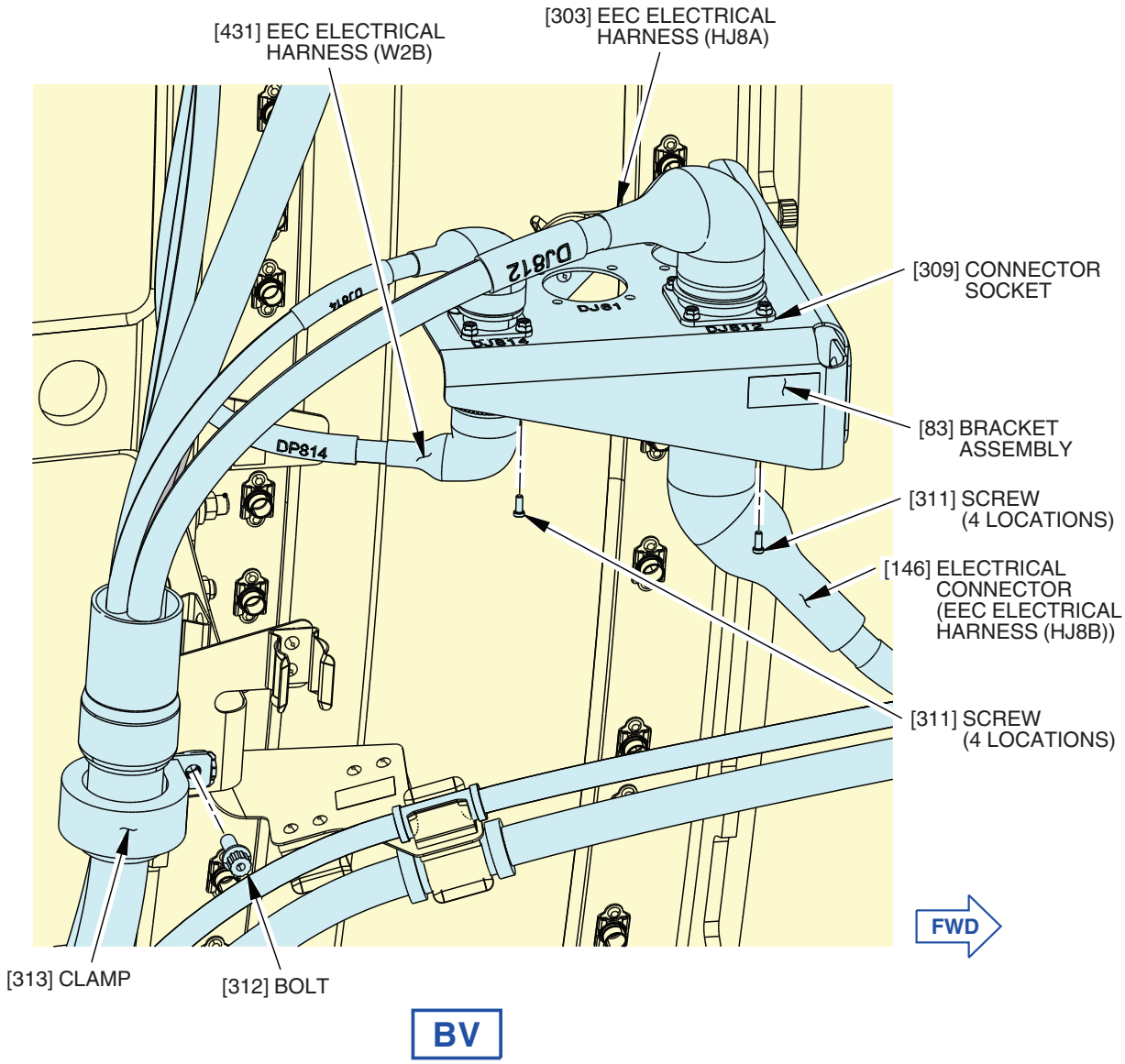
NOTE:

SOME COMPONENTS ARE HIDDEN FOR BETTER CLARITY.

ICNLEAP1B732201D5882800162A00101
3132215 S0000862091_V1

EEC Electrical Harness (HJ8A) Installation
Figure 412/73-22-01-990-858-G00 (Sheet 6 of 14)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

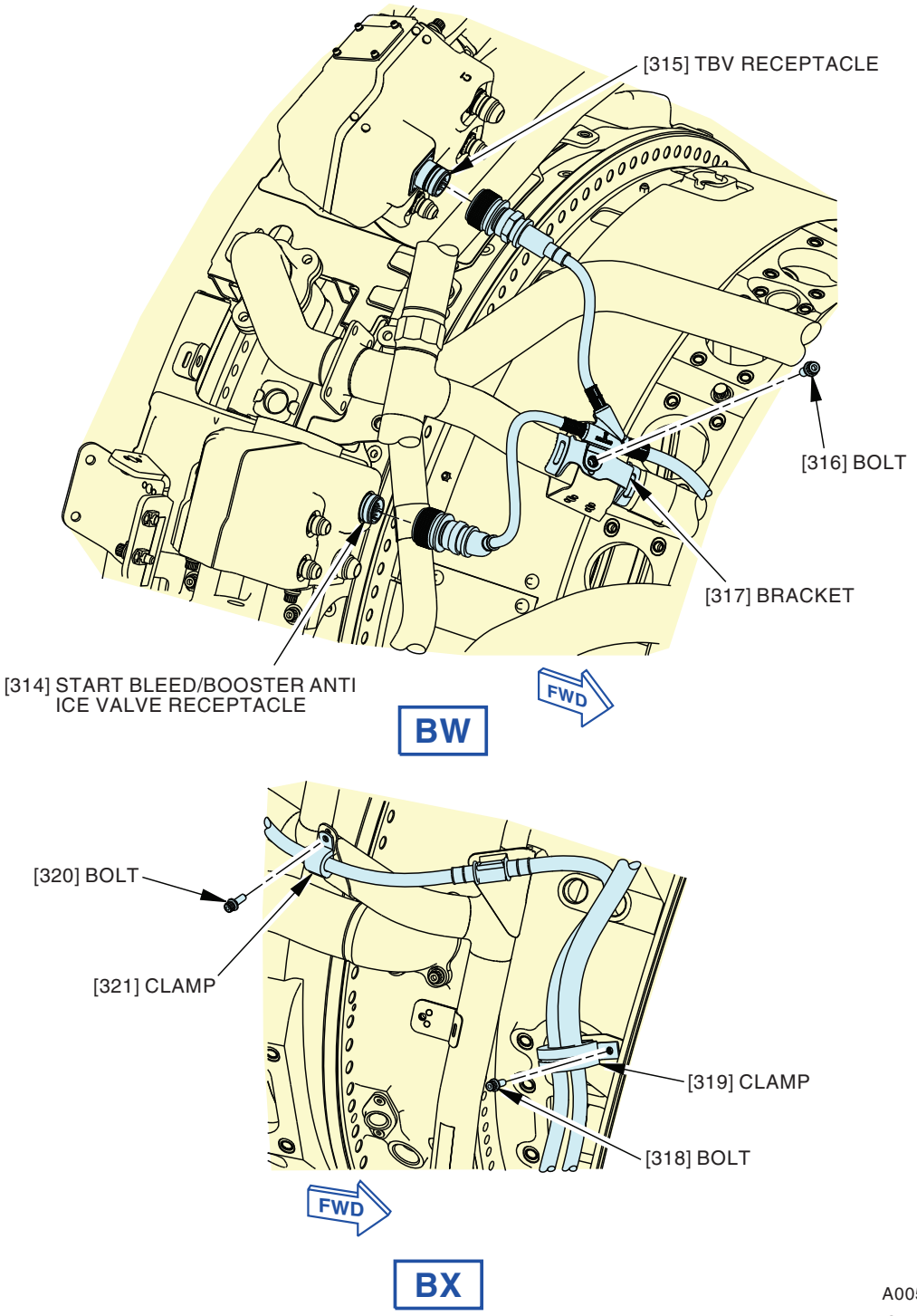


NOTE:
SOME COMPONENTS ARE HIDDEN FOR BETTER CLARITY.

3132216 S0000862092_V1

EEC Electrical Harness (HJ8A) Installation
Figure 412/73-22-01-990-858-G00 (Sheet 7 of 14)

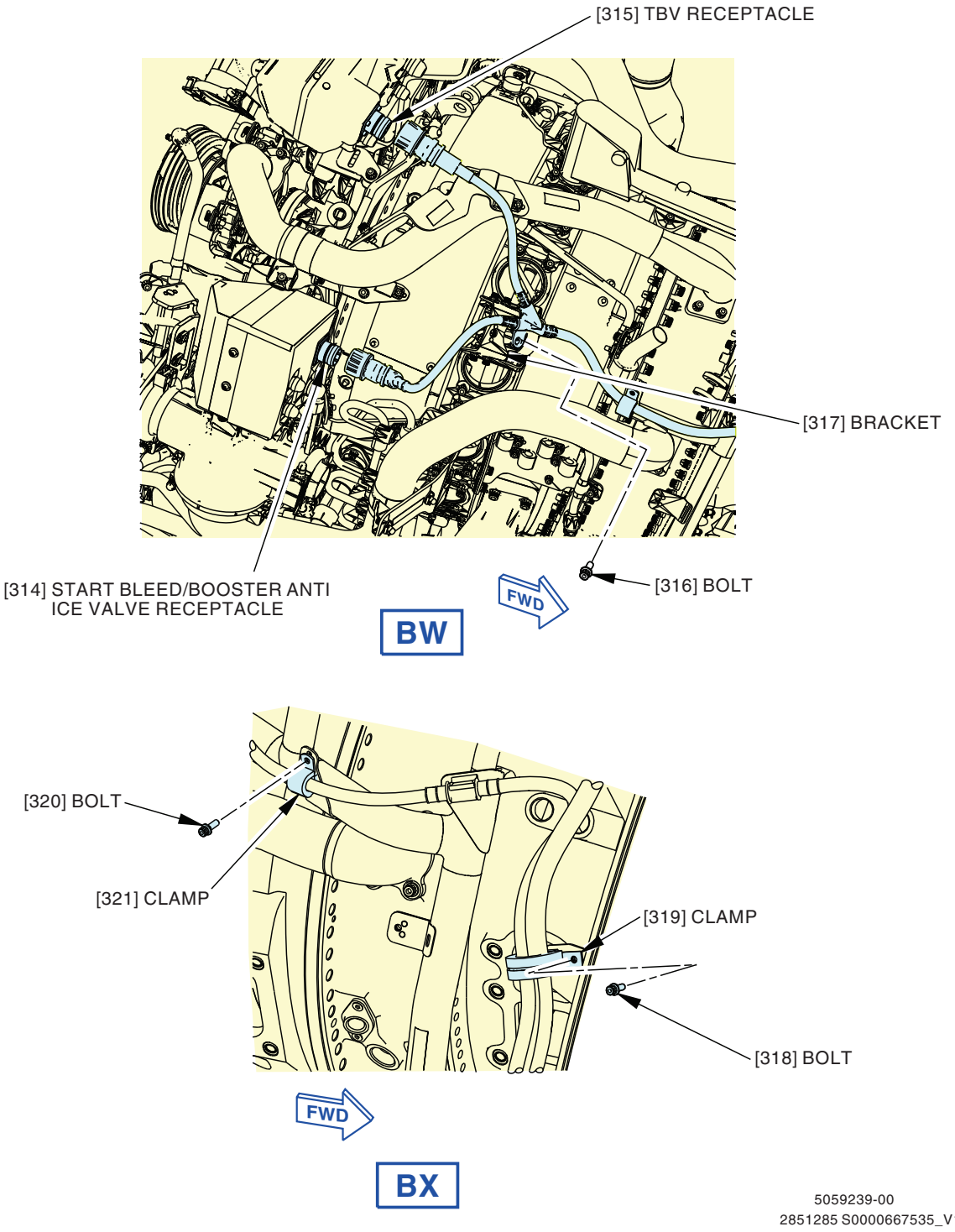
ECCN 9E991 BOEING PROPRIETARY - See title page for details



A0054-A-001-01
2354890 S00061516106_V2

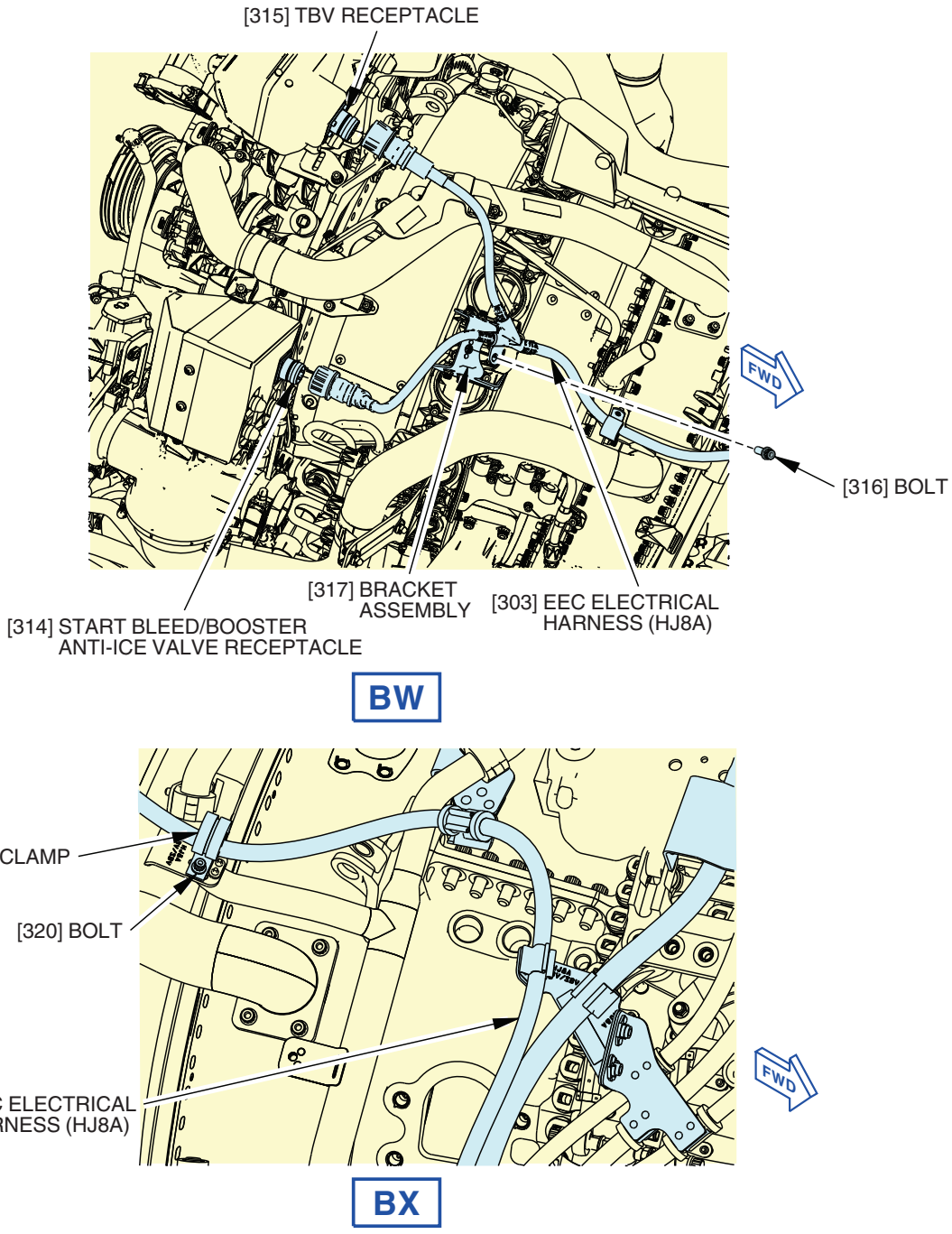
EEC Electrical Harness (HJ8A) Installation
Figure 412/73-22-01-990-858-G00 (Sheet 8 of 14)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



EEC Electrical Harness (HJ8A) Installation
Figure 412/73-22-01-990-858-G00 (Sheet 9 of 14)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

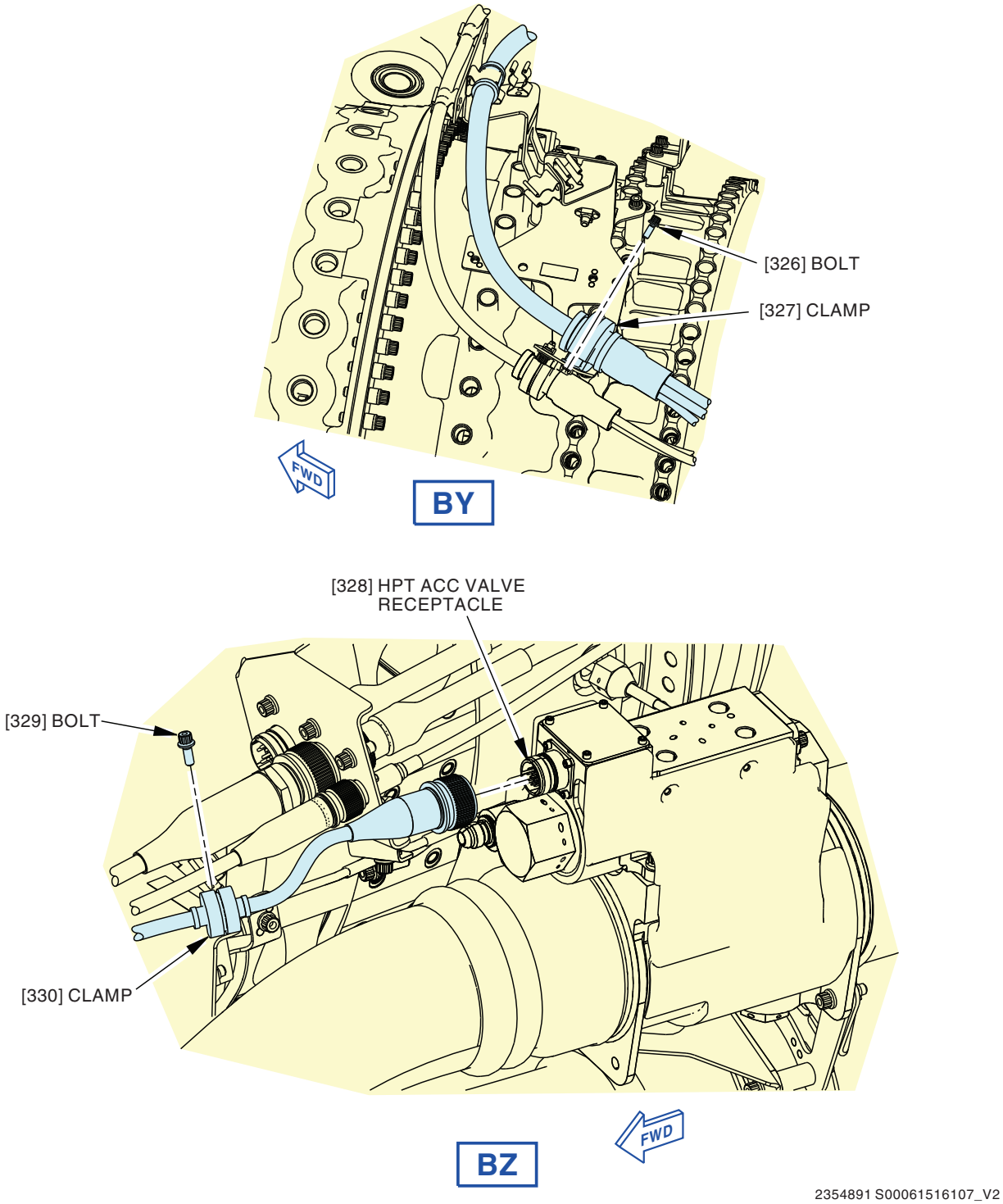
SOME COMPONENTS ARE HIDDEN FOR BETTER CLARITY.

ICNLEAP1B732201D5882800164A00101

3132221 S0000862093_V1

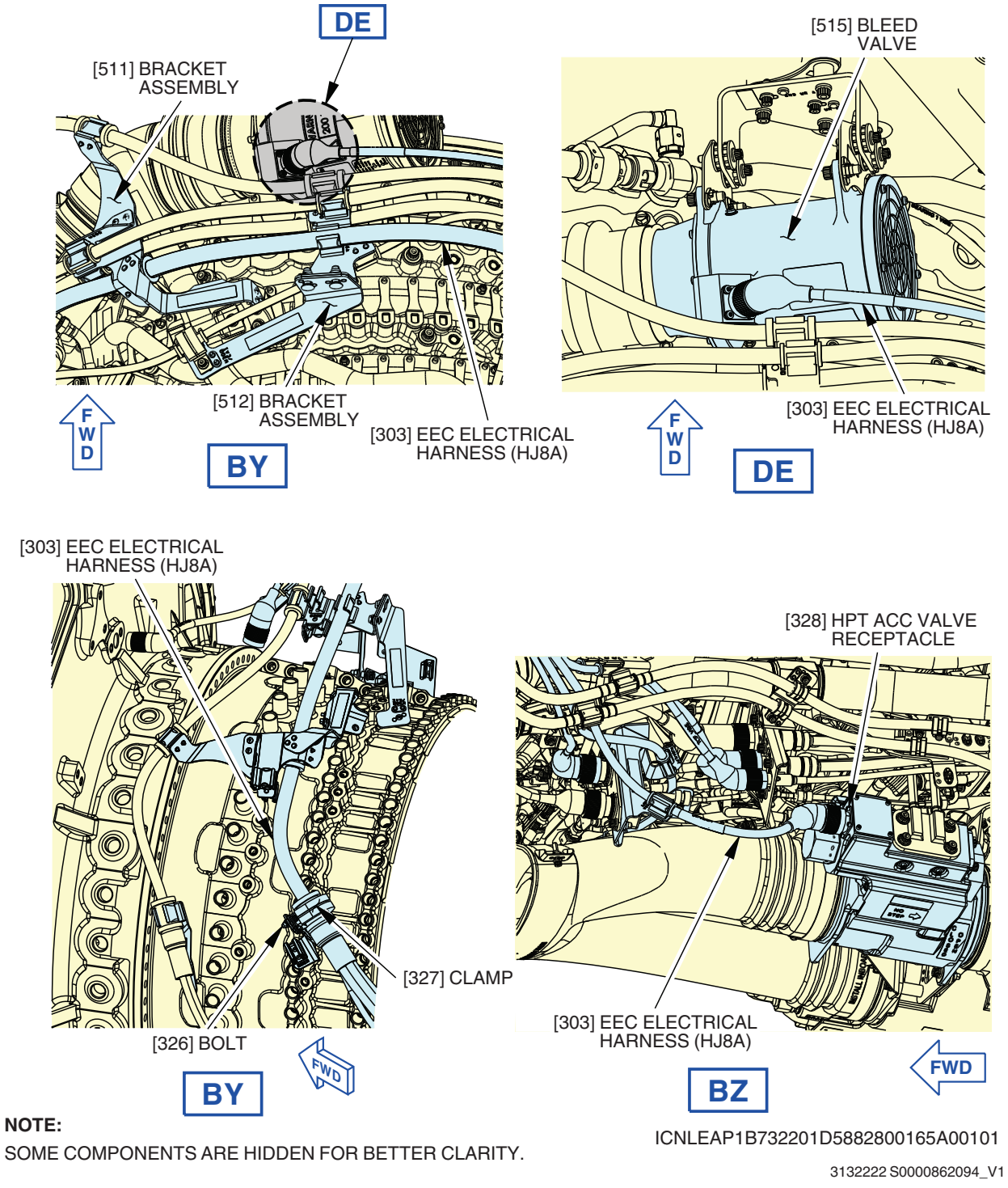
EEC Electrical Harness (HJ8A) Installation
Figure 412/73-22-01-990-858-G00 (Sheet 10 of 14)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



EEC Electrical Harness (HJ8A) Installation
Figure 412/73-22-01-990-858-G00 (Sheet 11 of 14)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

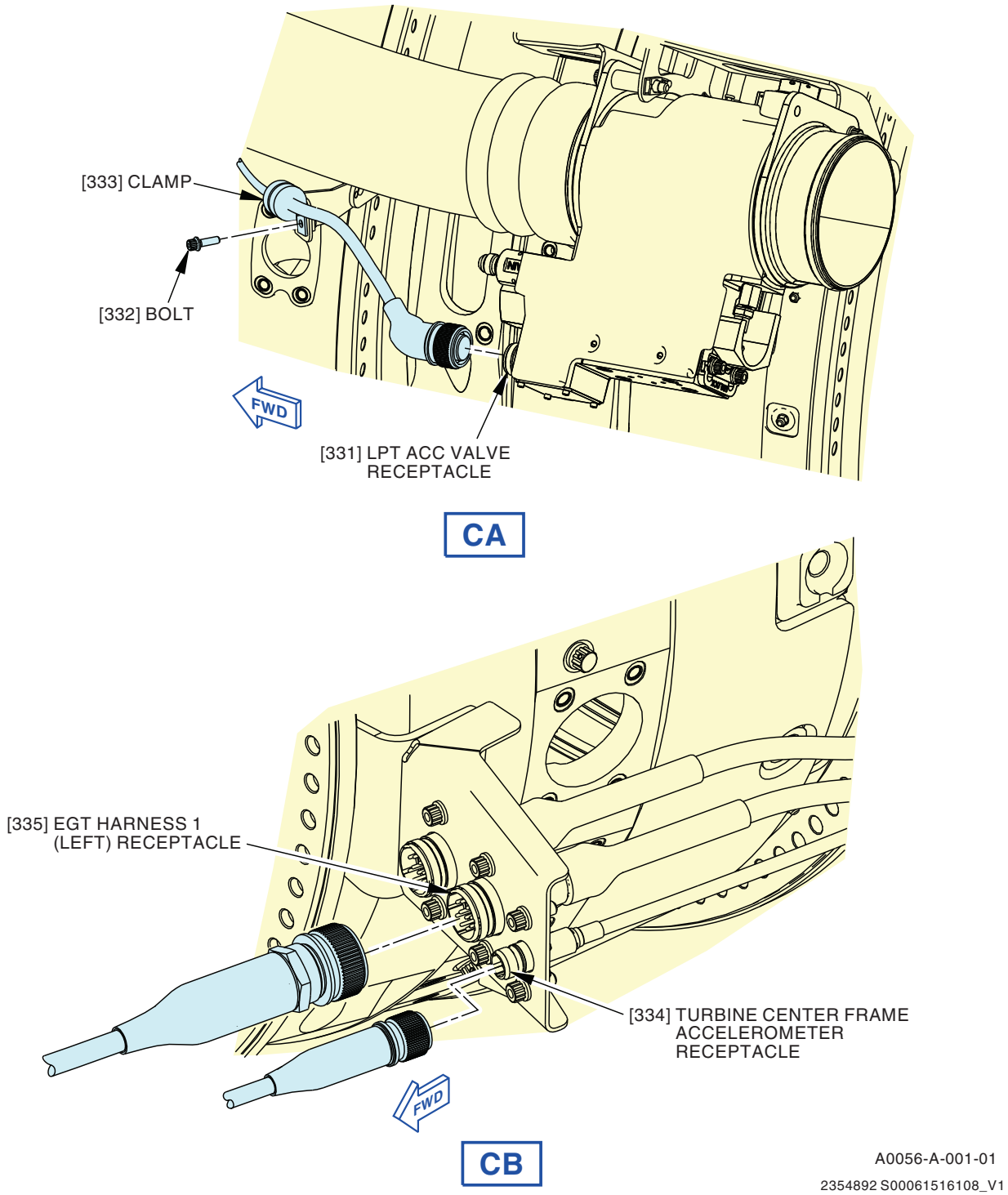
SOME COMPONENTS ARE HIDDEN FOR BETTER CLARITY.

ICNLEAP1B732201D5882800165A00101

3132222 S0000862094_V1

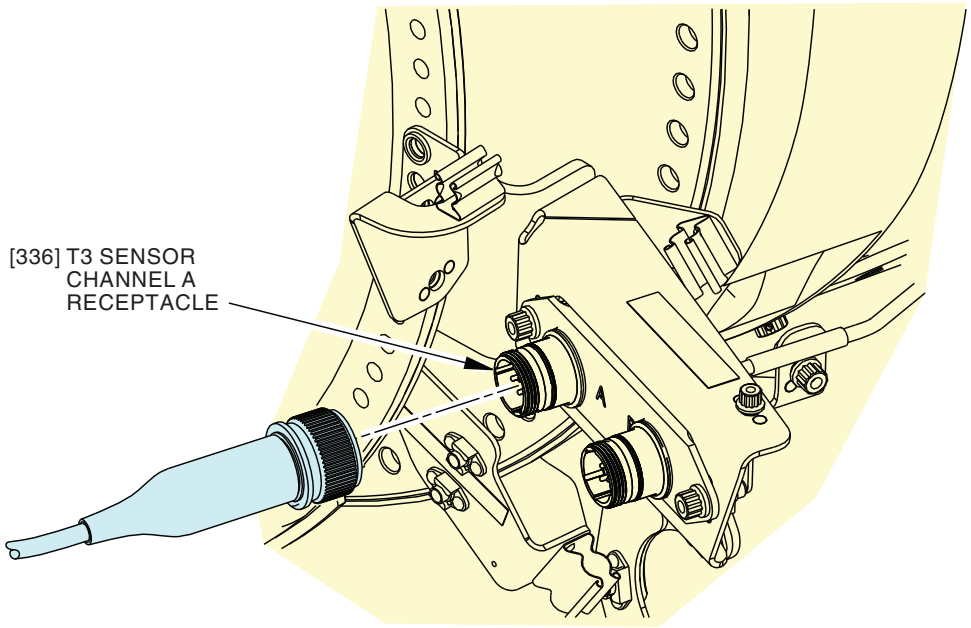
EEC Electrical Harness (HJ8A) Installation
Figure 412/73-22-01-990-858-G00 (Sheet 12 of 14)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



EEC Electrical Harness (HJ8A) Installation
Figure 412/73-22-01-990-858-G00 (Sheet 13 of 14)

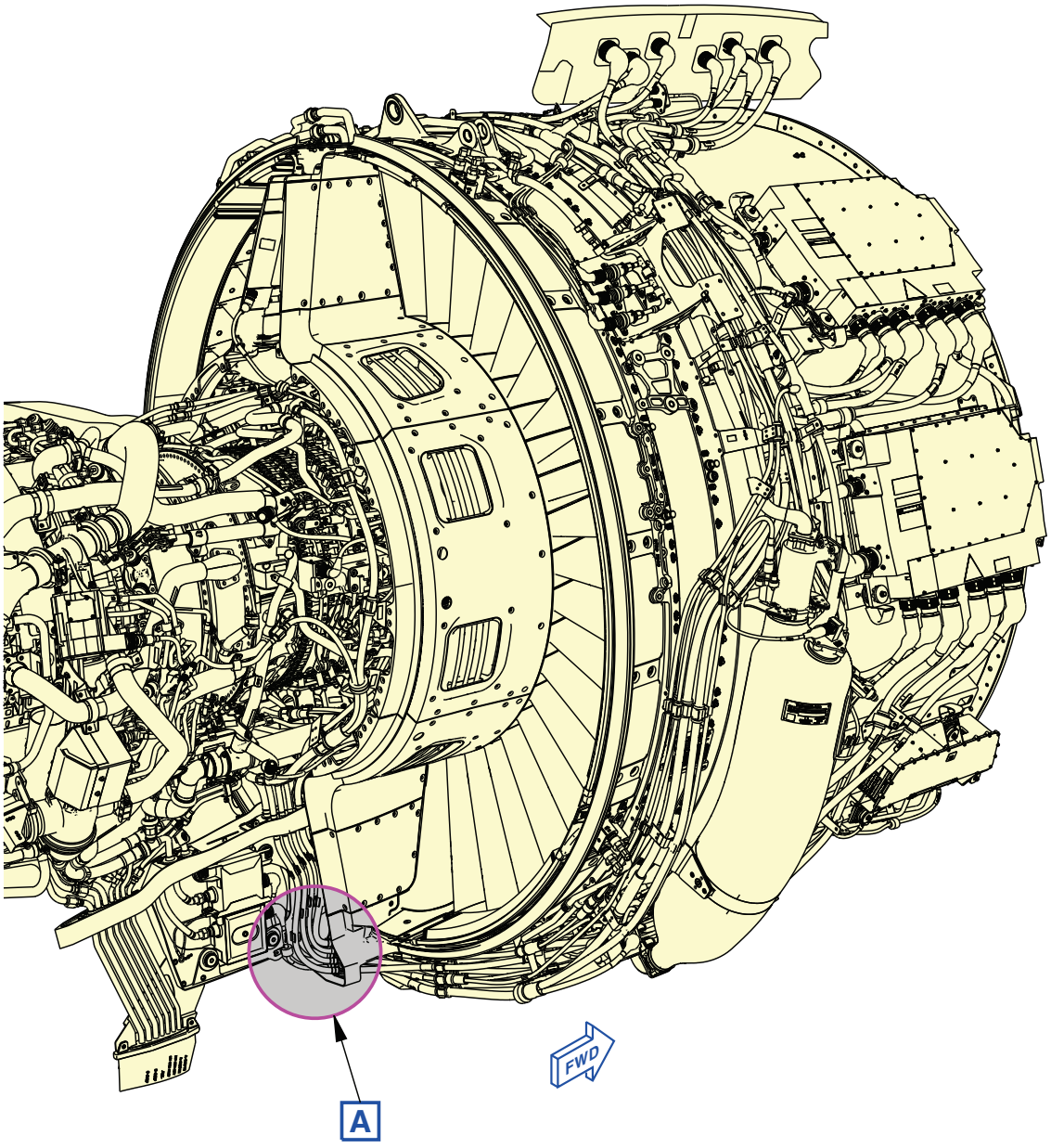
ECCN 9E991 BOEING PROPRIETARY - See title page for details



A0057-A-001-01
2354893 S00061516109_V2

EEC Electrical Harness (HJ8A) Installation
Figure 412/73-22-01-990-858-G00 (Sheet 14 of 14)

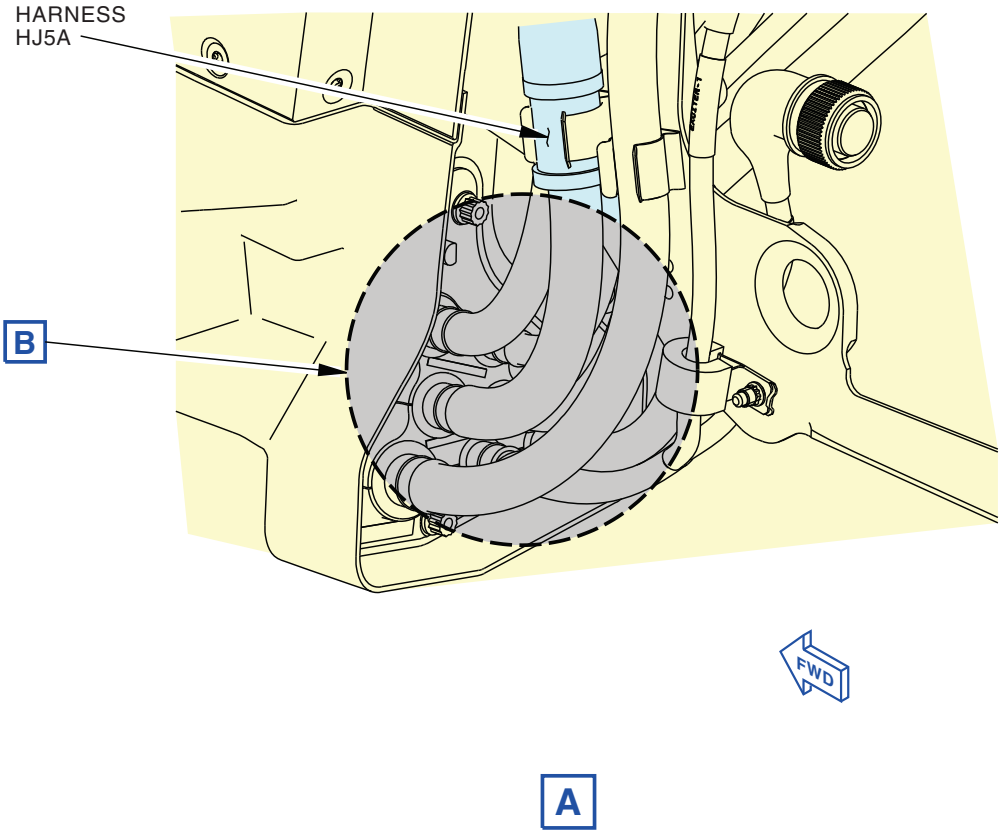
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2937794 S0000713383_V3

Strut Bracket Block Installation
Figure 413/73-22-01-990-874-G00 (Sheet 1 of 9)

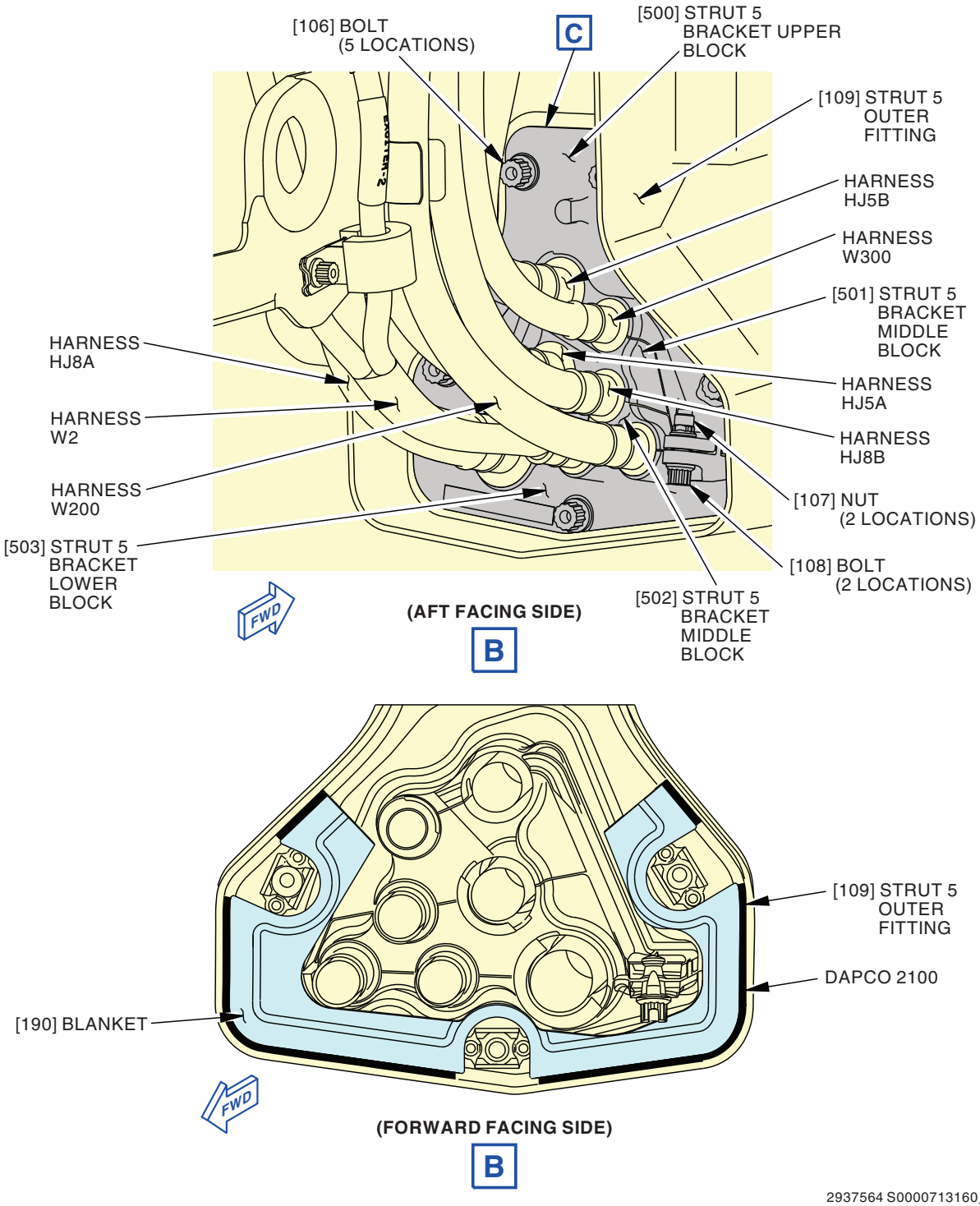
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2937795 S0000713384_V2

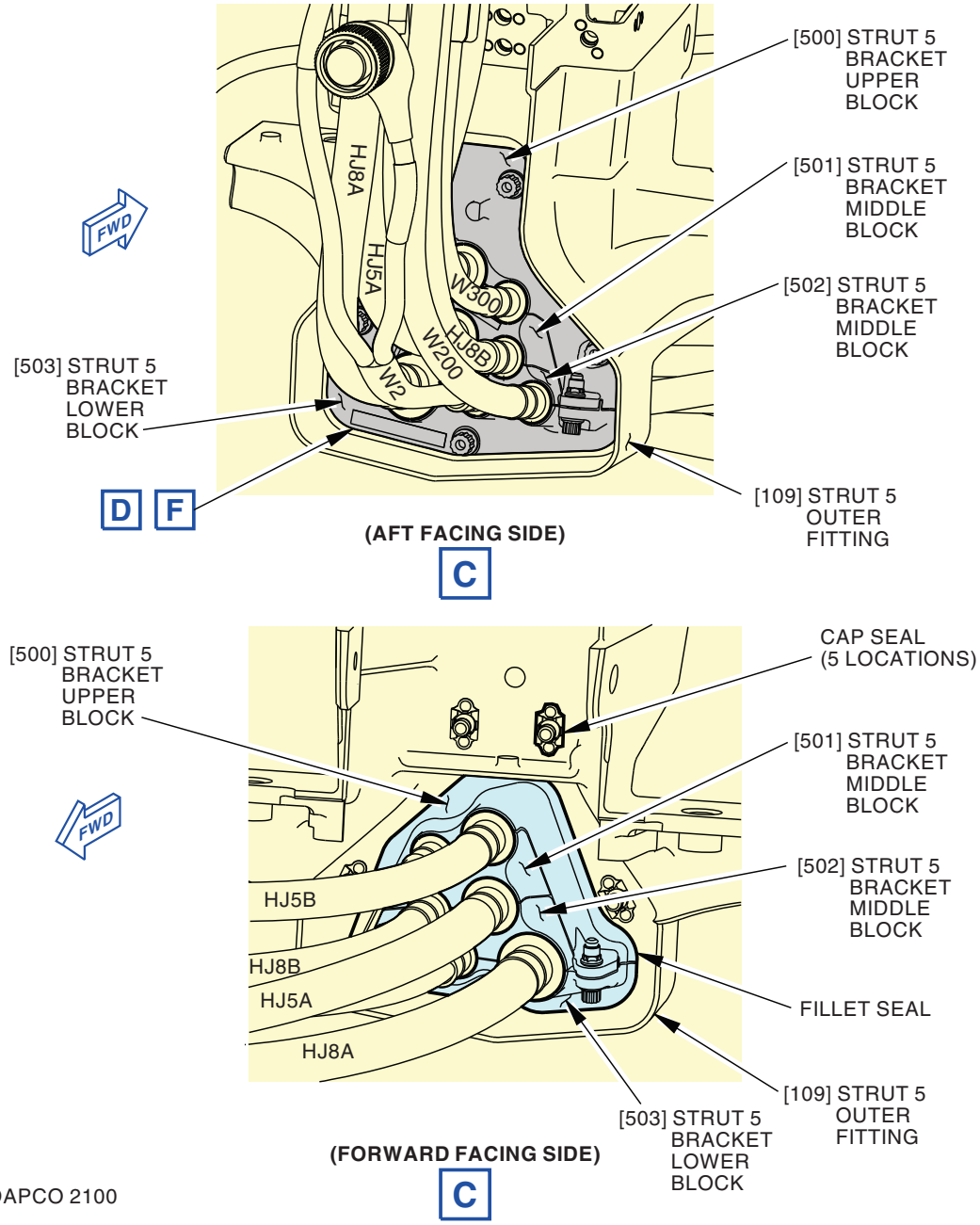
Strut Bracket Block Installation
Figure 413/73-22-01-990-874-G00 (Sheet 2 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Strut Bracket Block Installation
Figure 413/73-22-01-990-874-G00 (Sheet 3 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

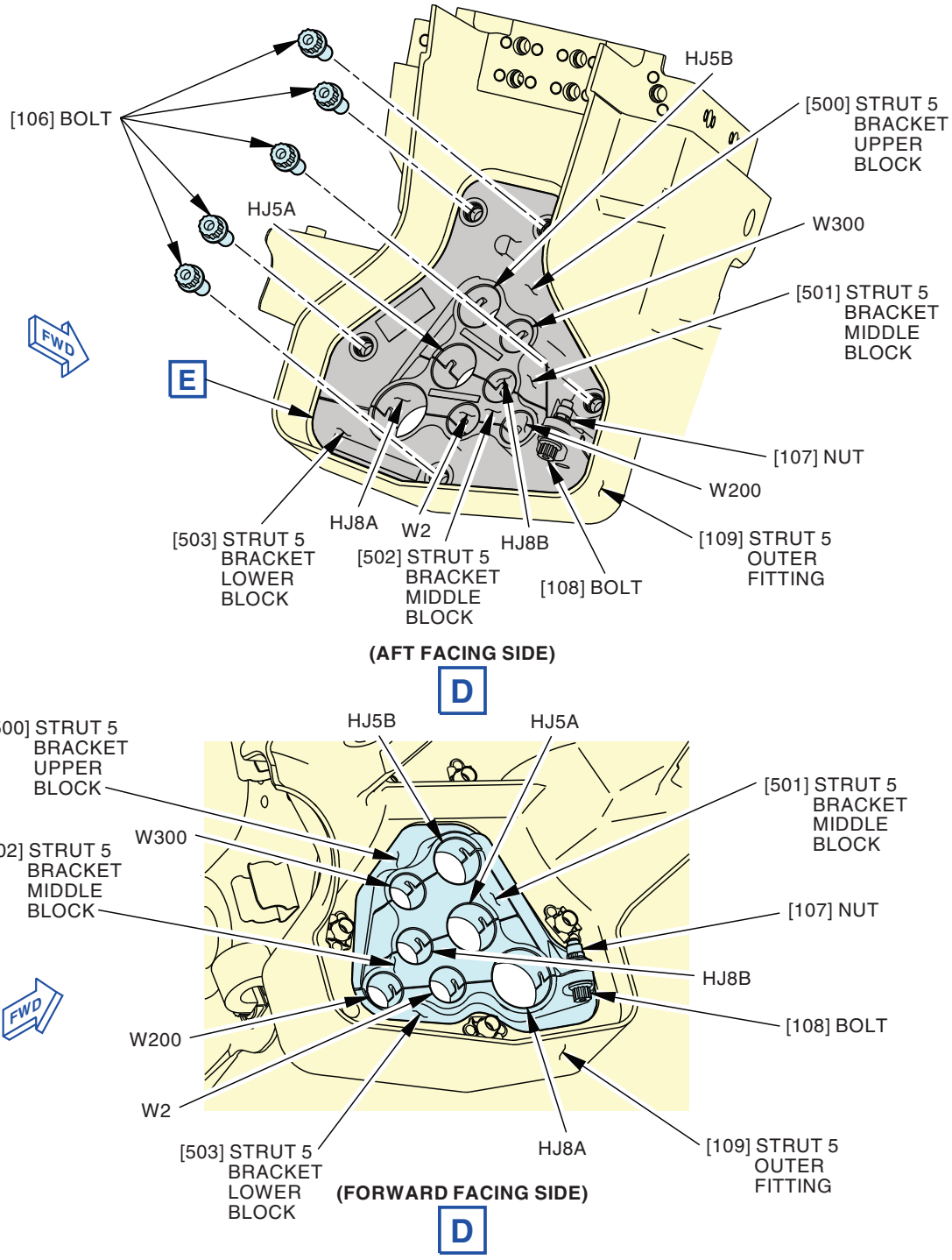
FAY SEAL PER AMM 51-22-00 ON INNER CONTOUR OF THE FITTING (AFT SIDE). SEALANT SQUEEZE-OUT SHOULD BE CONTINUOUS AND EVEN.

IT IS ACCEPTABLE TO APPLY THE FILLET SEAL ALONG INNER CONTOUR OF THE FITTING ON THE FORWARD SIDE.

2937801 S0000713390_V3

Strut Bracket Block Installation
Figure 413/73-22-01-990-874-G00 (Sheet 4 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



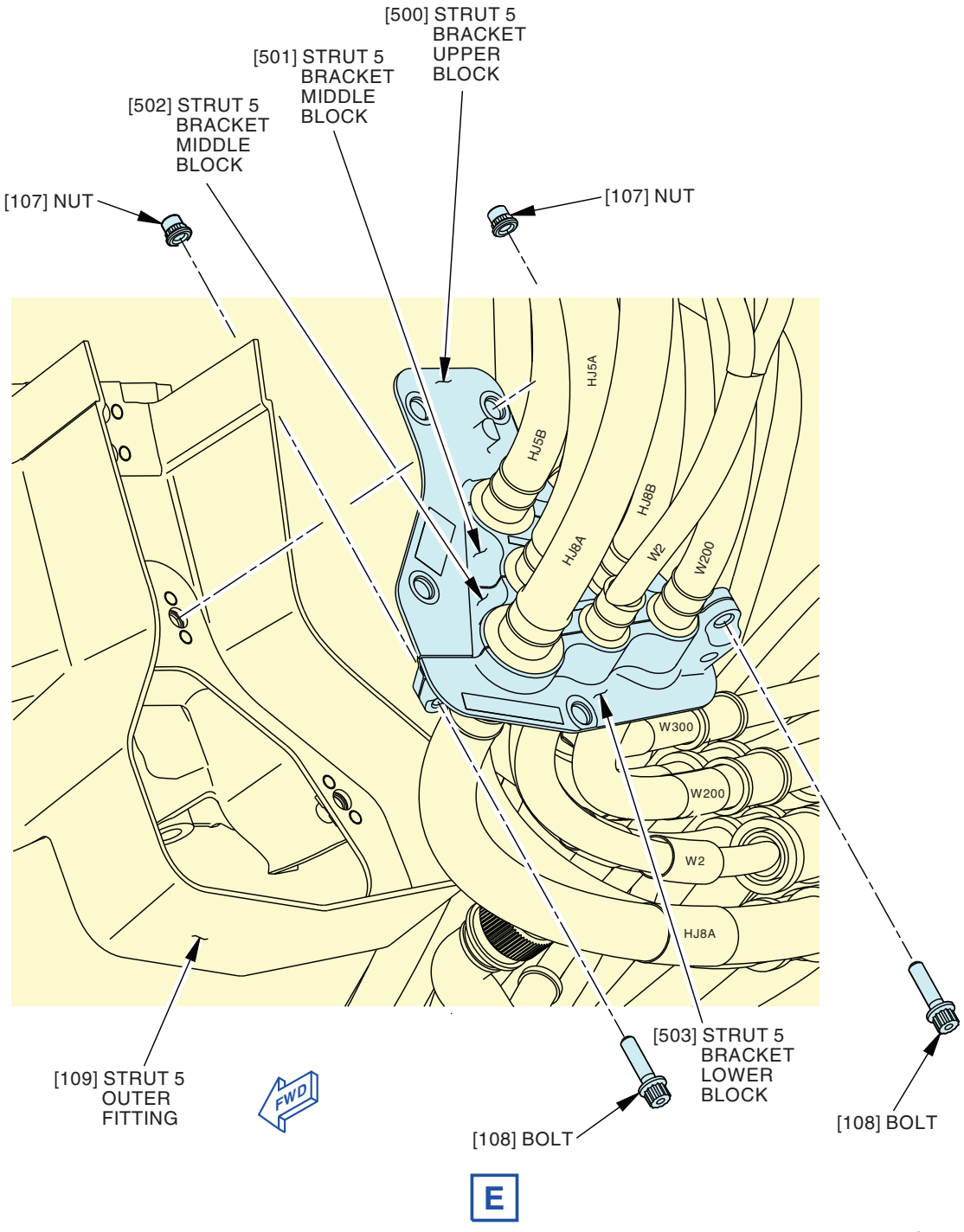
NOTE:

HARNESSES NOT SHOWN

2943980 S0000720369_V2

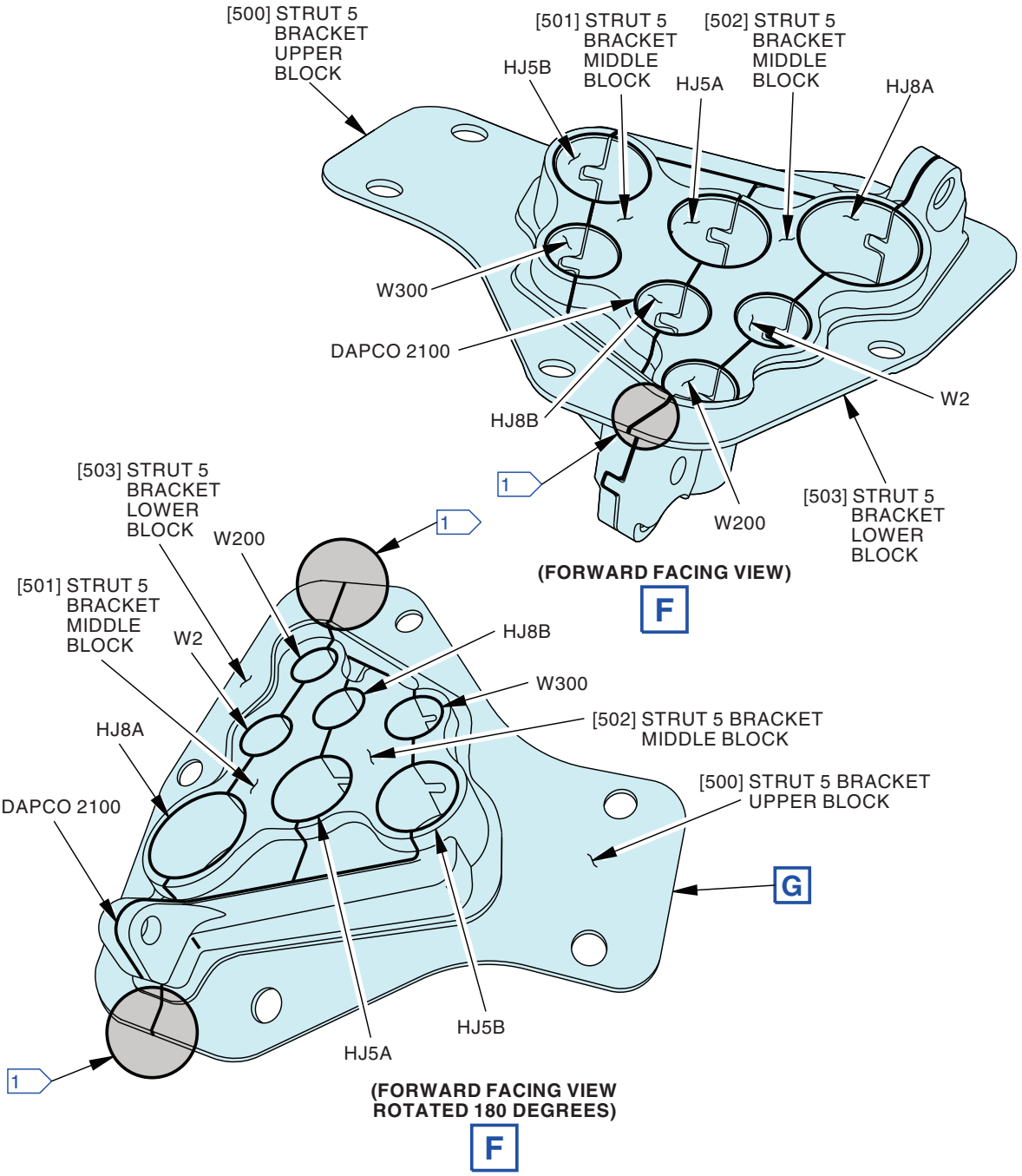
Strut Bracket Block Installation
Figure 413/73-22-01-990-874-G00 (Sheet 5 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Strut Bracket Block Installation
Figure 413/73-22-01-990-874-G00 (Sheet 6 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

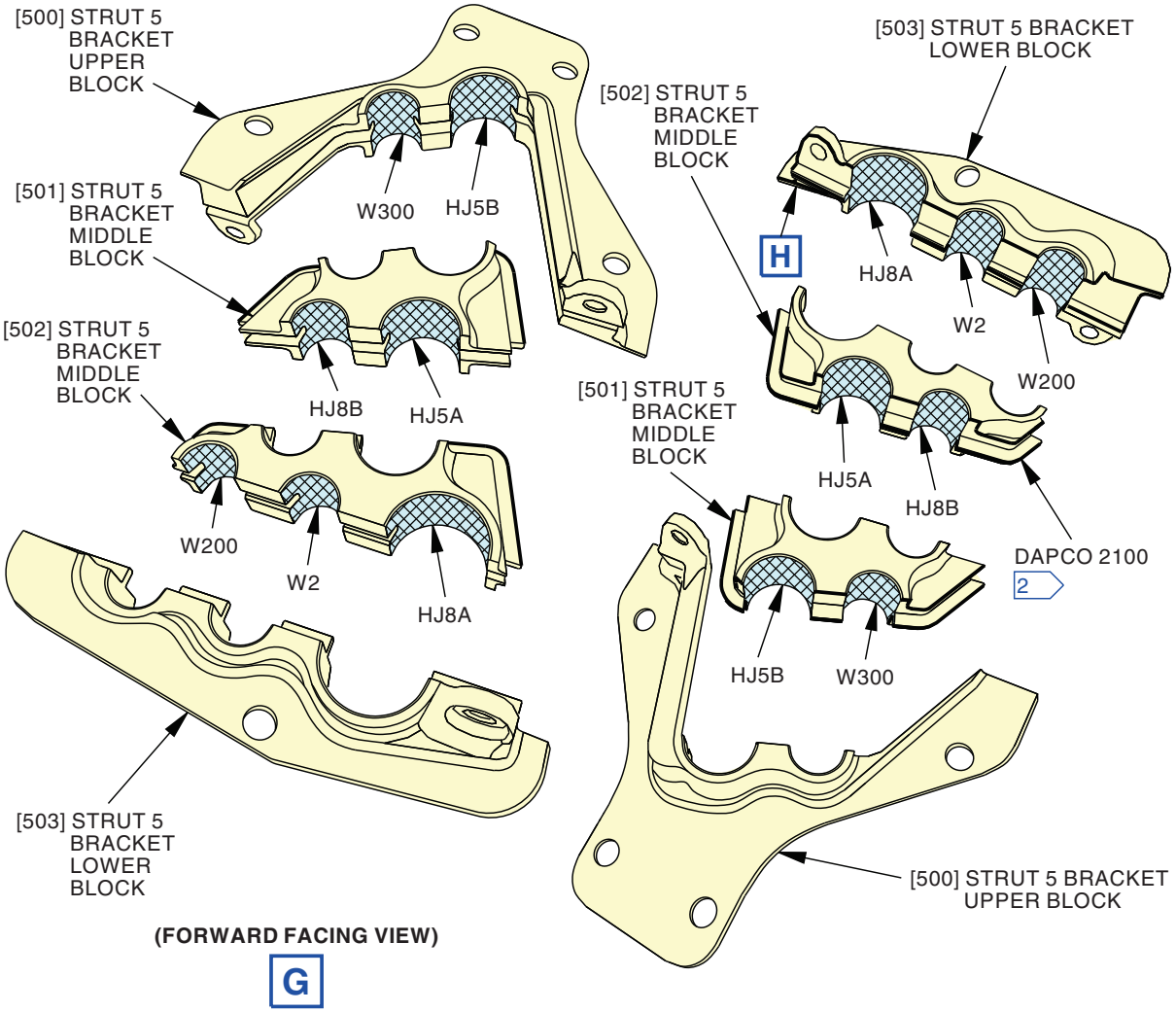
HARNESSES NOT SHOWN.

1 SEALANT SQUEEZE OUT IS FLUSH WITH SURFACE.

2937800 S0000713389_V3

Strut Bracket Block Installation
Figure 413/73-22-01-990-874-G00 (Sheet 7 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

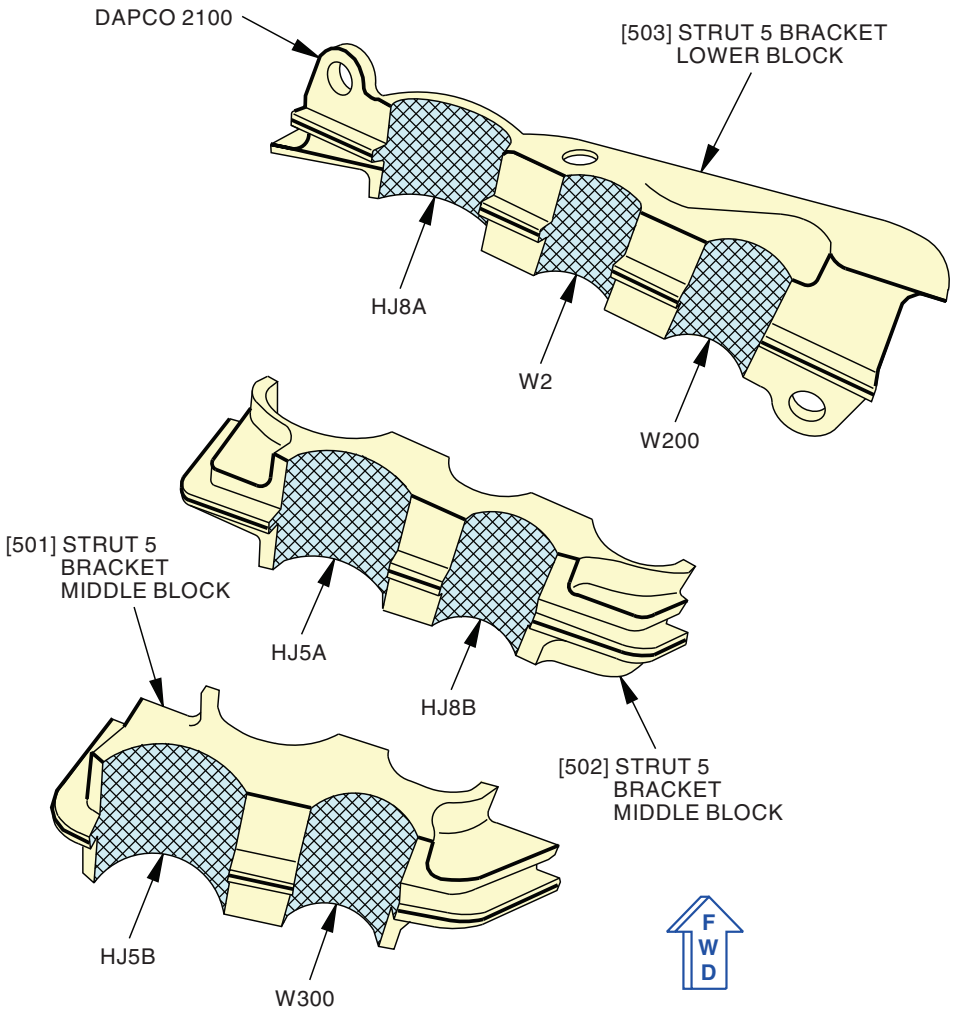
SEALANT PROVIDES FAY SEAL BETWEEN WIRE HARNESS AND BLOCKS AND SHOULD BE APPLIED TO THE BOTH SIDES OF BLOCKS.

2 SEALANT APPLIED TO ONE SIDE (TOP OR BOTTOM) OF THE BLOCKS.

2937799 S0000713388_V3

Strut Bracket Block Installation
Figure 413/73-22-01-990-874-G00 (Sheet 8 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

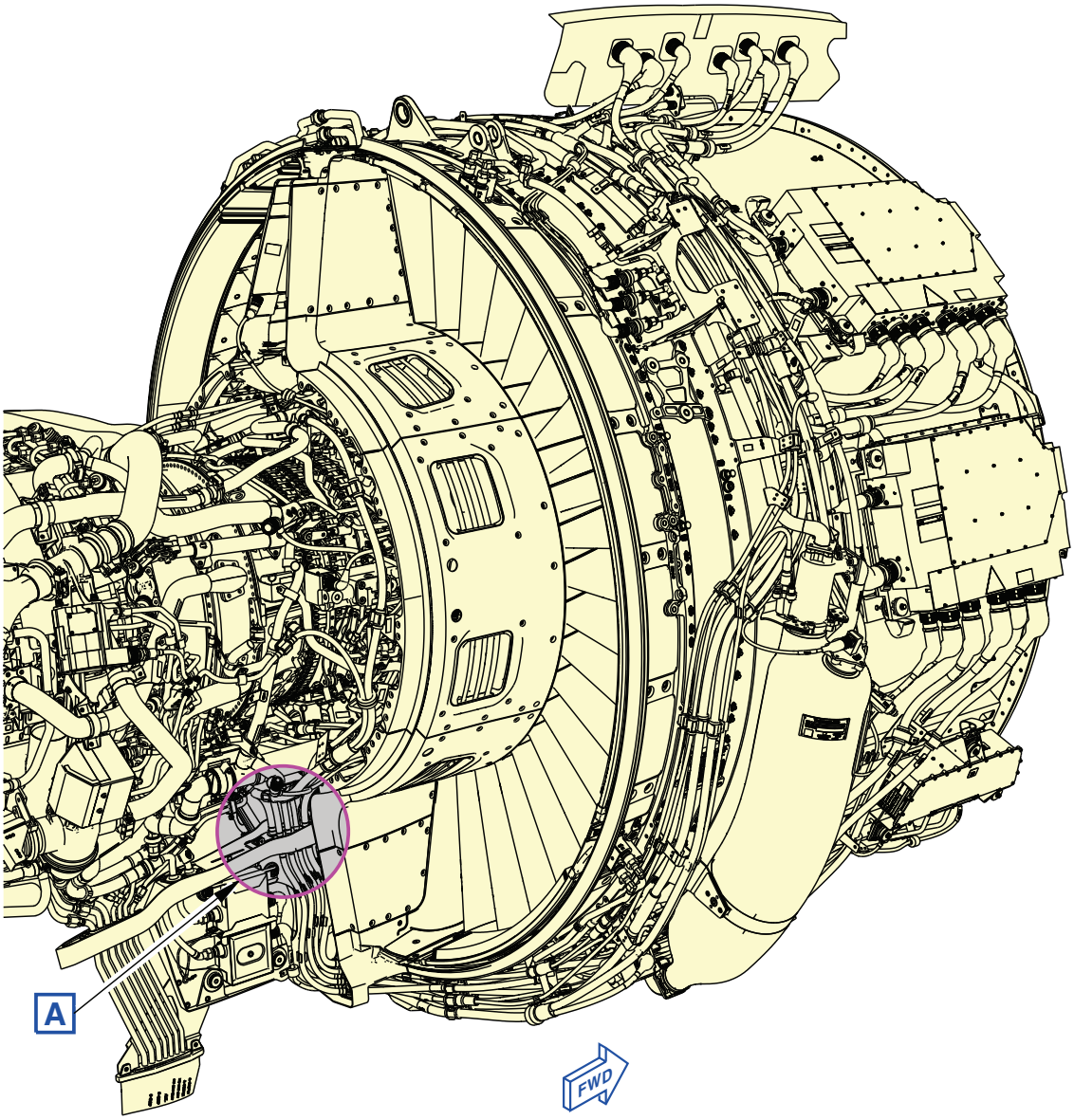


H

2949941 S0000726813_V1

Strut Bracket Block Installation
Figure 413/73-22-01-990-874-G00 (Sheet 9 of 9)

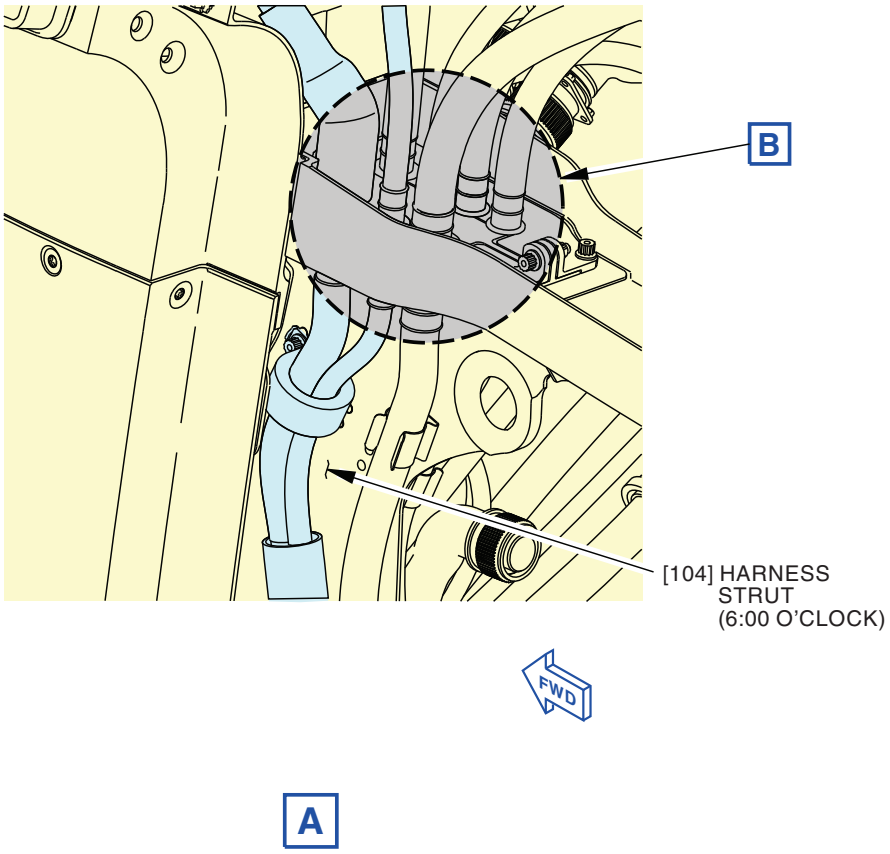
ECCN 9E991 BOEING PROPRIETARY - See title page for details



ICN-LEAP-1B-732201-D-58828-00045-A-001-01
2938818 S0000714836_V3

Harness Support Block Installation
Figure 414/73-22-01-990-875-G00 (Sheet 1 of 11)

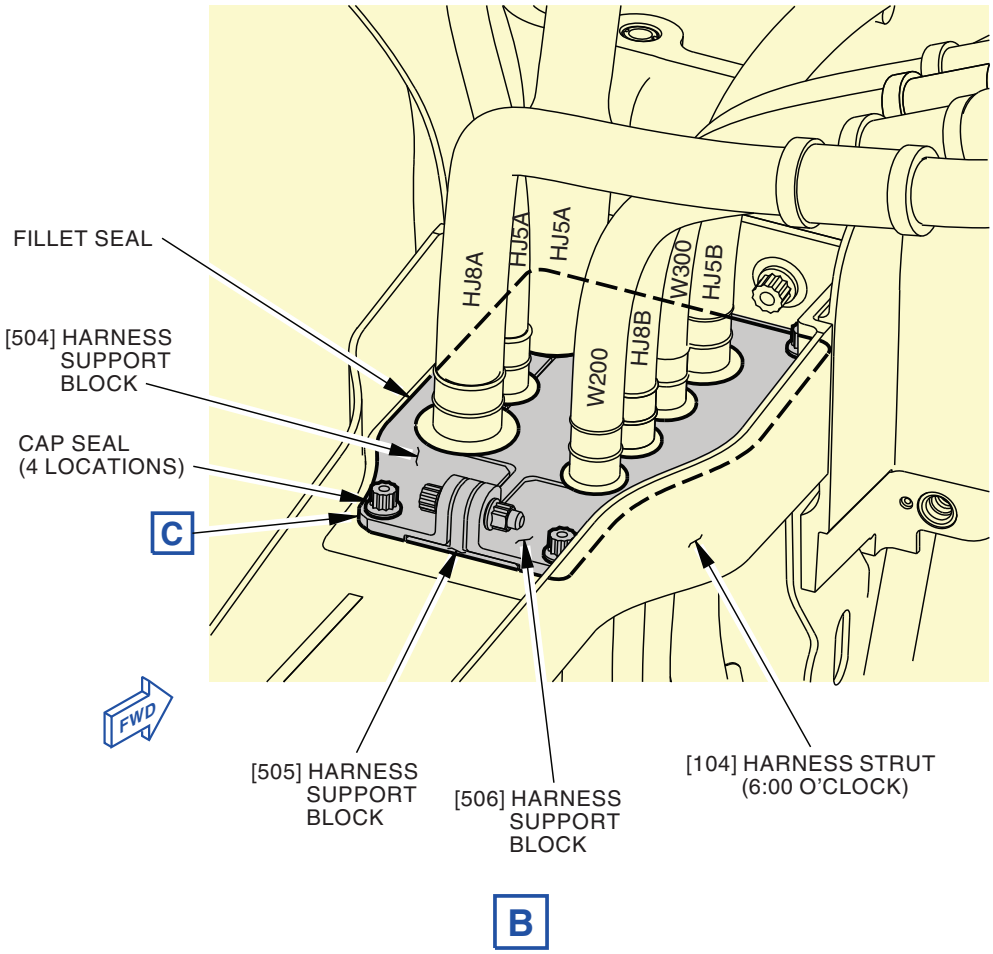
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2952925 S0000730238_V1

Harness Support Block Installation
Figure 414/73-22-01-990-875-G00 (Sheet 2 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEGEND:

——— DAPCO 2100

NOTE:

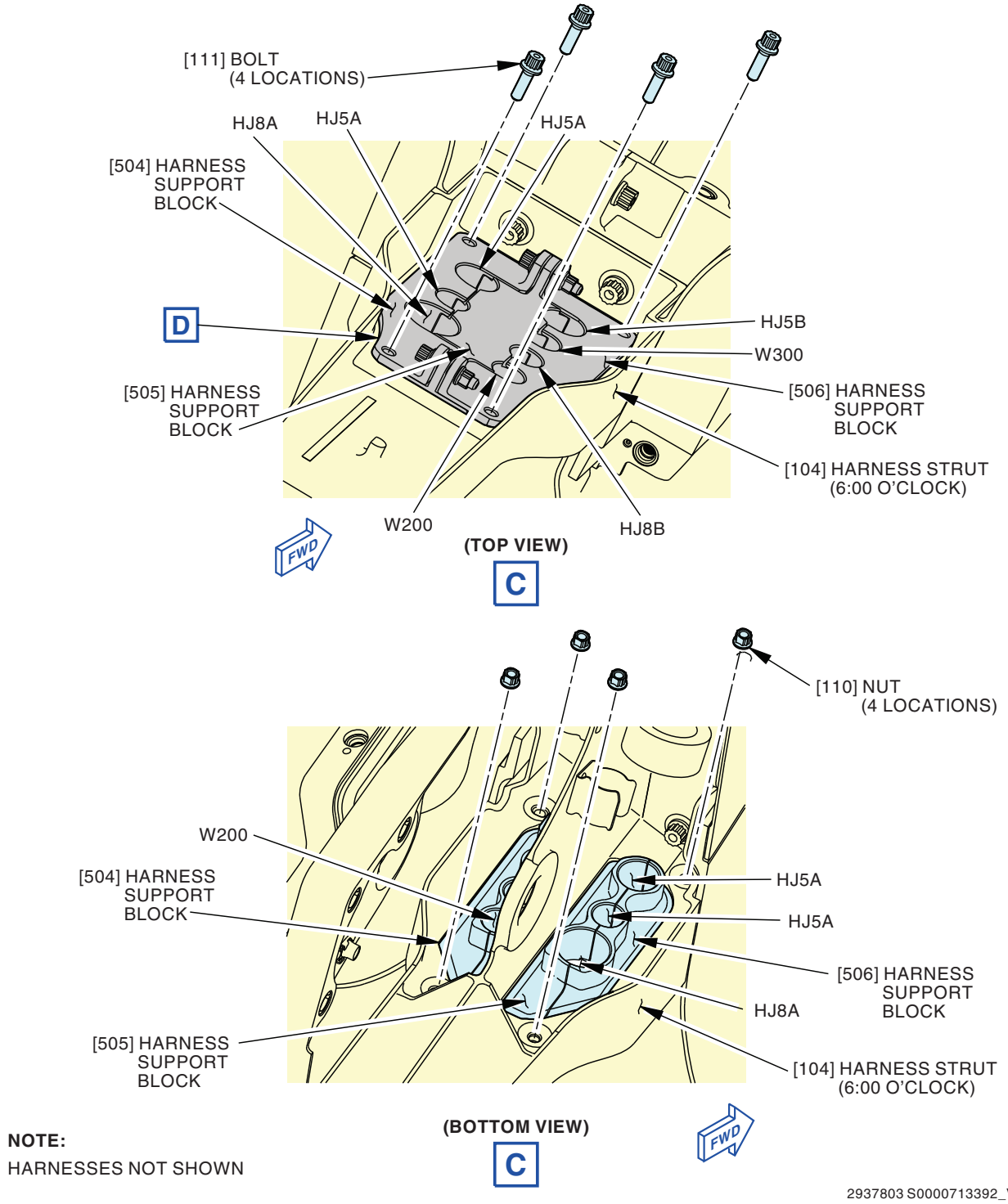
THE FILLET SEAL IS FINISH OF THE SQUEEZE OUT AROUND THE CONTOUR OF THE HARNESS SUPPORT BLOCK. IT IS ACCEPTABLE TO APPLY A FILLET SEAL AROUND THE CONTOUR OF THE BLOCK AFTER IT IS INSTALLED.

THIS VIEW SHOWS DAPCO 2100 APPLICATION WHEN ALL HARNESS SUPPORT BLOCKS HAVE BEEN REMOVED.

2937807 S0000713396_V4

Harness Support Block Installation
Figure 414/73-22-01-990-875-G00 (Sheet 3 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

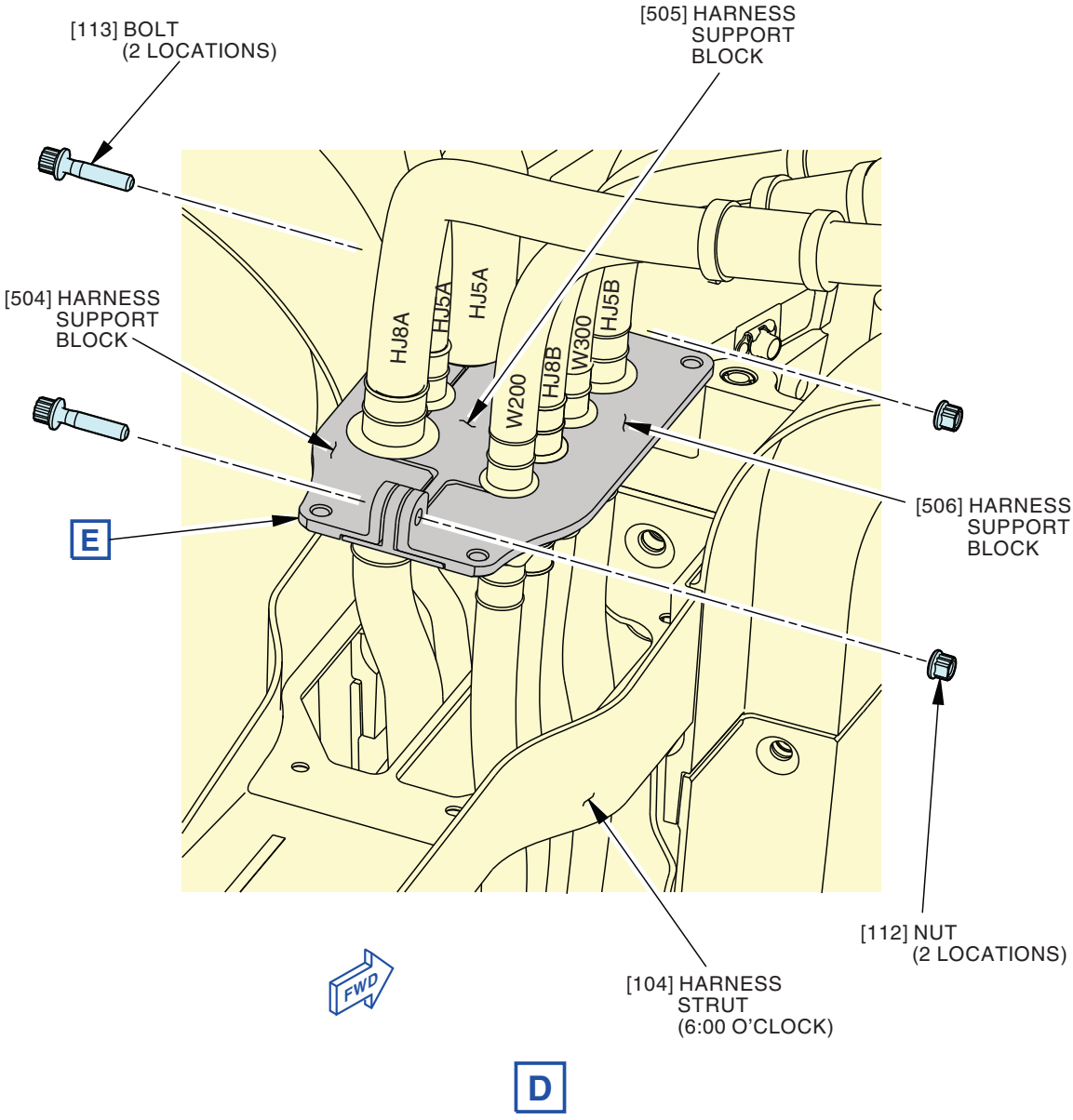


NOTE:
HARNESSES NOT SHOWN

2937803 S0000713392_V2

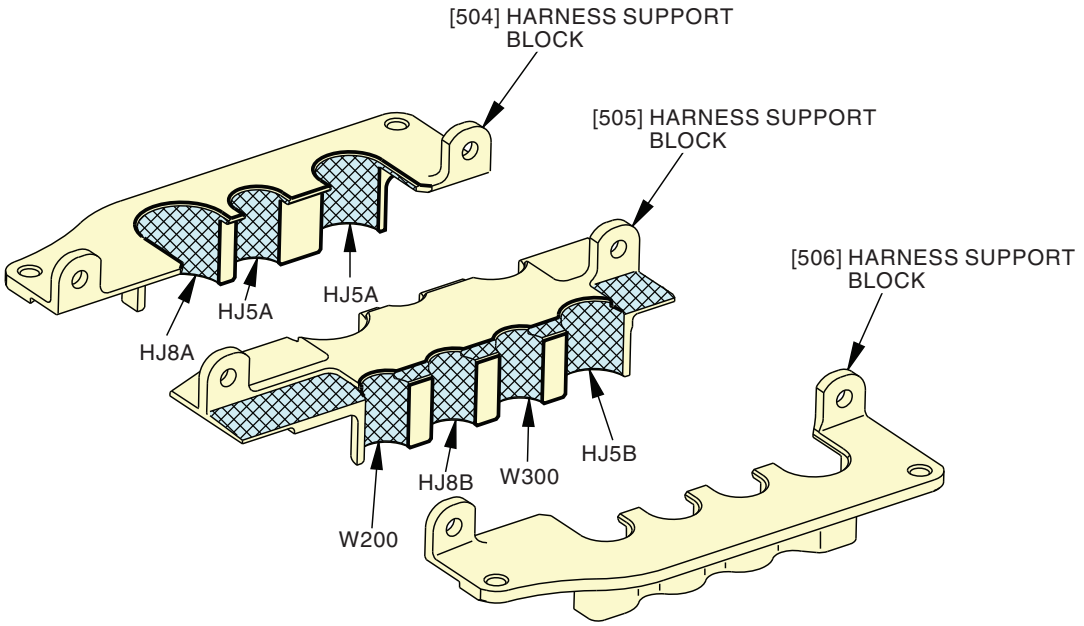
Harness Support Block Installation
Figure 414/73-22-01-990-875-G00 (Sheet 4 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



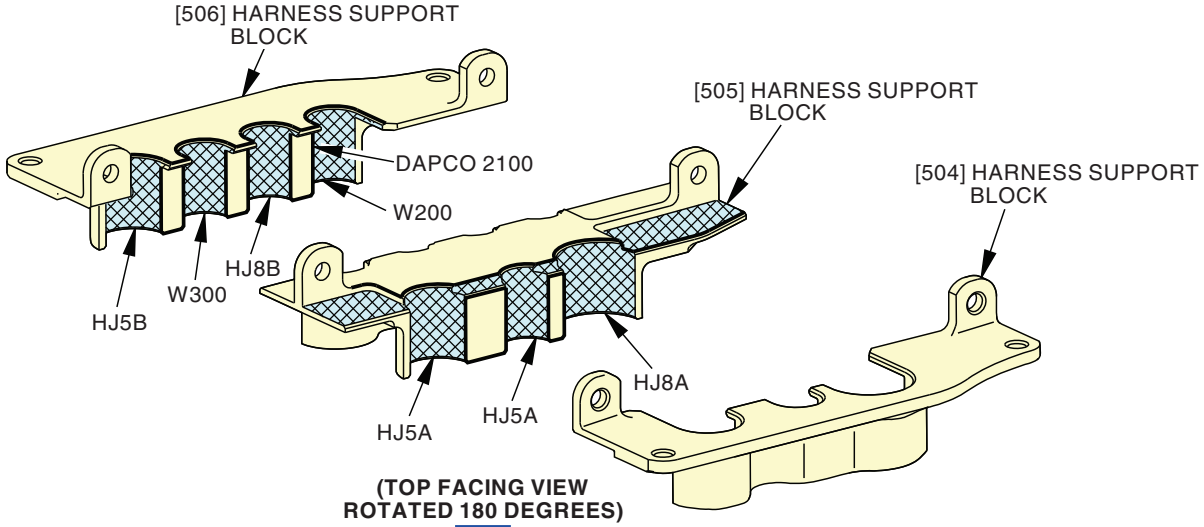
Harness Support Block Installation
Figure 414/73-22-01-990-875-G00 (Sheet 5 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(TOP FACING VIEW)

E



(TOP FACING VIEW ROTATED 180 DEGREES)

E

LEGEND:

DAPCO 2100

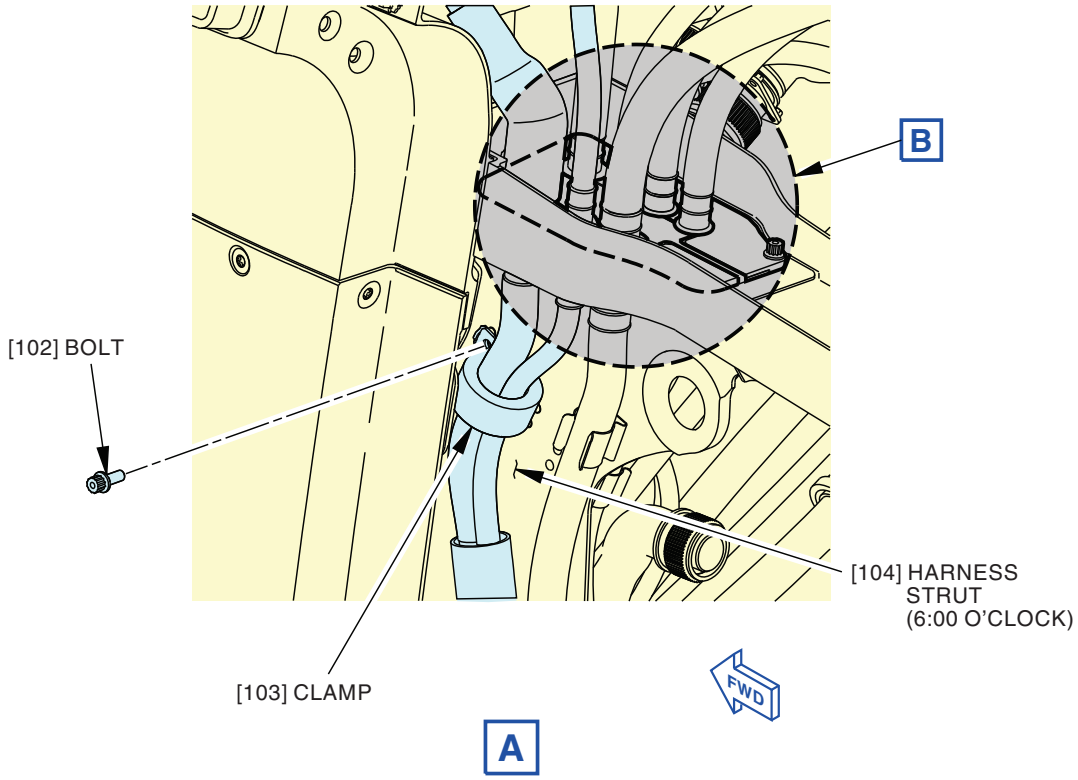
NOTE:

THIS VIEW SHOWS DAPCO 2100 APPLICATION WHEN ALL HARNESS SUPPORT BLOCKS HAVE BEEN REMOVED.

2937805 S0000713394_V4

Harness Support Block Installation
Figure 414/73-22-01-990-875-G00 (Sheet 6 of 11)

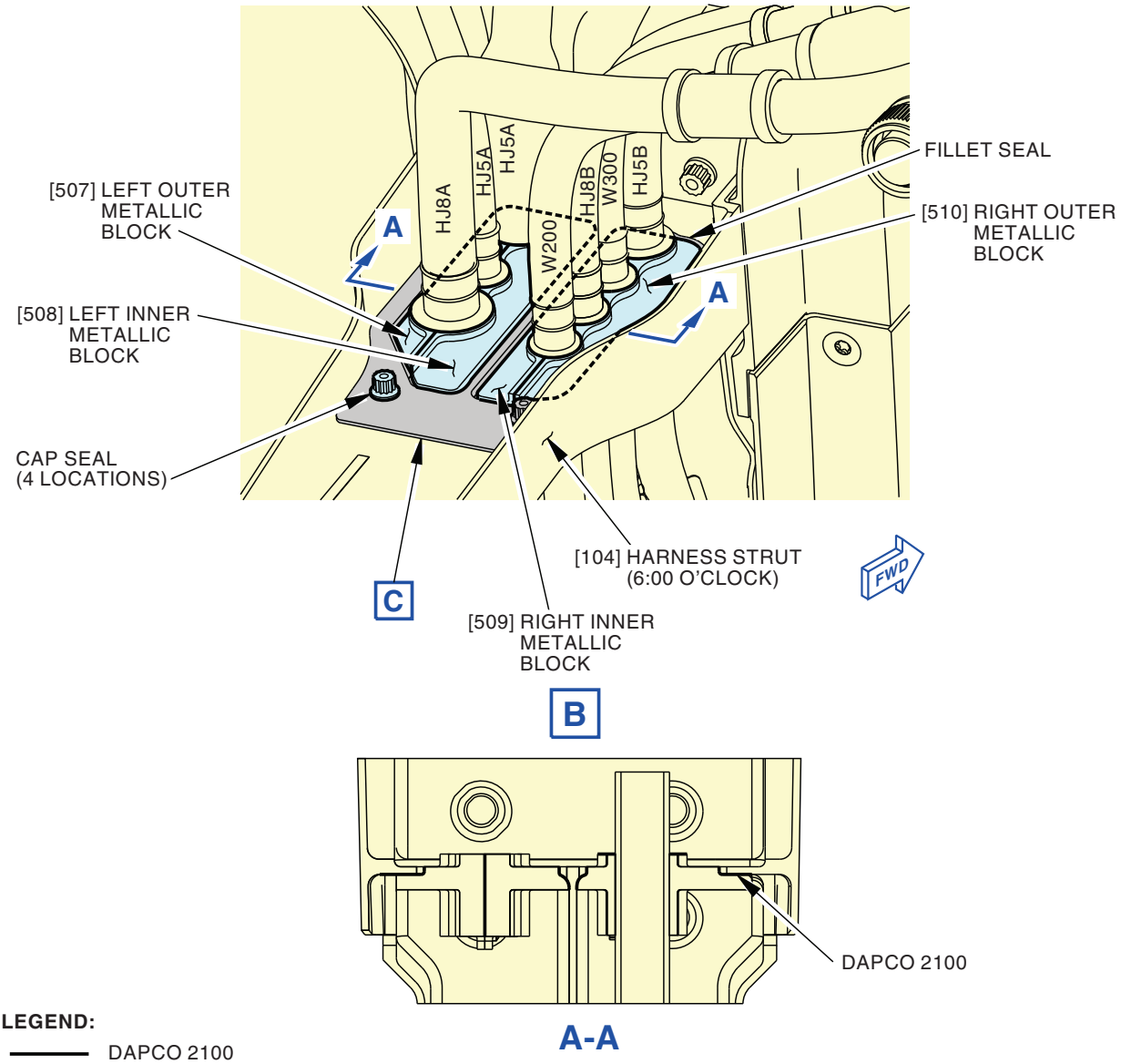
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2962027 S0000739197_V1

Harness Support Block Installation
Figure 414/73-22-01-990-875-G00 (Sheet 7 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEGEND:

DAPCO 2100

NOTE:

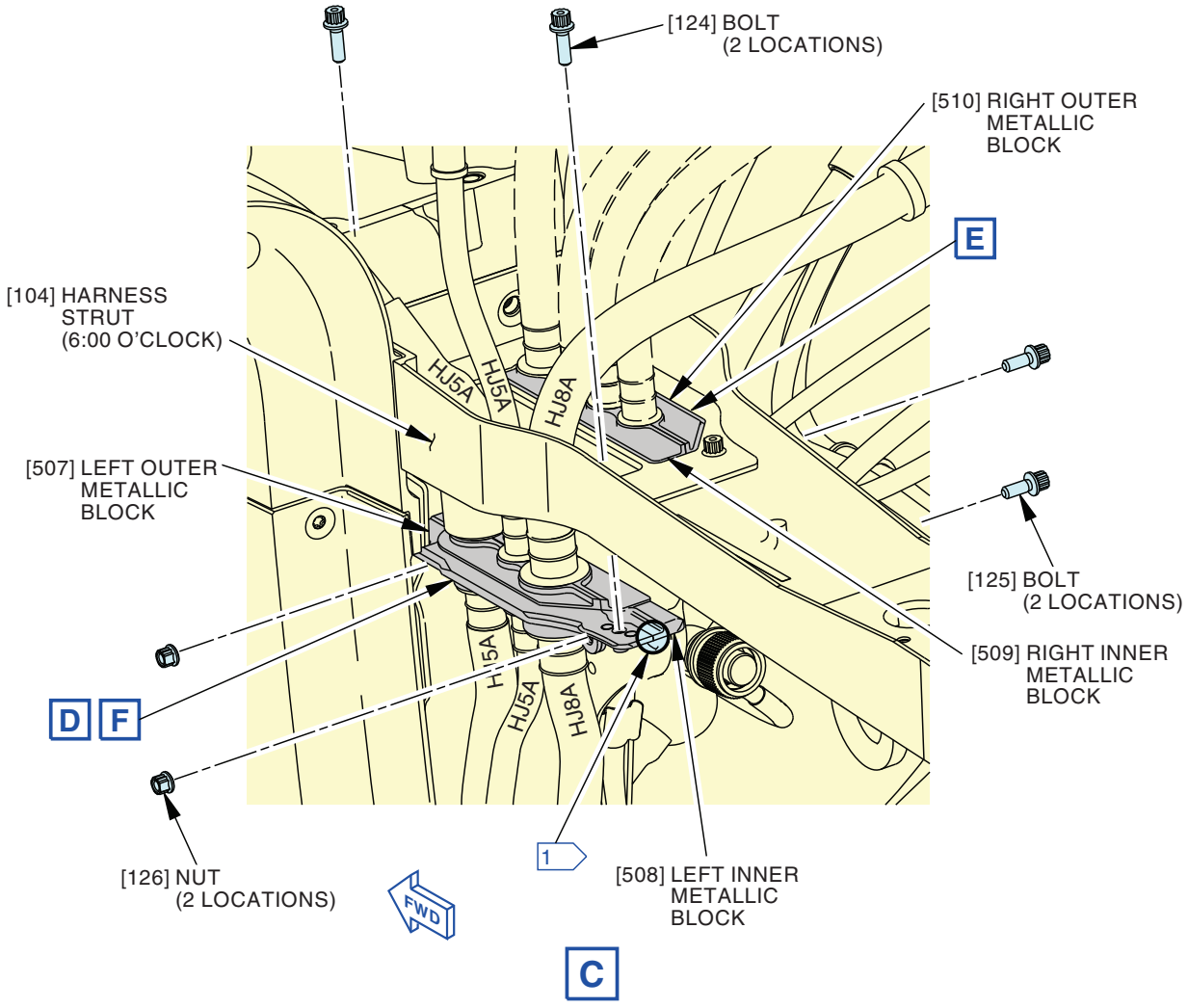
THE FILLET SEAL IS FINISH OF THE SQUEEZE OUT AROUND THE CONTOUR OF THE METALLIC BLOCKS. IT IS ACCEPTABLE TO APPLY A FILLET SEAL AROUND THE CONTOUR OF THE BLOCK AFTER IT IS INSTALLED.

THIS VIEW SHOWS DAPCO 2100 APPLICATION WHEN BOTH RIGHT AND LEFT METALLIC BLOCKS HAVE BEEN REMOVED.

2962029 S0000739198_V2

Harness Support Block Installation
Figure 414/73-22-01-990-875-G00 (Sheet 8 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

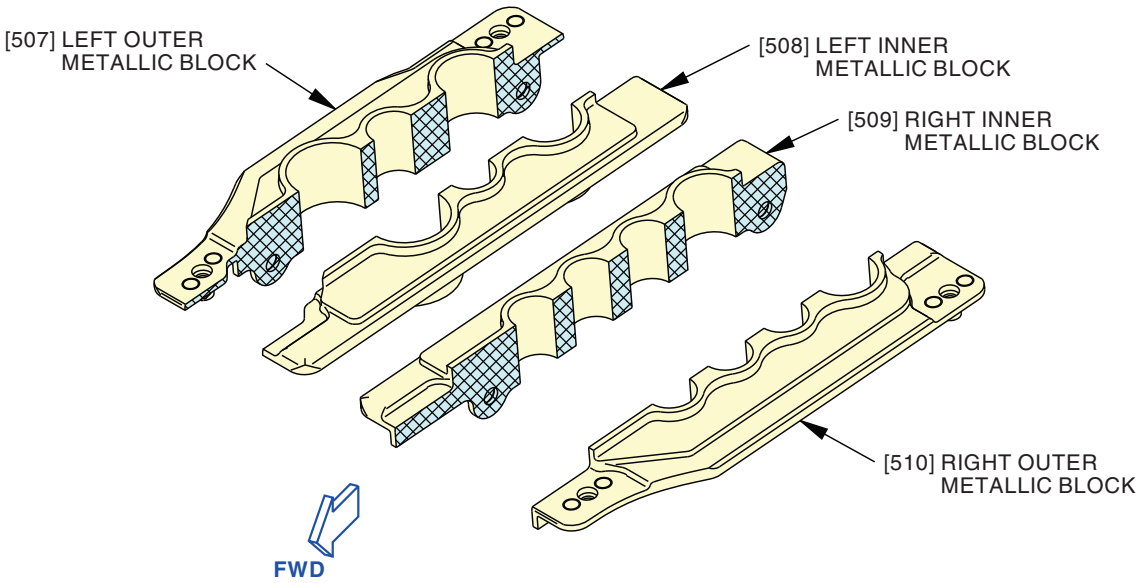
THIS VIEW IS SPECIFIC FOR LEFT INNER AND OUTER METALLIC BLOCKS. SAME LOGIC IS APPLIED FOR RIGHT INNER AND OUTER METALLIC BLOCKS.

1 SEALANT SQUEEZE OUT IS FLUSH WITH SURFACE.

ICNLEAP1B732201D5882800060A00101
2962028 S0000739199_V2

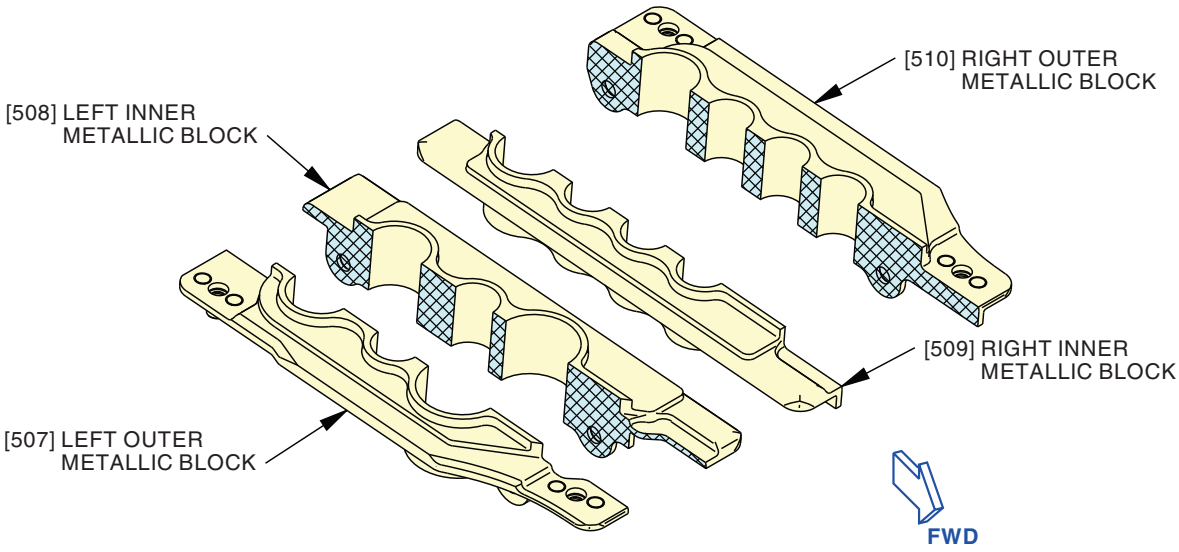
Harness Support Block Installation
Figure 414/73-22-01-990-875-G00 (Sheet 9 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(TOP FACING VIEW)

D



(TOP FACING VIEW ROTATED 90 DEGREES)

D

LEGEND:

DAPCO 2100

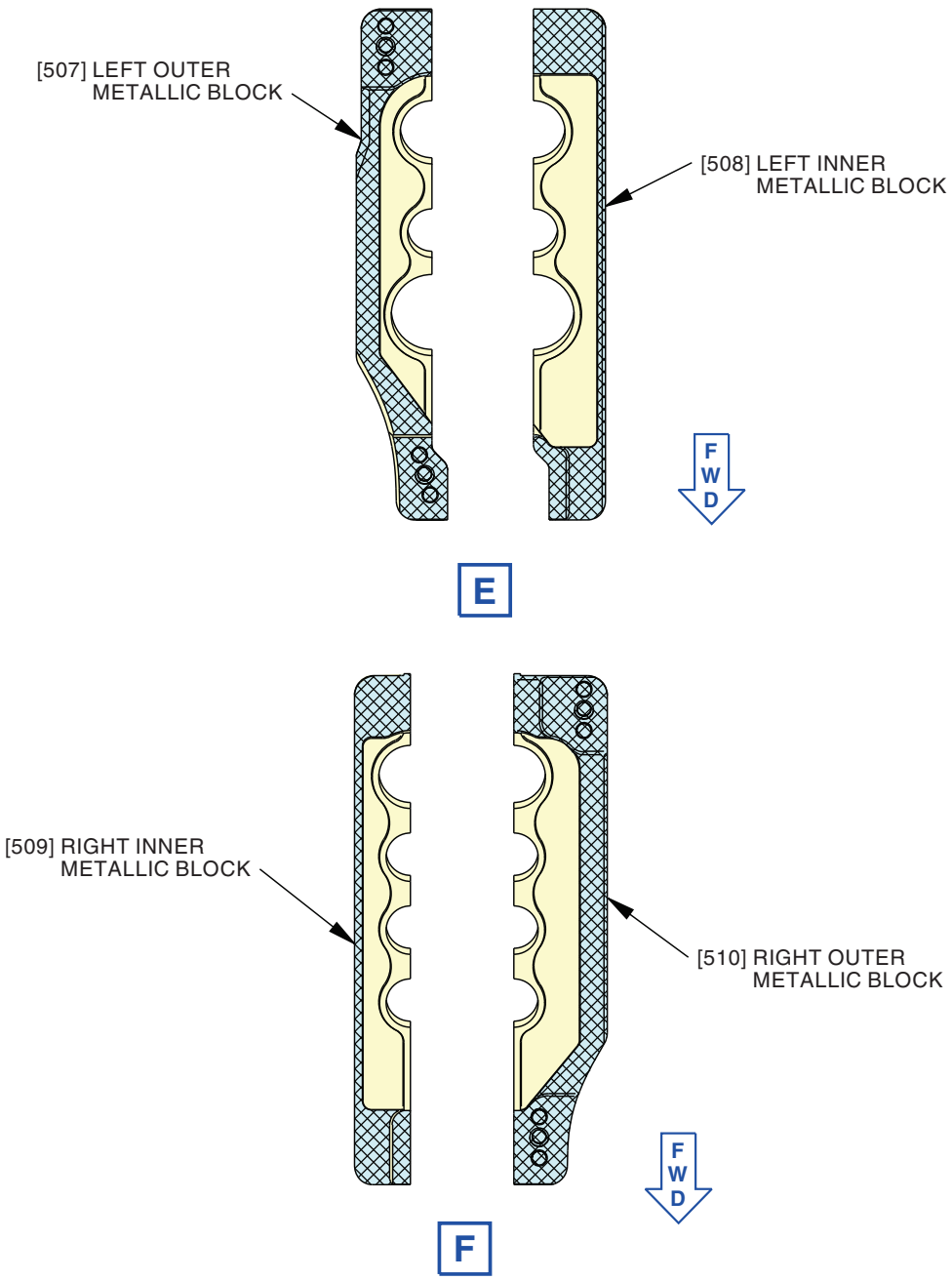
NOTE:

THIS VIEW SHOWS DAPCO 2100 APPLICATION WHEN ALL HARNESS SUPPORT BLOCKS HAVE BEEN REMOVED.

2994662 S0000771309_V1

Harness Support Block Installation
Figure 414/73-22-01-990-875-G00 (Sheet 10 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEGEND:

DAPCO 2100

NOTE:

THIS VIEW SHOWS DAPCO 2100 APPLICATION WHEN ALL HARNESS SUPPORT BLOCKS HAVE BEEN REMOVED.

2994663 S0000771310_V1

Harness Support Block Installation
Figure 414/73-22-01-990-875-G00 (Sheet 11 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 73-22-01-400-808-G00

17. Electronic Engine Control (EEC) Electrical Harness (HJ8A) Installation

(Figure 412, Figure 413, and Figure 414)

A. General

(1) This task gives the instructions to install the electronic engine control (EEC) electrical harness (HJ8A).

(2) The electronic engine control (EEC) electrical harness is referred to as the EEC electrical harness.

(3) You must do the tests that are given in the power plant test reference table after you install the applicable engine control electrical harnesses.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-23-04-400-801-G00	Fan Frame Module Segment Shrouds Installation (P/B 401)
73-22-01-390-802-G00	Electronic Engine Control (EEC) Electrical Harness (HJ8A) Tape Wrap (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
A50396	Sealant - Dapco 2200 Primerless Silicone Firewall Sealant	BMS5-63 Type II Class B-1/2
A50483 [CP2927]	Silicone Paste - Primerless, For Fire Resistance application (Dapco 2100 Firewall Sealant)	
D00599 [CP2442]	Oil - Engine	
D00601 [CP2101]	Vaseline - Graphite Mineral	
G51328 [CP2674]	Cloth - Lint-Free	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Electronic Engine Control (EEC) Electrical Harness (HJ8A) Installation

SUBTASK 73-22-01-210-005-G00

(1) During the installation, make sure that EEC electrical harness (HJ8A) [303] is not stressed or damaged.

SUBTASK 73-22-01-210-006-G00

(2) Make sure that all components are clean and free of old sealant prior to adding new sealant.

SUBTASK 73-22-01-390-007-G00

(3) For the EEC electrical harness (HJ8A) [303] P/N 364-083-811-0, do this task: Electronic Engine Control (EEC) Electrical Harness (HJ8A) Tape Wrap, TASK 73-22-01-390-802-G00.

NOTE: For the EEC electrical harness (HJ8A) [303] P/N 364-083-811-0, it is necessary to apply sealant and aluminum tape over the new wire bundle before installation. Other EEC electrical harness (HJ8A) [303] part numbers do not require the sealant and aluminum tape. The sealant and aluminum tape is required to prevent fluid leak from migrating down the wire harness through the firewall at the 6 o'clock location.

SUBTASK 73-22-01-420-015-G00

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

(4) Install the EEC electrical harness (HJ8A) [303] as follows (TASK 70-00-01-910-803-G00):

(a) Make sure that the mating faces are clean and free of unwanted materials.

(b) Remove the protective covers on the connectors and receptacles.

(c) Make sure to apply silicone sealant during an assembly of harnesses and all related parts.

1) Apply sealant on edges that are the interfaces of parts.

(d) Install the EEC electrical harness (HJ8A) [303] in a correct position with the clips [302] along its routing.

(e) Install the EEC electrical harness (HJ8A) [303] on the harness strut (6:00 o'clock) [104] as follows:

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0288

1) Install the harness support block [504], harness support block [505], and harness support block [506] on the harness strut (6:00 o'clock) [104] as follows:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0288 (Continued)

a) Apply graphite mineral vaseline, D00601 [CP2101], on the threads and pressure faces of the four bolts [111] and two nuts [112].

b) Apply Dapco 2100 sealant, A50483 [CP2927], on the harness support block [504], harness support block [505], harness support block [506], and around the harnesses (View E, Figure 414).

c) Make sure that the EEC electrical harness (HJ8A) [303] and harnesses are in the correct position on the harness support block [504], harness support block [505], harness support block [506], (View D, Figure 414).

d) Install the two bolts [113] and two nuts [112] to assemble the harness support block [504], harness support block [505], and harness support block [506] in the correct position with the harnesses.

e) Tighten the nuts [112] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

f) Remove the unwanted graphite mineral vaseline, D00601 [CP2101], from the nuts [112] with a lint-free cloth, G51328 [CP2674].

CAUTION MAKE SURE THAT YOU REMOVE UNWANTED SEALANT FROM THE BOTTOM SIDE OF THE ASSEMBLY. THIS CAN DECREASE THE RISK OF IGNITION DURING A FIRE. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

CAUTION MAKE SURE THAT DAPCO 2100 SEALANT, A50483 [CP2927], IS FLUSH BETWEEN ADJACENT SURFACES. THE SEALANT CAN DECREASE THE RISK OF IGNITION DURING A FIRE. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

g) Make sure that there is minimum amount of the squeeze-out on the lower side.

h) Install the harness support block [504], harness support block [505], and harness support block [506] in the correct position on the harness strut (6:00 o'clock) [104].

i) Apply Dapco 2100 sealant, A50483 [CP2927], between the assembly of the harness support block [504], harness support block [505], harness support block [506], and harness strut (6:00 o'clock) [104] (View B, Figure 414).

j) Install the four bolts [111] and four nuts [110] that attach the harness support block [504], harness support block [505], and harness support block [506] to the harness strut (6:00 o'clock) [104].

k) Tighten the nuts [110] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

l) Remove the unwanted graphite mineral vaseline, D00601 [CP2101], from the four bolts [111] with a lint-free cloth, G51328 [CP2674].

CAUTION MAKE SURE THAT YOU REMOVE UNWANTED SEALANT FROM THE BOTTOM SIDE OF THE ASSEMBLY. THIS CAN DECREASE THE RISK OF IGNITION DURING A FIRE. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

m) Make sure that there is minimum amount of the squeeze-out on the lower side.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0288 (Continued)

n) Apply more sealant to form a correct fillet seal around the contour of the harness support block [504], harness support block [505], harness support block [506], and harness strut (6:00 o'clock) [104].

CAUTION MAKE SURE THAT THE BOLTS ARE CLEAN BEFORE YOU APPLY SEALANT TO THEM. IF YOU DO NOT REMOVE UNWANTED MATERIALS, THE SEALANT WILL NOT BOND. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

o) Clean the four nuts [110] and four bolts [111] that will have sealant applied to them.

p) Apply Dapco 2100 sealant, A50483 [CP2927], to the heads of the four bolts [111].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0288

2) Remove any remaining of old sealant residue left on the harnesses or harness strut (6:00 o'clock) [104].

3) Install the left outer metallic block [507] and left inner metallic block [508] on the harness strut (6:00 o'clock) [104] as follows:

a) Apply graphite mineral vaseline, D00601 [CP2101], on the threads and pressure faces of the two bolts [124] and two nuts [126].

b) Apply Dapco 2100 sealant, A50483 [CP2927], on the mating faces between the left outer metallic block [507] and left inner metallic block [508].

NOTE: Dapco 2100 sealant, A50483 [CP2927], must be evenly applied in large quantity on each hatching area to make sure that there is enough Dapco 2100 sealant, A50483 [CP2927], between parts after the squeeze out when assembling and tightening the bolts and nuts.

c) Dapco 2100 sealant, A50483 [CP2927], on the mating faces between the left outer metallic block [507], left inner metallic block [508], and harness strut (6:00 o'clock) [104].

d) Install the two bolts [125] and two nuts [126] to assemble the left outer metallic block [507] and left inner metallic block [508] in the correct position with the harnesses.

e) Tighten the two nuts [126] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

f) Remove the unwanted graphite mineral vaseline, D00601 [CP2101], from the nuts [126] with a lint-free cloth, G51328 [CP2674].

g) Install the left outer metallic block [507] and left inner metallic block [508] in the correct position on the harness strut (6:00 o'clock) [104] by the bottom side.

h) Install the two bolts [124] that attach the left outer metallic block [507] and left inner metallic block [508] to the harness strut (6:00 o'clock) [104].

i) Tighten the two bolts [124] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

j) Remove the unwanted graphite mineral vaseline, D00601 [CP2101], from the bolts [124] with a lint-free cloth, G51328 [CP2674].

k) Remove the squeezed out excedent and remainings of Dapco 2100 sealant, A50483 [CP2927], using a profiler tool, a swipe or with fingers using protective gloves.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CAUTION: MAKE SURE THAT YOU REMOVE UNWANTED SEALANT FROM THE BOTTOM SIDE OF THE ASSEMBLY. THIS CAN DECREASE THE RISK OF IGNITION DURING A FIRE. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

l) Make sure that there is minimum amount of the squeeze-out on the bottom side.

<1> Make sure that the sealant squeeze-out is finished flush with the surface.

<2> Make sure that the visible faces of the bolts [124] and harness strut (6:00 o'clock) [104] are clean and clear of graphite mineral vaseline, D00601 [CP2101].

m) Apply Dapco 2100 sealant, A50483 [CP2927], fillet around the head of the two bolts [124], in contact with the harness strut (6:00 o'clock) [104].

n) Apply Dapco 2100 sealant, A50483 [CP2927], fillet around each electrical harness, between the metal bobbin of each wire harness and top side of left outer metallic block [507] and left inner metallic block [508].

4) Install the new right inner metallic block [509] and right outer metallic block [510] on the harness strut (6:00 o'clock) [104] as follows:

a) Apply graphite mineral vaseline, D00601 [CP2101], on the threads and pressure faces of the two bolts [124] and two nuts [126].

b) Apply Dapco 2100 sealant, A50483 [CP2927], on the mating faces between the right inner metallic block [509] and right outer metallic block [510].

NOTE: Dapco 2100 sealant, A50483 [CP2927], must be evenly applied in large quantity on each hatching area to make sure that there is enough Dapco 2100 sealant, A50483 [CP2927], between parts after the squeeze out when assembling and tightening the bolts and nuts.

c) Apply Dapco 2100 sealant, A50483 [CP2927], on the mating faces between the right inner metallic block [509], right outer metallic block [510], and harness strut (6:00 o'clock) [104].

d) Install the two bolts [125] and two nuts [126] to assemble the right inner metallic block [509] and right outer metallic block [510] in the correct position with the harnesses.

e) Tighten the two nuts [126] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

f) Remove the unwanted graphite mineral vaseline, D00601 [CP2101], from the nuts [126] with a lint-free cloth, G51328 [CP2674].

g) Install the right inner metallic block [509] and right outer metallic block [510] in the correct position on the harness strut (6:00 o'clock) [104] by the bottom side.

h) Install the two bolts [124] that attach the right inner metallic block [509] and right outer metallic block [510] to the harness strut (6:00 o'clock) [104].

i) Tighten the two bolts [124] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

j) Remove the unwanted graphite mineral vaseline, D00601 [CP2101], from the bolts [124] with a lint-free cloth, G51328 [CP2674].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0288 (Continued)

k) Remove the squeezed out excedent and remainings of Dapco 2100 sealant, A50483 [CP2927], using a profiler tool, a swipe or with fingers using protective gloves.

CAUTION MAKE SURE THAT DAPCO 2100 SEALANT, A50483 [CP2927], IS FLUSH BETWEEN ADJACENT SURFACES. THE SEALANT CAN DECREASE THE RISK OF IGNITION DURING A FIRE. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

CAUTION MAKE SURE THAT YOU REMOVE UNWANTED SEALANT FROM THE BOTTOM SIDE OF THE ASSEMBLY. THIS CAN DECREASE THE RISK OF IGNITION DURING A FIRE. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

l) Make sure that there is minimum amount of the squeeze-out on the bottom side.

<1> Make sure that the visible faces of the two bolts [124] and harness strut (6:00 o'clock) [104] are clean and clear of graphite mineral vaseline, D00601 [CP2101].

m) Apply Dapco 2100 sealant, A50483 [CP2927], fillet around the head of the two bolts [124], in contact with the harness strut (6:00 o'clock) [104].

n) Apply Dapco 2100 sealant, A50483 [CP2927], fillet around each electrical harness, between the metal bobbin of each wire harness and top side of right inner metallic block [509] and right outer metallic block [510].

PTF ALL

5) Install the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503] on the strut 5 outer fitting [109] as follows:

a) Apply graphite mineral vaseline, D00601 [CP2101], on the threads and pressure faces of the five bolts [106] and two nuts [107].

b) Apply Dapco 2100 sealant, A50483 [CP2927] to the one side (either top or bottom) of the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503].

NOTE: Fay seal applies to the two sides of the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503].

c) Make sure that the EEC electrical harness (HJ8A) [303] and harnesses are in the correct position on the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503] (View E, Figure 413).

d) Install the two bolts [108] and two nuts [107] to assemble the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503].

e) Tighten the nuts [107] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

f) Remove the unwanted graphite mineral vaseline, D00601 [CP2101], from the nuts [107] with a lint-free cloth, G51328 [CP2674].

g) Make sure that the fillet seal of the squeeze out is on the forward facing side.

CAUTION MAKE SURE THAT YOU REMOVE UNWANTED SEALANT FROM THE AFT SIDE OF THE ASSEMBLY. THIS CAN DECREASE THE RISK OF IGNITION DURING A FIRE. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

h) Make the sealant squeeze-out flush on the outer edges of the strut 5 bracket upper block [500] and strut 5 bracket lower block [503] (View F, Figure 413).

i) Install the assembly of the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503] in a correct position on the strut 5 outer fitting [109].

j) Apply Dapco 2100 sealant, A50483 [CP2927], between the assembly of the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 outer fitting [109] (View C, Figure 413).

k) Install the five bolts [106] that attach the assembly of the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503] on the strut 5 outer fitting [109].

l) Tighten the bolts [106] to 67.45 in-lb (7.6 N·m) - 74.55 in-lb (8.4 N·m).

m) Remove the unwanted graphite mineral vaseline, D00601 [CP2101], from the bolts [106] with a lint-free cloth, G51328 [CP2674].

CAUTION MAKE SURE THAT YOU REMOVE UNWANTED SEALANT FROM THE AFT SIDE OF THE ASSEMBLY. THIS CAN DECREASE THE RISK OF IGNITION DURING A FIRE. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

n) Make sure that there is a minimum amount of the squeeze-out on the aft side around the edges of the block assembly and that the sealant is flush with the block assembly (View C, Figure 413).

o) Make sure that the squeeze-out on the forward facing side is finished as a fillet seal.

CAUTION MAKE SURE THAT THE BOLTS ARE CLEAN BEFORE YOU APPLY SEALANT TO THEM. IF YOU DO NOT REMOVE UNWANTED MATERIALS, THE SEALANT WILL NOT BOND. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

p) Clean the nut plate side of the bolts [106] and nut plates.

q) Apply Dapco 2100 sealant, A50483 [CP2927], to the nut plate side of the bolts [106] and nut plates.

(f) Attach blanket [190] to the strut 5 outer fitting [109].

1) Apply Dapco 2100 sealant, A50483 [CP2927], between the blanket [190] and strut 5 outer fitting [109] (View B, Figure 413).

NOTE: Dapco 2200 firewall sealant, A50396, may use as optional. If this consumable meets the local regulatory requirements of the operator, it is acceptable as a chemical sealant for further usage.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0410

(g) Apply engine oil, D00599 [CP2442], on the threads and pressure faces of the bolt [332], bolt [329], bolt [326], bolt [320], bolt [318], and bolt [316].

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0410

(h) Apply engine oil, D00599 [CP2442], on the threads and pressure faces of the bolt [332], bolt [326], bolt [320], and bolt [316].

PTF ALL

(i) Install the bolt [332] and clamp [333] that attach the EEC electrical harness (HJ8A) [303] to the forward stator case bracket.

(j) Tighten the bolt [332] to 33.25 in-lb (3.8 N·m) - 36.75 in-lb (4.2 N·m).

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0410

(k) Install the bolt [329] and clamp [330] that attach the EEC electrical harness (HJ8A) [303] to the forward stator case bracket.

(l) Tighten the bolt [329] to 33.25 in-lb (3.8 N·m) - 36.75 in-lb (4.2 N·m).

PTF ALL

(m) Install the bolt [326] and clamp [327] that attach the EEC electrical harness (HJ8A) [303] to the forward stator case bracket.

(n) Tighten the bolt [326] to 33.25 in-lb (3.8 N·m) - 36.75 in-lb (4.2 N·m).

(o) Install the bolt [320] and clamp [321] that attach the EEC electrical harness (HJ8A) [303] to the forward stator case bracket.

(p) Tighten the bolt [320] to 33.25 in-lb (3.8 N·m) - 36.75 in-lb (4.2 N·m).

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0410

(q) Install the bolt [318] and clamp [319] that attach the EEC electrical harness (HJ8A) [303] on the forward stator case bracket.

(r) Tighten the bolt [318] to 33.25 in-lb (3.8 N·m) - 36.75 in-lb (4.2 N·m).

PTF ALL

(s) Install the bolt [316] that attaches the EEC electrical harness (HJ8A) [303] to the bracket [317].

(t) Tighten the bolt [316] to 109.31 in-lb (12.35 N·m) - 120.81 in-lb (13.65 N·m).

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

(u) Use the teflon-jawed pliers, STD-664, to connect the EEC electrical harness (HJ8A) [303] as follows (TASK 70-00-01-910-803-G00):

1) Connect the EEC electrical harness (HJ8A) [303] connector to the T3 sensor channel A receptacle [336].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) Connect the EEC electrical harness (HJ8A) [303] connector to the EGT harness 1 (left) receptacle [335].

3) Connect the EEC electrical harness (HJ8A) [303] connector to the TCF accelerometer receptacle [334].

4) Connect the EEC electrical harness (HJ8A) [303] connector to the LPTACC valve receptacle [331].

5) Connect the EEC electrical harness (HJ8A) [303] connector to the HPTACC valve receptacle [328].

6) Connect the EEC electrical harness (HJ8A) [303] connector to the TBV receptacle [315].

7) Connect the EEC electrical harness (HJ8A) [303] connector to the start bleed/booster anti-ice valve receptacle [314].

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0410

8) Connect the EEC electrical harness (HJ8A) [303] connector to the bleed valve [515].

PTF ALL

(v) Connect the connector socket [309] of the EEC electrical harness (HJ8A) [303] as follows:

1) Install the four screws [311] that attach the connector socket [309] to the bracket [83].

2) Tighten the four screws [311] to 10.2 in-lb (1.15 N·m) - 11.1 in-lb (1.25 N·m).

3) Connect the HJ8B engine wiring harness electrical connector [146] (at 5:00 o'clock) to the connector socket [309].

NOTE: If the HJ8B engine wiring harness has been installed, do not do this step.

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0410

(w) Connect the connector socket of the EEC electrical harness (HJ8A) [303] to the bracket [83] as follows:

1) Install the four screws [311] that attach the connector socket of the EEC electrical harness (HJ8A) [303] to the bracket [83].

2) Tighten the four screws [311] to 10.2 in-lb (1.15 N·m) - 11.1 in-lb (1.25 N·m).

3) Connect the EEC electrical harness (W2B) [431] to connector socket of the EEC electrical harness (HJ8A) [303] and manually tighten the connector until you cannot see the blue mark on the receptacle.

PTF ALL

(x) Install the bolt [312] and clamp [313] that attach the EEC electrical harness (HJ8A) [303] to the fan case bracket.

(y) Tighten the bolt [312] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

(z) Install the bolt [307] and clamp [308] that attach the EEC electrical harness (HJ8A) [303] to the fan case bracket.

(aa) Tighten the bolt [307] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0410

(ab) Put the overlength of the EEC electrical harness (HJ8A) [303] between the bracket assembly [511] and bracket assembly [512].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0410 (Continued)

(ac) Put the overlength of the EEC electrical harness (HJ8A) [303] between the bracket assembly [513] and bracket assembly [514].

PTF ALL

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

(ad) Use the teflon-jawed pliers, STD-664, to connect the EEC electrical harness (HJ8A) [303] connector to the EEC channel A J8 receptacle [306].

(ae) Make sure that the work area is clean of all tools and other items.

SUBTASK 73-22-01-420-016-G00

(5) Install the segment shroud 5 [305] (TASK 72-23-04-400-801-G00).

SUBTASK 73-22-01-420-017-G00

(6) Install the segment shroud 2 [68] (TASK 72-23-04-400-801-G00).

SUBTASK 73-22-01-420-018-G00

(7) Install the segment shroud 1 [304] (TASK 72-23-04-400-801-G00).

H. Put the Airplane Back to Its Usual Condition

SUBTASK 73-22-01-410-029-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 73-22-01-860-061-G00

(2) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	4	C04008	ENGINE 1 ALTN PWR CHAN B
A	5	C04007	ENGINE 1 ALTN PWR CHAN A
B	3	C01312	ENGINE 1 RUN/PWR

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	5	C01313	ENGINE 2 RUN/PWR
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	3	C04018	ENG 1 STBY ALTN PWR RELAY CHAN A
E	4	C04019	ENG 1 STBY ALTN PWR CHAN A

SUBTASK 73-22-01-860-062-G00

(3) For the right engine, remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 73-22-01-860-063-G00

(4) For the left engine, remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

SUBTASK 73-22-01-860-064-G00

(5) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

I. Electronic Engine Control Electrical (EEC) Electrical Harness (HJ8A) Installation Test

SUBTASK 73-22-01-730-008-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

TASK 73-22-01-000-809-G00

18. Electronic Engine Control (EEC) Electrical Harness (HJ8B) Removal

(Figure 415, Figure 416, and Figure 417)

A. General

(1) This task gives the instructions on how to remove the electronic engine control (EEC) electrical harness (HJ8B).

(2) The electronic engine control (EEC) electrical harness is referred to as the EEC electrical harness.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
72-23-04-000-801-G00	Fan Frame Module Segment Shrouds Removal (P/B 401)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Removal

SUBTASK 73-22-01-860-065-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-22-01-860-066-G00

(2) Open these applicable circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	4	C04008	ENGINE 1 ALTN PWR CHAN B
A	5	C04007	ENGINE 1 ALTN PWR CHAN A
B	3	C01312	ENGINE 1 RUN/PWR

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	5	C01313	ENGINE 2 RUN/PWR
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	3	C04018	ENG 1 STBY ALTN PWR RELAY CHAN A
E	4	C04019	ENG 1 STBY ALTN PWR CHAN A

SUBTASK 73-22-01-860-067-G00

(3) For the right engine, open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 73-22-01-860-068-G00

(4) For the left engine, open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

SUBTASK 73-22-01-010-031-G00

(5) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Electronic Engine Control (EEC) Electrical Harness (HJ8B) Removal

SUBTASK 73-22-01-020-019-G00

(1) Remove the segment shroud 1 [304] (TASK 72-23-04-000-801-G00).

SUBTASK 73-22-01-020-020-G00

(2) Remove the segment shroud 2 [68] (TASK 72-23-04-000-801-G00).

SUBTASK 73-22-01-020-021-G00

(3) Remove the segment shroud 5 [305] (TASK 72-23-04-000-801-G00).

SUBTASK 73-22-01-020-022-G00

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(4) Use the teflon-jawed pliers, STD-664, to disconnect the EEC electrical harness (HJ8B) [378] as follows (TASK 70-00-01-910-803-G00):

(a) Disconnect the EEC electrical harness (HJ8B) [378] connector from the EEC channel B J8 receptacle [379].

(b) Disconnect the EEC electrical harness (HJ8B) [378] (at 5:00 o'clock) from the EEC HJ8A harness receptacle [145].

NOTE: If the HJ8A engine wiring harness has been removed, do not do this step.

(c) Remove the bolt [380] from the clamp [381] that attaches the EEC electrical harness (HJ8B) [378] to the fan case bracket.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(d) Disconnect the EEC electrical harness (HJ8B) [378] connector from the No. 1 bearing acclerometer receptacle [382].

(e) Disconnect the EEC electrical harness (HJ8B) [378] connector from the EGT harness 2 (right) receptacle [386].

(f) Disconnect the EEC electrical harness (HJ8B) [378] connector from the T3 sensor channel B receptacle [387].

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0410

(g) Remove the bolt [384] from the clamp [385] that attaches the EEC electrical harness (HJ8B) [378] to the forward stator case bracket.

PTF ALL

(h) Remove the EEC electrical harness (HJ8B) [378] from the clips [377] along its routing.

(i) Remove blanket [190] from the strut 5 outer fitting [109] (View B, Figure 416).

(j) Remove the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503] from strut 5 outer fitting [109] as follows:

1) Remove the five bolts [106] that attach the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503] to the strut 5 outer fitting [109].

2) Cut and remove the sealant between the assembly of the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], strut 5 bracket lower block [503], and strut 5 outer fitting [109].

3) Move the assembly of the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503] rearward from the electrical harnesses.

4) Remove the two nuts [107] and two bolts [108] to disassemble the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503].

5) Cut and remove the sealant on the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], strut 5 bracket lower block [503] and around the harnesses.

6) Remove the harnesses from the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503].

7) Remove any remaining sealant residue.

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0288

(k) Remove the harness support block [504], harness support block [505], and harness support block [506] from the harness strut (6:00 o'clock) [104] as follows:

1) Remove the four bolts [111] and four nuts [110] that attach the harness support block [504], harness support block [505], and harness support block [506] to the harness strut (6:00 o'clock) [104].

2) Cut and remove the sealant between the assembly of the harness support block [504], harness support block [505], harness support block [506], and harness strut (6:00 o'clock) [104].

3) Move the harness support block [504], harness support block [505], and harness support block [506] upward and away from the harness strut (6:00 o'clock) [104].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0288 (Continued)

4) Remove the two nuts [112] and two bolts [113] to disassemble the harness support block [504], harness support block [505], and harness support block [506].

5) Cut and remove the sealant on the harness support block [504], harness support block [505], harness support block [506], and around the harnesses.

6) Remove the harnesses from the harness support block [504], harness support block [505] and the harness support block [506].

7) Remove any remaining sealant residue.

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0288

(l) Remove the left outer metallic block [507], left inner metallic block [508], right inner metallic block [509], and right outer metallic block [510] from the harness strut (6:00 o'clock) [104] as follows:

1) Remove the four bolts [124] that attach the left outer metallic block [507], left inner metallic block [508], right inner metallic block [509], and right outer metallic block [510] to the harness strut (6:00 o'clock) [104].

2) Cut and remove the sealant between the assembly of the left outer metallic block [507], left inner metallic block [508], right inner metallic block [509], right outer metallic block [510], and harness strut (6:00 o'clock) [104].

3) Move the assembly of the left outer metallic block [507], left inner metallic block [508], right inner metallic block [509], and right outer metallic block [510] upward and away from the harness strut (6:00 o'clock) [104].

4) Remove the four nuts [126] and four bolts [125] to disassemble the left outer metallic block [507], left inner metallic block [508], right inner metallic block [509], and right outer metallic block [510].

5) Cut and remove the sealant on the left outer metallic block [507], left inner metallic block [508], right inner metallic block [509], right outer metallic block [510], and around the harnesses.

6) Remove the harnesses from the left outer metallic block [507], left inner metallic block [508], right inner metallic block [509], and right outer metallic block [510].

7) Remove any remaining sealant residue.

PTF ALL

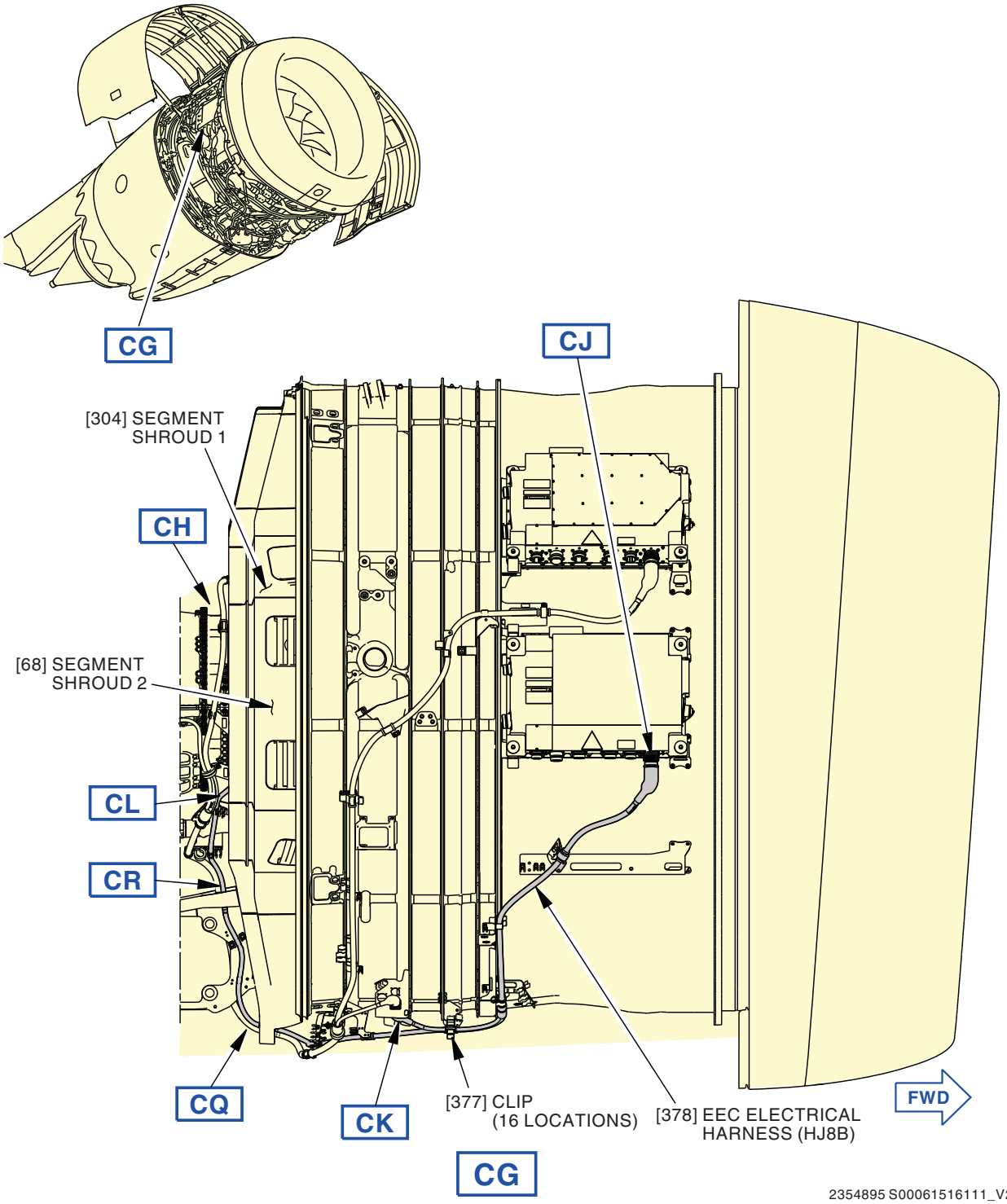
(m) Remove the EEC electrical harness (HJ8B) [378] from the harness strut (6:00 o'clock) [104].

(n) Carefully remove the EEC electrical harness (HJ8B) [378] from the engine.

(o) Install the protective covers on all of the harness connectors and the component receptacles.

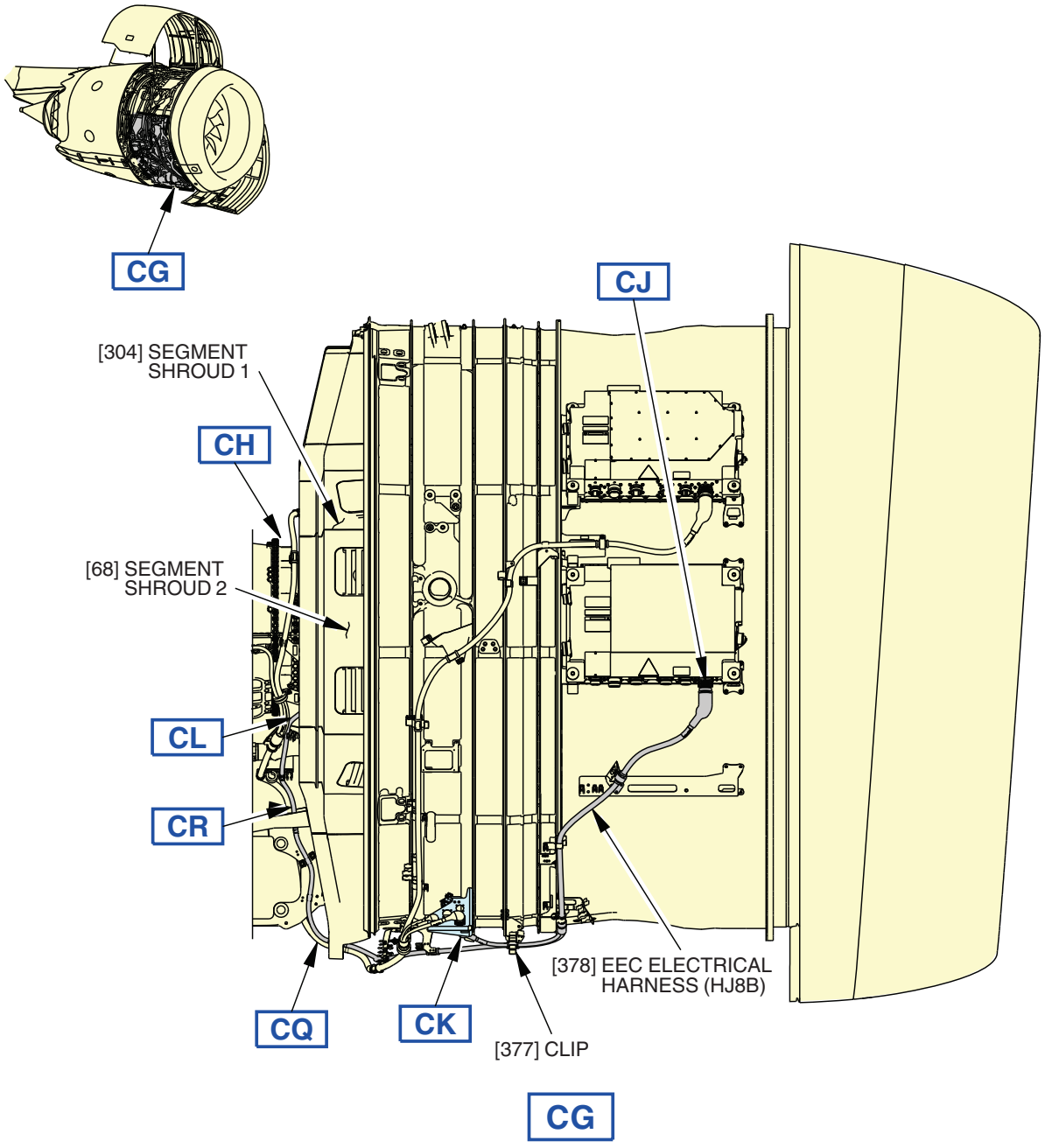
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



EEC Electrical Harness (HJ8B) Installation
Figure 415/73-22-01-990-859-G00 (Sheet 1 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

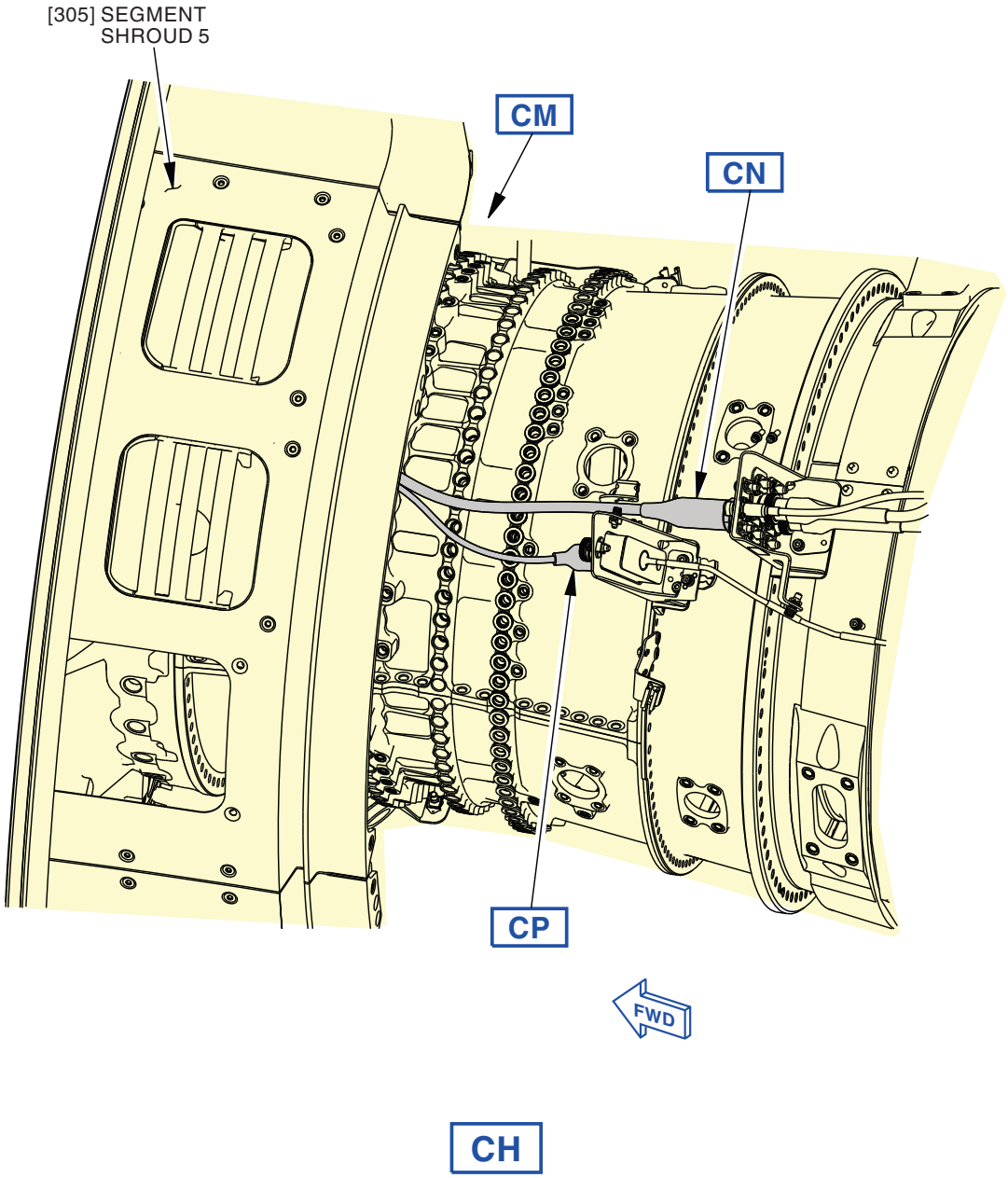
SOME COMPONENTS ARE HIDDEN FOR BETTER CLARITY.

ICNLEAP1B732201D5882800155A00101

3132227 S0000862163_V1

EEC Electrical Harness (HJ8B) Installation
Figure 415/73-22-01-990-859-G00 (Sheet 2 of 8)

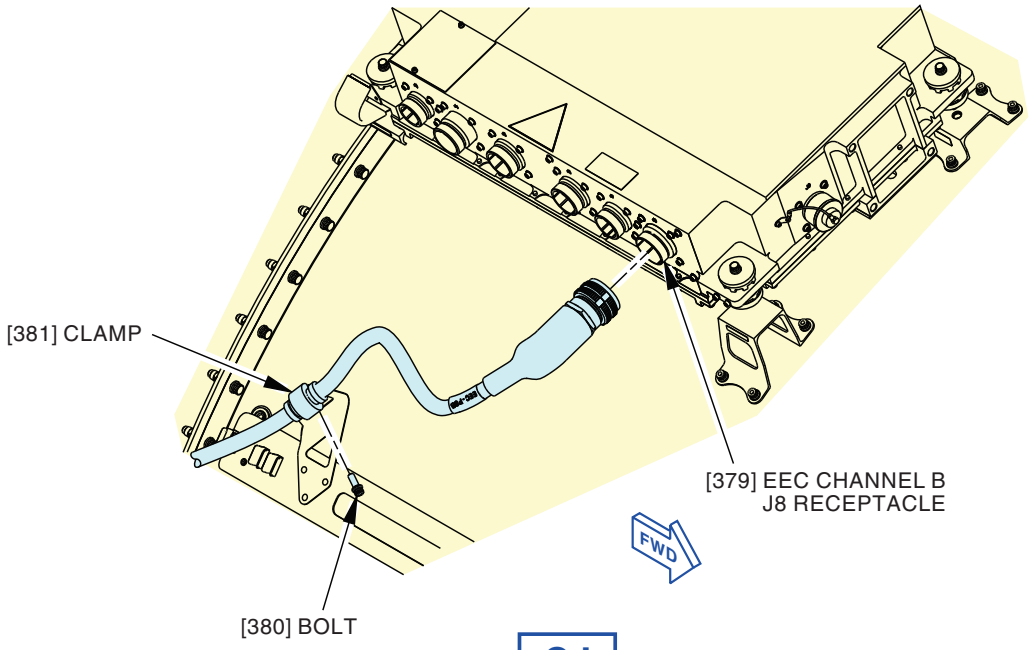
ECCN 9E991 BOEING PROPRIETARY - See title page for details

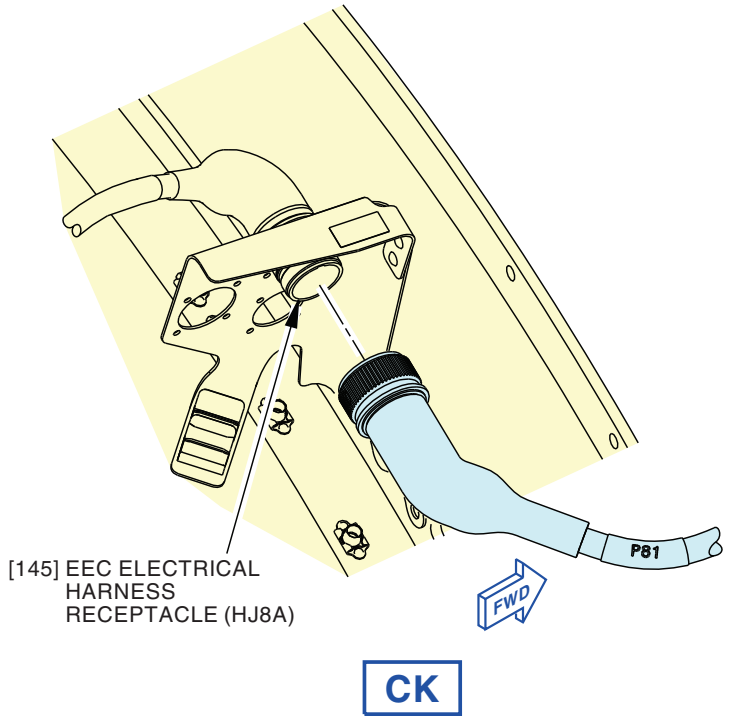


A0060-A-001-01
2420974 S0000557454_V1

EEC Electrical Harness (HJ8B) Installation
Figure 415/73-22-01-990-859-G00 (Sheet 3 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

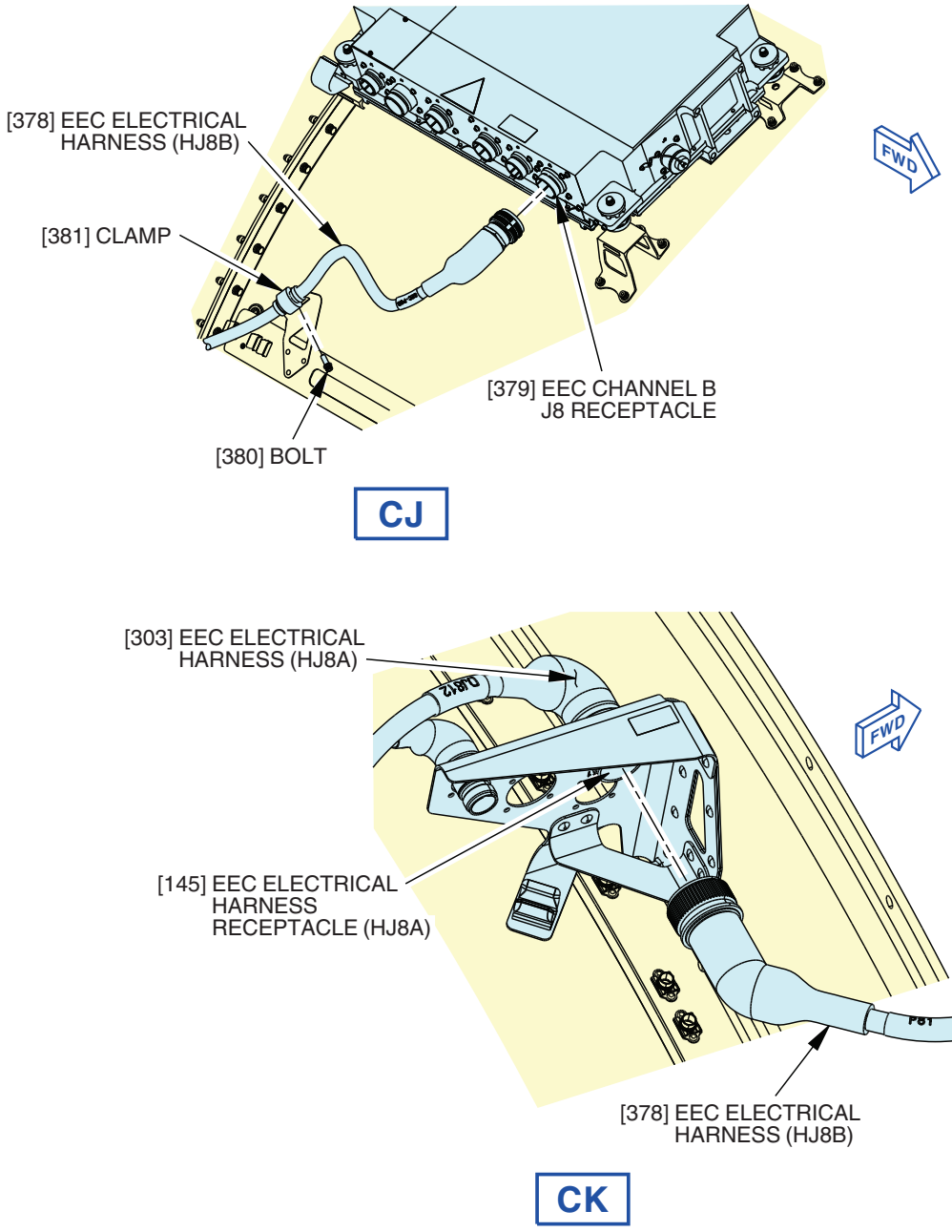




A0061-A-001-01
2420988 S0000557455_V1

EEC Electrical Harness (HJ8B) Installation
Figure 415/73-22-01-990-859-G00 (Sheet 4 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

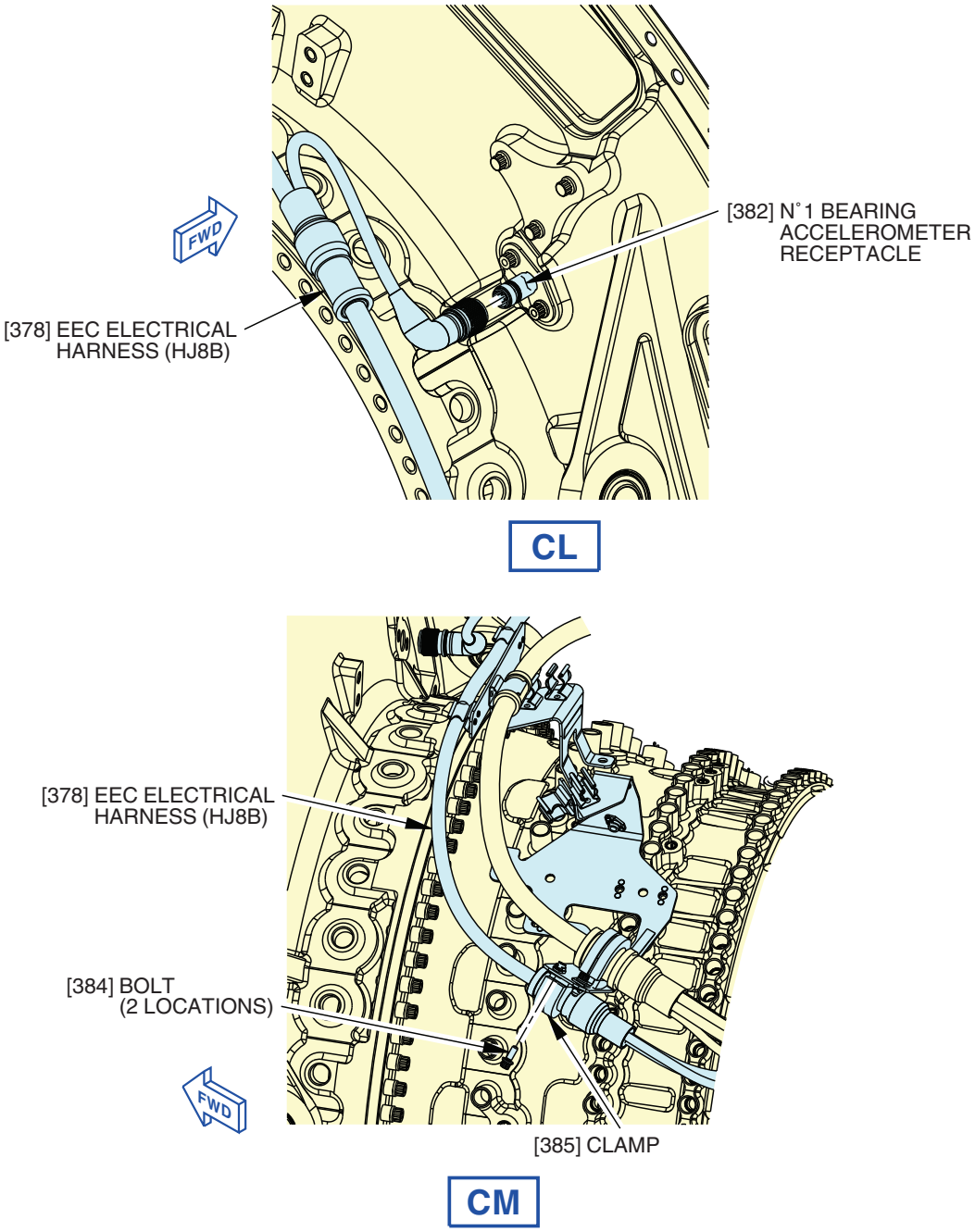
SOME COMPONENTS ARE HIDDEN FOR BETTER CLARITY.

ICNLEAP1B732201D5882800156A00101

3132229 S0000862164_V1

EEC Electrical Harness (HJ8B) Installation
Figure 415/73-22-01-990-859-G00 (Sheet 5 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

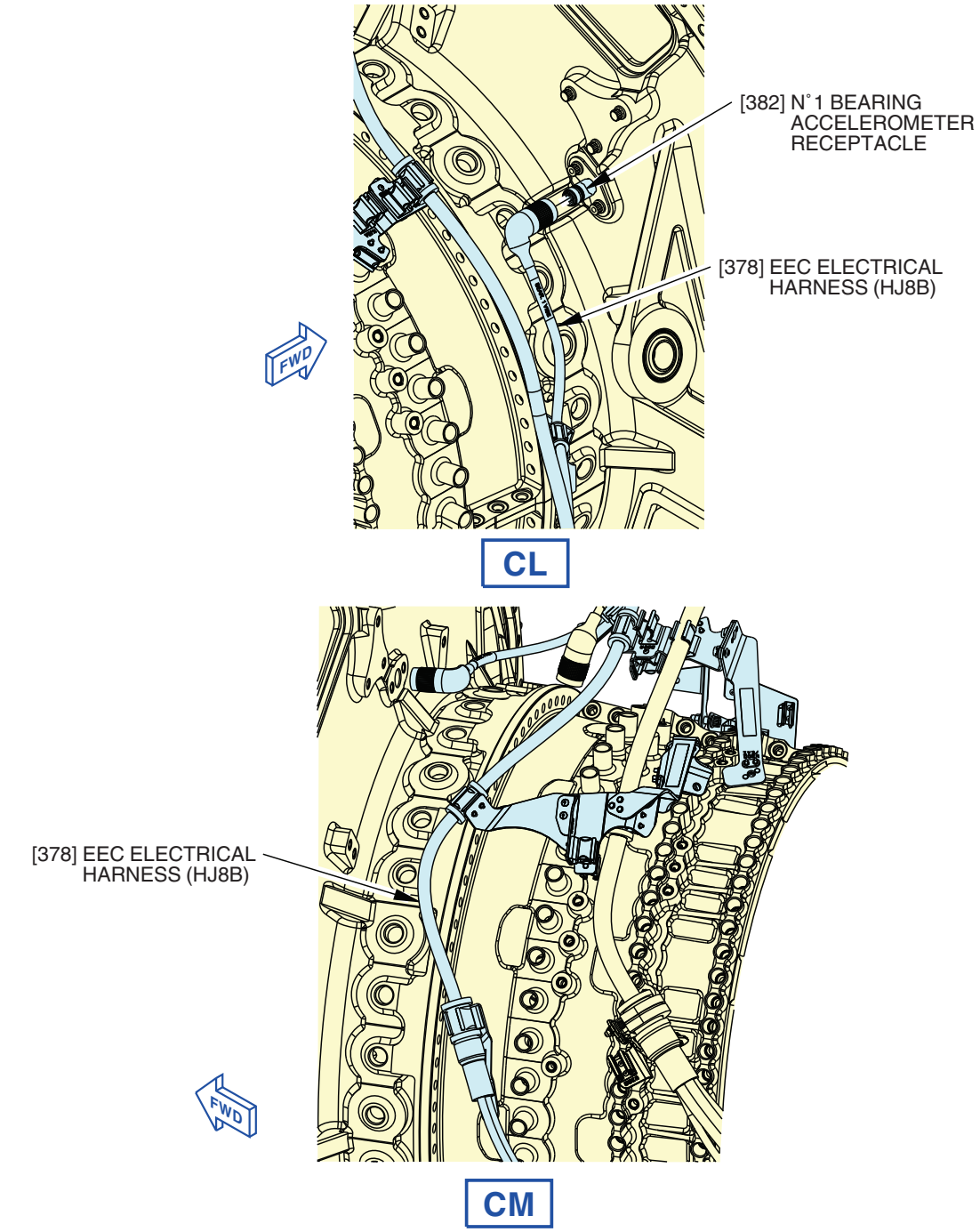
SOME COMPONENTS ARE HIDDEN FOR BETTER CLARITY.

ICNLEAP1B732201D5882800157A00101

3132231 S0000862165_V1

EEC Electrical Harness (HJ8B) Installation
Figure 415/73-22-01-990-859-G00 (Sheet 6 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

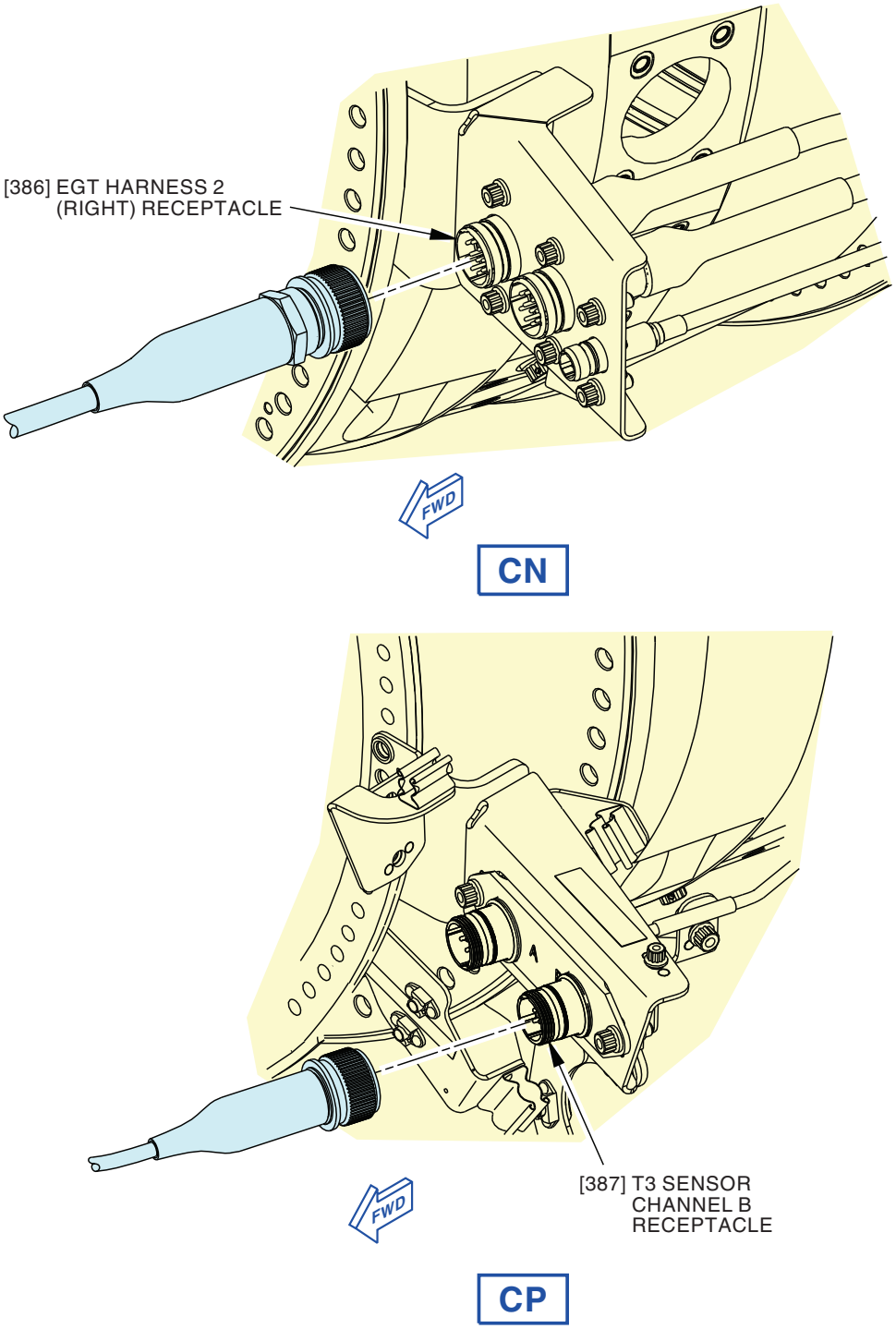
SOME COMPONENTS ARE HIDDEN FOR BETTER CLARITY.

ICNLEAP1B732201D5882800158A00101

3132234 S0000862166_V1

EEC Electrical Harness (HJ8B) Installation
Figure 415/73-22-01-990-859-G00 (Sheet 7 of 8)

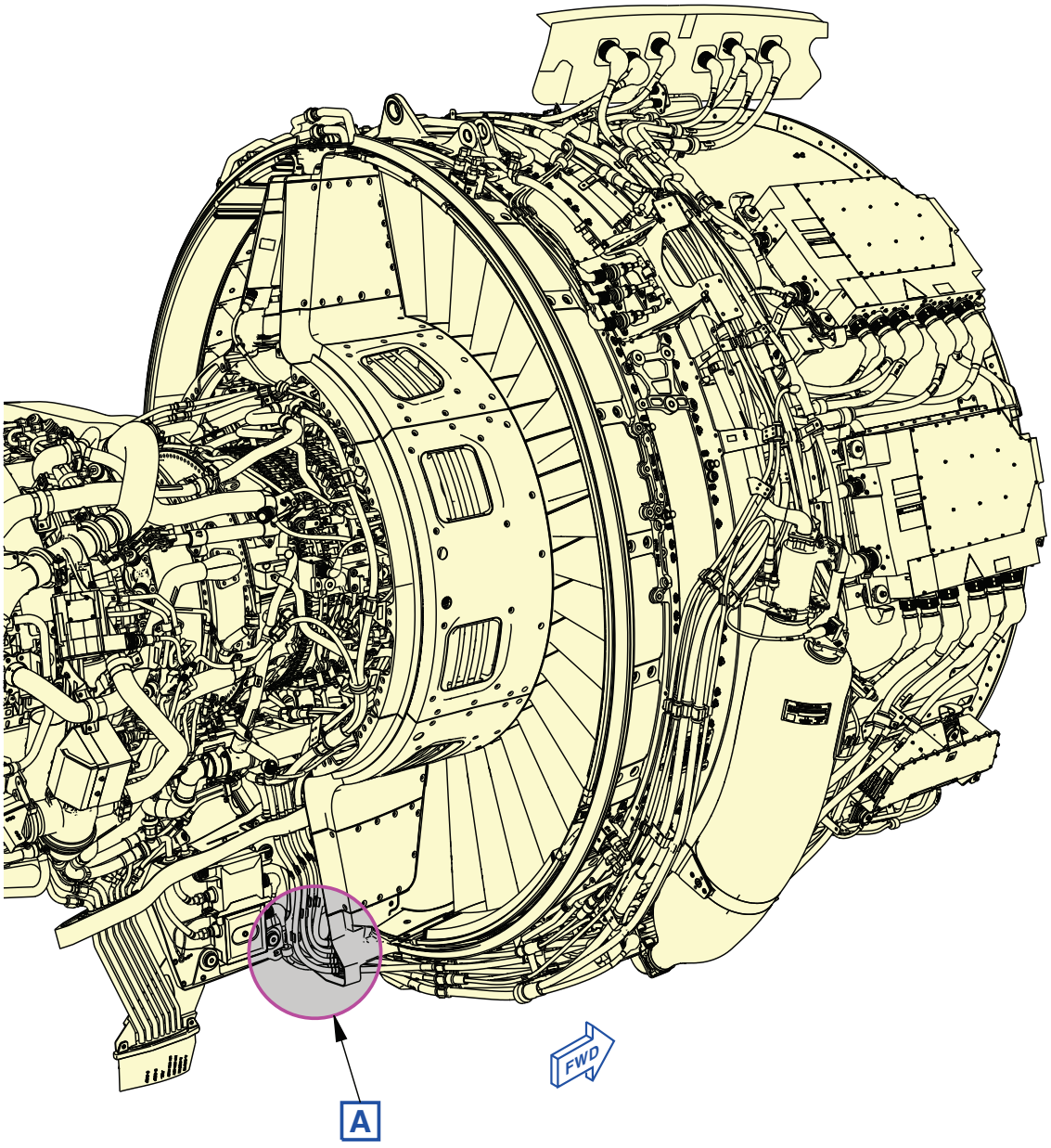
ECCN 9E991 BOEING PROPRIETARY - See title page for details



A0063-A-001-01
2421019 S0000557457_V1

EEC Electrical Harness (HJ8B) Installation
Figure 415/73-22-01-990-859-G00 (Sheet 8 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2937794 S0000713383_V3

Strut Bracket Block Installation
Figure 416/73-22-01-990-878-G00 (Sheet 1 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details